



KENYATTA UNIVERSITY

DIGITAL SCHOOL OF VIRTUAL & OPEN LEARNING

IN COLLABORATION WITH

DEPARTMENT OF ZOOLOGY

i

SZL 200/210: VERTEBRATE ZOOLOGY

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LESSON 1: VERTEBRATE CHARACTERISTICS

1.0 Introduction



Welcome to lesson 1. In this lesson you will learn how to distinguish vertebrates from other animals. You will also be introduced to the milestones in the evolution vertebrates that form the bases for their classification.

Apart from the diagrams in this module illustrating the characteristics of vertebrates, arrangements will be made for you to observe these characteristics in preserved specimens. The department of zoology has a museum where these preserved specimens are stored.

1.1 Objectives



At the end of this lesson you should be able to

1. List the characteristics of vertebrates
2. Explain the relationship between vertebrates and their invertebrate relatives in Phylum Chordata.
3. Outline the criteria used in classifying: vertebrates



1.2 Position of Vertebrates in the Animal Kingdom

What is a vertebrate?

For a long time vertebrates were defined as animals with a backbone. It is now known that there are vertebrates that lack this feature. But all vertebrate have a cranium. I will explain what a cranium is shortly.

Vertebrates belong to Phylum Chordata. This is one of the most successful phyla of the animal kingdom occupying most habitats. Members include two subphyla of animals without a backbone or cranium, and the vertebrates that have cranium and most have a backbone. The former are

termed acraniata or invertebrate chordates and the latter as craniata or vertebrates. The invertebrate chordates belong to subphyla Urochordata and Cephalochordate. We will learn more about these two subphyla in lesson 2. How comes that animals that have vertebral column and others without are put in the same phylum?

This is because they;

- Share certain similarities in their embryonic development
- Display at some point in their life cycle the three diagnostic characteristic of the phylum.
- They have a notochord, a stiffening rod Located along the back. The notochord is a flexible rod of cells along the dorsal region of the body. All chordates have a notochord at least as embryos. It provides support.
- They have a dorsal hollow nerve cord that lies dorsal to the notochord.
- The wall of the throat region (pharynx) has clefts usually referred to as gill slits. These are openings that allow water to pass through from the throat region to the outside. In fish and other aquatic chordates they allow the movement of water over the gills.

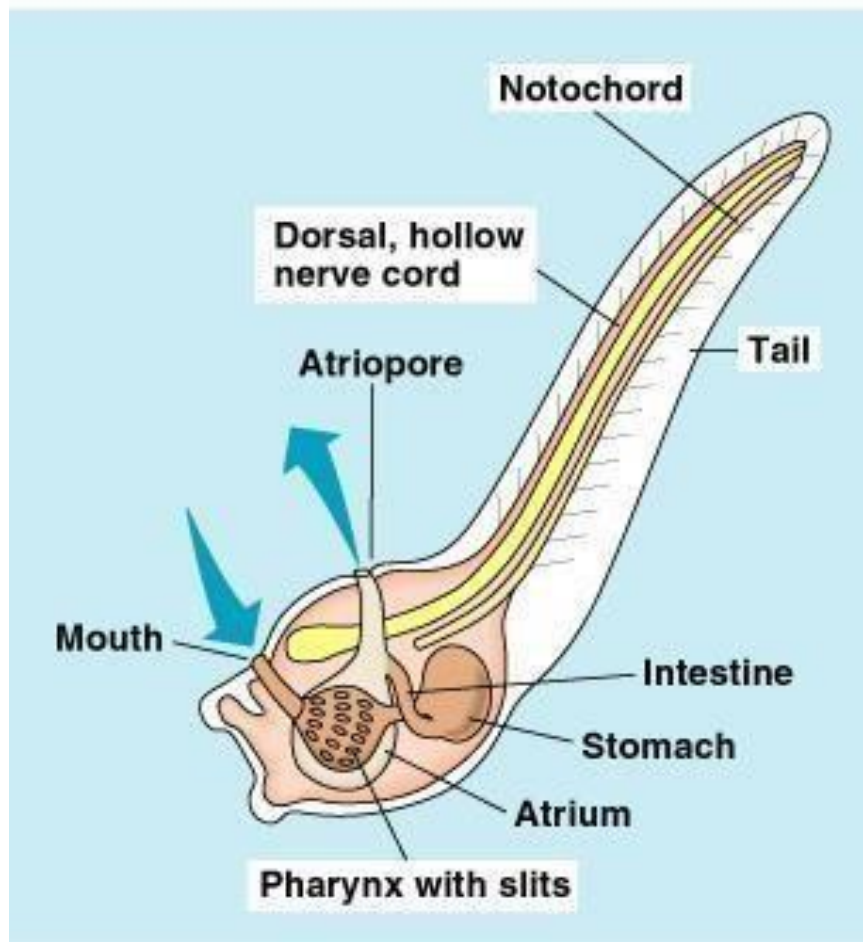


Fig 1.1 Urochordate (Ascidia) larva showing the characteristics of chordates Activity 1.1



Study Figure 1.1 and note the

- The anterior and posterior end of the animal
- Position of the notochord
- Position of the nerve cord
- The pharyngeal gill slits

We will compare the structure of this larva with those of other chordates later

In many chordates these features are modified during early development. Thus notochord is present in embryos of urochordates but lost in adults. It persists through life in Cephalochordates (Amphioxus) and in vertebrates it is present in embryos but replaced by a series of vertebrae that form the backbone (vertebral column) in adults. The nerve cord is replaced by the spinal cord in adult vertebrates and is enclosed by the vertebral column. In terrestrial vertebrates (reptiles, birds and mammals) the gill slits are present in the embryo but never perforate the neck and they disappear before birth. The gill slits resemble similar structures in hemichordates (reason, previously hemichordates were put in the same phylum with chordates).

You will learn more about other members of phylum Chordata but at this point I would like us to define what vertebrates are.



1.3 Definition of a Vertebrate

The term vertebrate comes from the vertebrae that form the backbone in most vertebrates. So from a very early stage we define vertebrates as animals that possess a backbone. True, most of them do but not all animals included in this subphylum have vertebrae. Hagfishes and lampreys that are normally classified as vertebrates do not.

So then is there a unifying feature that is unique to vertebrates and is shared by all? Yes

- The cranium. All vertebrates have a brain case that encloses the brain. It may be made of bone, cartilage or fiber-like material. Thus many people would rather refer to this subphylum as Craniata rather than as Vertebrata.
- Another unique characteristic of vertebrates is the development of embryonic tissue known as neural crest that form many new structures in vertebrates especially in the head region including: peripheral nervous system, central nervous system, several endocrine glands, part of circulatory system, the skeleton of the head.
- A third important distinguishing characteristic of vertebrates is the enlarged and structurally complex brain.

There are other characteristics of vertebrates that distinguish vertebrates from invertebrate chordates but are not necessarily unique to them. Vertebrates differ from invertebrate chordates in size (most are much bigger) and their level of activity. Most vertebrates are many times bigger than the invertebrate chordates and more active.

One reason why vertebrates are very active is that they can support a burst of activity using anaerobic metabolism (metabolism that does not use oxygen). They do so by metabolizing carbohydrates into lactic acid that is then metabolized aerobically. Perhaps this is the reason why they developed an internal skeleton made of calcium phosphate instead of calcium carbonate, as is the case with invertebrates that have an external skeleton. Calcium phosphate does not dissolve in lactic acid.

Another adaptation for increased activity is the replacement of the notochord with a segmented

Also, vertebrates are bilaterally symmetrical. The symmetry of an animal seems to correlate with the degree of activity. Animals that are bilaterally symmetrical are more active than those that have radial symmetry or are asymmetrical.

Activity 1.2



1. List two examples of animals with radial and bilateral symmetry
2. List them in order of their degree of activity.

Vertebrates have the sense organs and nervous tissue concentrated at the front of the body (Cephalization) as in other active groups of animals such as insects.

To support the increased level of activity, more food and a high rate of metabolism is necessary. Consequently, vertebrates had to evolve efficient methods of gathering food and an efficient digestive system. Thus, the gut became more muscularized and differentiated into regions specialized to perform specific functions in the digestion process—mouth, pharynx, esophagus, stomach, and intestines. In some lower vertebrates, the esophagus is absent, while in mammals, the gut terminates in the anus. In many groups, the last segment of the gut is the cloaca, a common chamber into which the intestine and urinary and genital ducts lead.

The internal transport system (circulatory and excretory) became more efficient.

Note:



The simple definition of a vertebrate uses only one of the many characteristics of these animals. However, there are many characteristics which when used in combination give a clearer picture of what is a vertebrate.

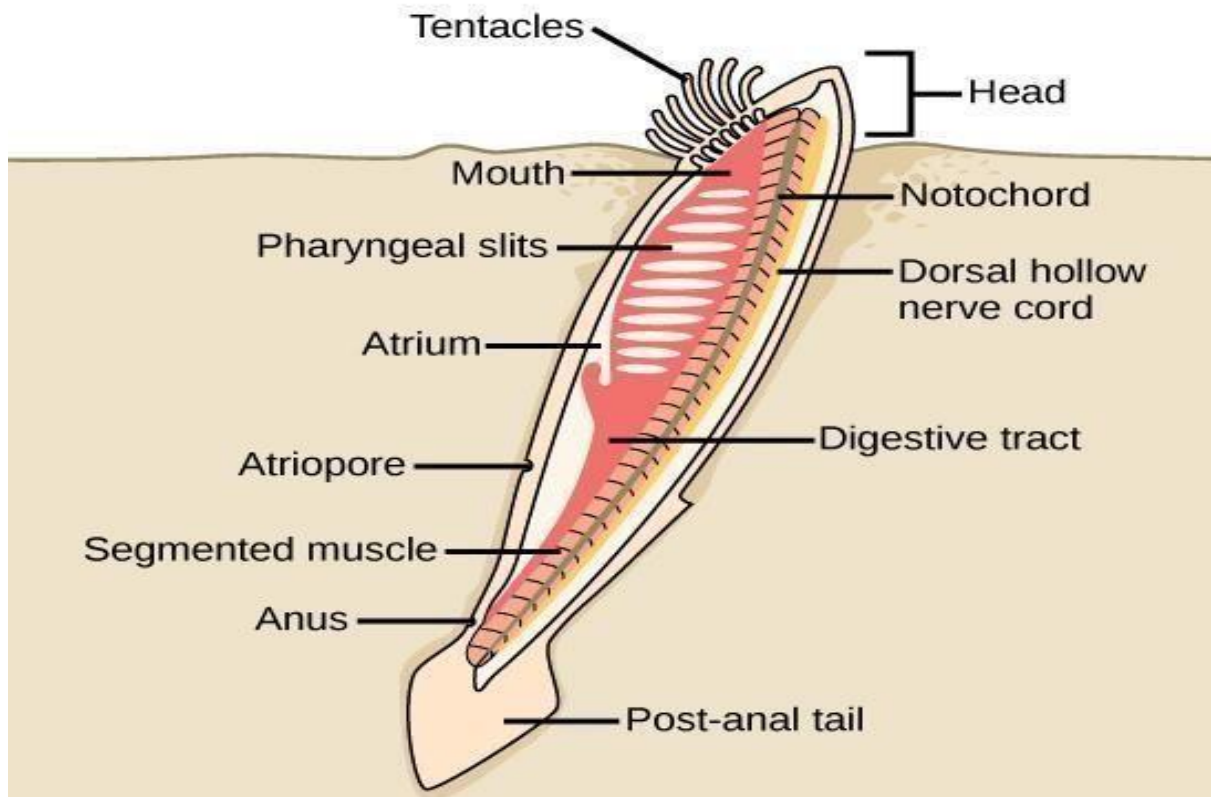


Figure 1.2 The distinctive features of vertebrates.



1.4 Diversity and Classification of Vertebrates

The features discussed above were acquired early in the evolution of vertebrates and enabled them to diversify in size, mode of life, range of habitats e.tc.

Activity 1.3



List twenty vertebrates, fish, frogs, lizards, birds and mammals. For each animal indicate

- Where it lives
- What it feeds on
- The time of day it is active
- Size

From this simple exercise you must have noticed that vertebrates feed on different types of food, occupy different habitats and are active at different times of the day. It means that the vertebrates in your area are diversified in these respects.

There are about 50000 known species of vertebrates. The smallest weigh a few grams and the largest several tons. They are found in almost all habitats- water, trees, deserts, polar areas. They have a variety of feeding methods- herbivores, carnivores, filter feeders, nectar feeders etc. They have a wide range of breeding behaviour. Some pair for life just like humans, others pair only during the breeding season, some form no bonds. Some take care of their young while others don't. In some it's the male that cares for the young, in others it is the female and in some it is both parents. In most the basic vertebrate body plan is modified to meet the requirements of a particular habitat, mode of life and so on. Since we cannot study every vertebrate for us to learn, a brief classification can be given.

Classification of Chordata

PHYLUM CHORDATA

Group ACRANIA (=PROTOCHORDATA)

(Primitive chordates without head and vertebral column)

Subphylum **HEMICHORDATA**, *Balanoglossus*, *Cephalodiscus*, *Rhabdopleura*, primitive and doubtful chordates, now classified under non-chordates after echinoderms.

Subphylum **UROCHORDATA**, *Herdmania*, *Salpa*, *Doliolum*, *Pyrosoma*, *Oikopleura* sedentary or planktonic tunicates in which chordate characters manifest in the larval stage.

Subphylum **CEPHALOCHORDATA**, *Amphioxus*, *Asymmetron*, typical chordates having chordate characters in the larval as well as adult stage.

Group **CRANIATA (=EUCHORDATA)**

(Chordates with skull, with 54,000 species of true chordates)

Subphylum **VERTEBRATA**, chordates with head, brain and vertebral column.

Superclass **AGNATHA**, 90 species of paraphyletic group of jawless fishes, which were also the first vertebrates. Living forms are elongated, scaleless, slimy parasites and scavengers that include lampreys and hagfishes. They have no paired fins.

Class **OSTRACODERMI**, extinct shelled jawless fishes of Ordovician period. *Cephalaspsis*.

Class **CYCLOSTOMATA**, jawless fishes of today, without scales and paired fins.

Order **Myxinoidea**: the hagfishes, 40 species. *Myxine*, *Bdellostoma*, *Eptatretus*.

Order **Petromyzontia**: lampreys, 41 species, parasitic on other fishes. *Petromyzon*.

Superclass **GNATHOSTOMATA**, vertebrates with jaws that are modified gill arches and paired appendages. They include cartilaginous fishes, bony fishes and tetrapods.

FISHES

Class **PLACODERMI**, extinct group of spiny sharks. *Climatius*.

Class **CHONDRICHTHYES** – cartilaginous fishes that have cartilaginous skeleton, ventral mouth, placoid scales, heterocercal tail fin and 5 pairs of gill slits.

Subclass **Elasmobranchii**— 850 species of sharks, rays and skates.

Subclass **Holocephali**— 30 species of ratfish (*Chimaeras*).

Class **OSTEICHTHYES** – 20,000 species, bony fishes. Skeleton contains bone, four pairs of gills, covered with operculum. Possess swim bladder or lung.

Subclass **Actinopterygii**, ray-finned fishes.

Superorder **Chondrostei**, 25 species of sturgeons, bichirs and paddlefish.

Superorder **Holostei**, which includes *Lepistosteus* 7 species and *Amia* 1 species.

Superorder **Teleostei**, includes 20,000 species of bony fishes, such as tarpon, herring, perch, etc.

Subclass **Crossopterygii**, includes 2 species of coelacanth (*Latimeria*).

Class **CHOANICHTHYS (=DIPNOI)**, has 6 species of lungfishes under three genera, namely, *Protopterus*, *Lepidosiren* and *Neoceratodus*.

TETRAPODS

Class AMPHIBIA

Order **Gymnophiona** (Apoda) —165 species of burrowing caecilians, elongated bodies, limbless, dermal scales embedded in annular folds of skin.

Order **Caudata** (Urodela) —425 species of salamanders, tailed amphibians, usually with two pairs of limbs.

Order **Anura** (Salientia) —4300 species of frogs and toads, tail-less amphibians, long hind limbs for jumping, head and trunk fused. They have sound producing and hearing organs.

Class **REPTILIA**, 7800 species, turtles, crocodiles, lizards, snakes, etc.

They have internal fertilization and produce large cleioid eggs with leathery shells and are ectotherms. Body covered with epidermal scales, vertebrae procoelous.

Order **Chelonia**, 300 species of turtles and tortoises having bony shell on the body.

Superorder **Lepidosauria**, lizard-like with acrodont or pleurodont dentition.

Order **Rhynchocephalia (=Sphenodontia)**, 2 species of tuatara (*Sphenodon*) in New Zealand. They have acrodont teeth, amphicoelous vertebrae and a parietal eye.

Order **Squamata**, with pleurodont teeth, procoelous vertebrae, without third eye.

Suborder **Lacertilia**, includes 4000 species of lizards.

Suborder **Ophidia**, includes 2700 species of snakes.

Superorder **Archosauria**, includes modern crocodiles and extinct dinosaurs.

Order **Crocodylia**, 23 species of alligators, crocodiles and gavials

Class **AVES**, 9100 species.

Birds being feathered bipeds have internal fertilization and lay hard-shelled eggs and are endotherms. Nearly every anatomical feature is related to ability to fly. They are the only animals with feathers that are modified from reptilian scales.

Class **MAMMALIA**, 4,500 species.

Mammals evolved in the late Triassic, the time dinosaurs first appeared and diversified greatly following the extinction of dinosaurs during the Coenozoic. Characteristics include hairs for protection and from heat loss; mammary glands; heterodont teeth; endothermy; 4 chambered heart etc.

Subclass **PROTOTHERIA**, **Order Monotremata**, egg laying mammals having 6 species in Australia and New Zealand. No teeth and true mammary glands.

Subclass **METATHERIA**, **Order Marsupialia**, 275 species of marsupials that have brief gestation period after which the embryo develops in a pouch. They have prolonged lactation and parental care. Marsupials include: opossum, kangaroo, koala, Tasmanian devil, wombat, etc.

Subclass **EUTHERIA**, **16 orders** which include 4700 species of placental mammals that are truly viviparous, with a placenta for gas and nutrient exchange between the mother and foetus. They also have true mammary glands.

Revision questions



- How appropriate is the term vertebrate
- List the major milestones in the evolution of vertebrates
- In one sentence explain what is a vertebrate

1.5 Summary



You have learnt that vertebrates are distinguished from other organisms by a set of characteristic features. The group is diverse resulting in modification of the basic vertebrate plan to cope with challenges of specific environment.

Further reading



1. Storer, T.I., R.L. Usinger, R.C. Stebbins and J.W. Nybakken (1975), Zoology 5th Ed, Tata McGraw-Hill Publishing Co Ltd, New Delhi.
2. Harris C.L. (1996). Concepts in Zoology, Second edition. HarperCollins College Publishers,
3. Hildebrand M. (1995). Analysis of Vertebrate Structures, fourth edition. John Wiley & Sons, New York

LESSON 2: THE ORIGIN OF VERTEBRATES 2.0 Introduction



In the previous lesson you learnt about the characteristics that define vertebrates. You also learnt that vertebrates are grouped in the same phylum with invertebrates believed to be their closest relatives. In this lesson we will learn more about the non-vertebrates chordates and the evolutionary phylogeny relationship between them and vertebrates. Unless you come from the coast, animals described in this lesson may not be familiar to you. However you will be able to see preserved specimens in the laboratory (practical 1)

2.1 Objectives



At the end of this lesson you should be able to;

1. Outline the evolutionary sequence of chordates
2. Compare the features of the three subphyla of chordates
3. Explain the phylogenetic significance of hemichordates.



2.2 Chordate Evolution

Chordates share certain characteristics with some invertebrate phyla. For example they share the feature of metameric segmentation (**division of animal into similar segments or metemeres**) with annelids and insects although in the vertebrates it is confined to the embryonic stage. For this reason these groups were considered to be the ancestors of chordates.

However, on closer look, the resemblance is superficial for the following reasons;

- In annelids and insects, the segmentation divides the body almost into complete rings and the internal organs are affected while in chordates it is not the case. The annelids and insects are usually referred to as annulates (meaning animals composed of rings, from the Latin word *anulus* meaning ring).
- Also the nerve cord in the annulates is ventral to the gut while in the chordates it is dorsal.

Also the plan of limbs is different.

So because of these differences, the theory of annulates being the ancestors of chordates was discarded.

Animals that have similar embryonic development are likely to be phylogenically closer (phylogeny is the sequence of events involved in the evolution of a species). Early chordate development resembles that for members of Phyla Echinodermata and Hemichordata in that;

- They have indeterminate cleavage, if a blastomere is removed during early cleavage it can develop into a full animal. This is in contrast to determinate cleavage where a blastomere will only develop into that part of the animal that it would have become had it not been separated.
- Also the mouth does not arise from the blastopore of the gastrula as is the case in other phyla.

Instead, the blastopore develops into the anus and a new opening develops to form the mouth i.e. they are deuterostomes (two openings).

It is believed that these three phyla have a common ancestor. However, the evolutionary link between them is not clear. However, hemichords seem to be more closely related to chordates than echinoderms are. In fact, at one time hemichordates were put in phylum chordate.

Note



It is not known the exact invertebrate phylum from which Phylum Chordata evolved. We will now examine the characteristics of invertebrates closely related to the vertebrates and the events leading to the evolution of vertebrates



2.3 Phylum Hemichordate

Hemichordates were once classified as chordates but were later removed and put in their own phylum because they lack one of the most important characteristics of chordates, the notochord. However, they are of great significance because they may provide clue as to the origin of chordates. They have some features that they share with other invertebrates especially echinoderms and others they share with or are similar to those of chordates. Thus, they seem to form an evolutionary link between invertebrates' phyla and Phylum Chordata.

- The hemichordates trochophore larvae and the veliger larvae of echinoderms resemble each other in being transparent and having almost identical ciliated bands and similar number of coelomic cavities.
- They share with chordates the following features: having pharyngeal gill slits and dorsal nerve cord.

Hemichordates are marine 0.02-2 m worm-like organisms. They are divided into two classes that differ greatly as adults but very similar in their developmental stages;

Class Enteropneusta- acorn worms

Class Cephalochordata

Acorn Worms

Have worm-like body divided into three regions: proboscis, collar and trunk. These regions are shown in Figure 2.1

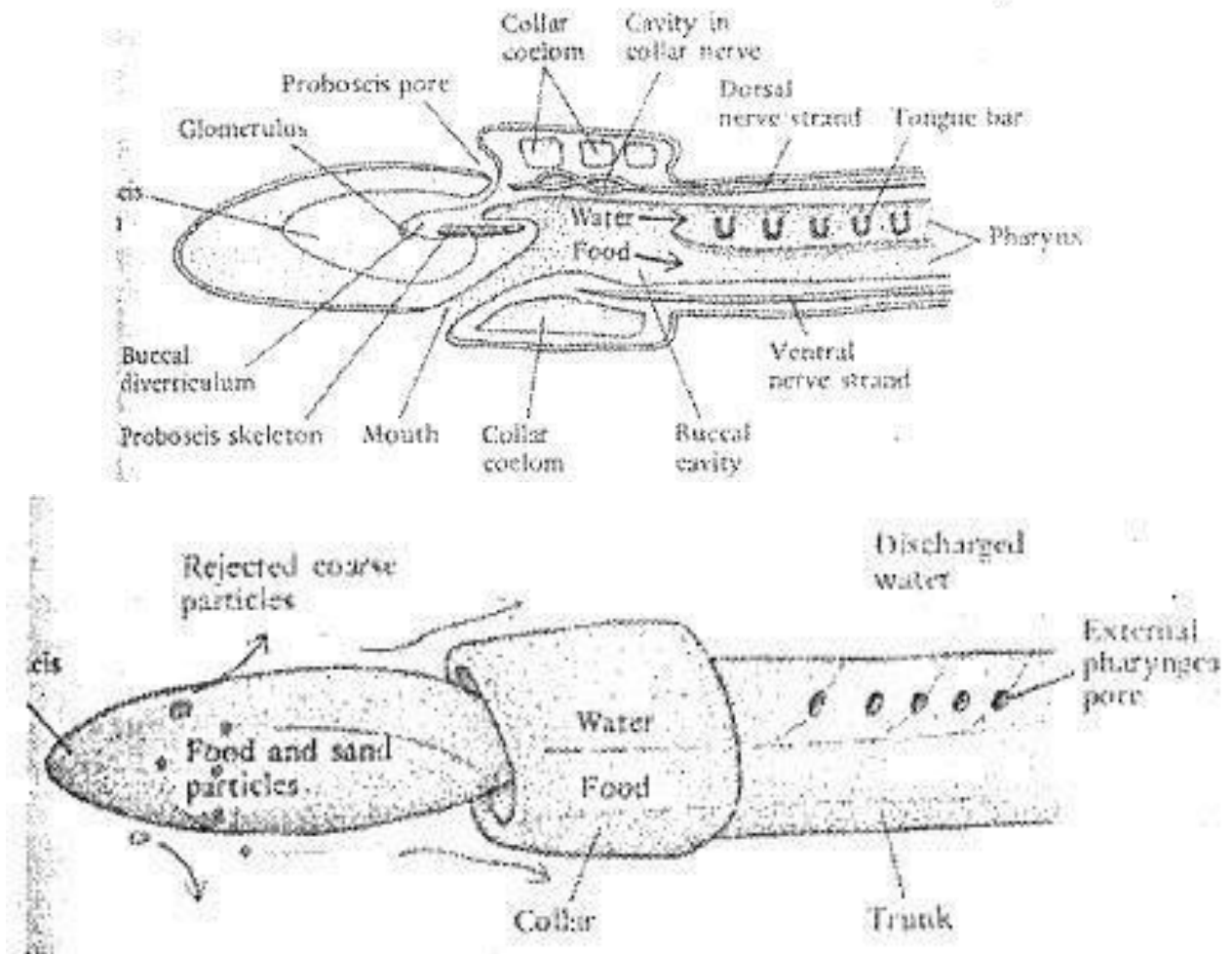


Figure 2.1 Structure of acorn worm

Activity 2.1



Study figure 2.1 List the diagnostic feature of chordates that this animal has in common with the ascidians larva in Figure 1.1.

2.4 Subphylum urochordata (tunicates)

Tunicates are widely distributed and are found from the polar region to the tropics. They vary in size from microscopic to about 30cm in length. Some are solitary while others are colonial. There are about 2000 species belonging to three classes: Ascidia, Thaliacea and Larvacea. The bestknown

tunicates are the ascidians or sea squirts (because they squirt water from the body openings when touched).

At first glance, adults of this subphylum resemble sponges more than they resemble chordates. Like sponges they are sessile, attached to the substratum, and have a body that is basically a sac with two opening and no coelom. Their bodies are covered in sac-like covering known as tunic from which they get their name. The larvae of tunicates are free living and look very different from the adults. Unlike the adults they possess the three characteristics that distinguish Phylum Chordata from all other phyla: notochord, dorsal hollow nerve and pharyngeal slits.

Adult ascidia

The adult is covered by a tunic made of fibers of tunicin, a rare form of cellulose, and is lined by a membranous mantle containing muscle fibers and blood vessels. It has two openings the incurrent or buccal siphon and the excurrent siphon or atrial aperture. Study Figure 2.2 and note the position of the structures described below.

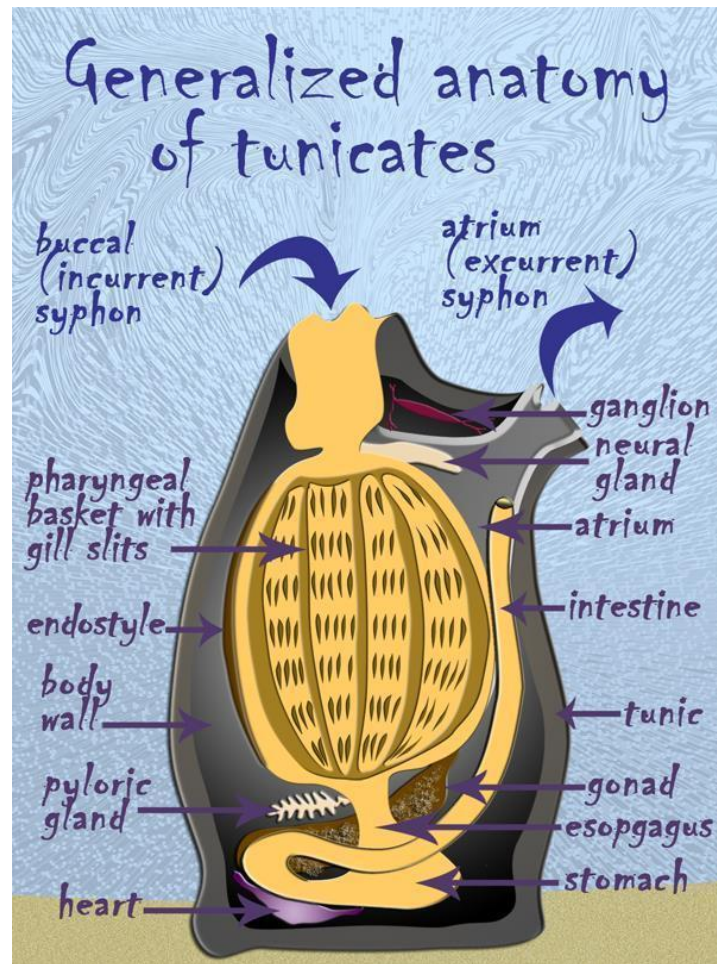


Fig 2.2 Structure of adult ascidia

Digestive system

Water enters the incurrent siphon then into the pharyngeal basket or branchial sac. The food in the water is trapped in a sheet of mucous secreted by endostyle. The endostyle is a groove situated along mid-ventral wall of the branchial sac. From the base of the branchial sac is a short esophagus leading to the stomach. The stomach connects to the intestine. The intestines end in anus just below the excurrent siphon. The wall of the branchial sac has many gill slits lined by ciliated cells. Movements of the cilia cause water to move from within the sac to the atrial cavity. From the atrial cavity, it flows to the excurrent siphon and as it does it carries with it fecal material from the anus.

Circulatory system

The circulatory system consists of a tubular heart near the stomach. On each end is a large vessel, aorta. One of the vessels goes to the stomach and the tunic. The other goes to the opposite side of the branchial sac. The direction of blood flow reverses every few minutes.

Nervous system

This consists of a ganglion in the mantle between the two siphons. It sends nerves to various body parts. Near the ganglion is the neural gland that is endocrine in nature.

Reproductive system

Ascidians are hermaphrodite. Some reproduce asexually by budding but most have sexual reproduction. They have a single ovary and testis. The oviduct and sperm duct open into the atrial cavity near the anus fertilization is external in some species. In others, it takes place within the atrial cavity and the eggs are brooded there to early embryonic stages. The fertilized egg develops into an elongate embryo and hatches into a tadpole-shaped free-swimming larva. The tail of the tadpole has a notochord (a longitudinal rod of turgid cells encased in a fibrous sheath), nerve cord and segmental muscles. The notochord cannot be compressed and it cannot be shortened by contraction of the longitudinal muscles. So the contraction of these muscles cause bending or undulations of the tail and propel the larva. On the head of the tadpoles three adhesive glands, a mouth leading to a pharynx. The pharynx is perforated by gill slits. The slits open into a chamber, the atrium just like in the adults. Some gaseous exchange may occur as water passes through the gill slits but their main function is feeding just as in adults. The nerve cord is single, hollow and dorsal. Its main function is to coordinate the muscle activities. There is also a sensory structure in the head known as "brain" and a trunk ganglion.

Metamorphosis

As you read this section keep on referring to Figure 2.3. After a few hours or days, the larva attaches itself to the substratum with the adhesive glands. Metamorphosis (transformation) then follows. The tail is partly reabsorbed and partly cast off. The notochord, nerve cord and muscles are withdrawn into the body and absorbed. The ganglion however remains to become the ganglion in adults. Stomach and intestine grow. The part of the body between the point of attachment and the mouth grow rapidly so that the mouth shifts through 180° to the nonattached end. The gonads and ducts form in the mesoderm between the stomach and the intestine. The adhesive glands disappear and the tunic grows upwards enclosing the whole animal.

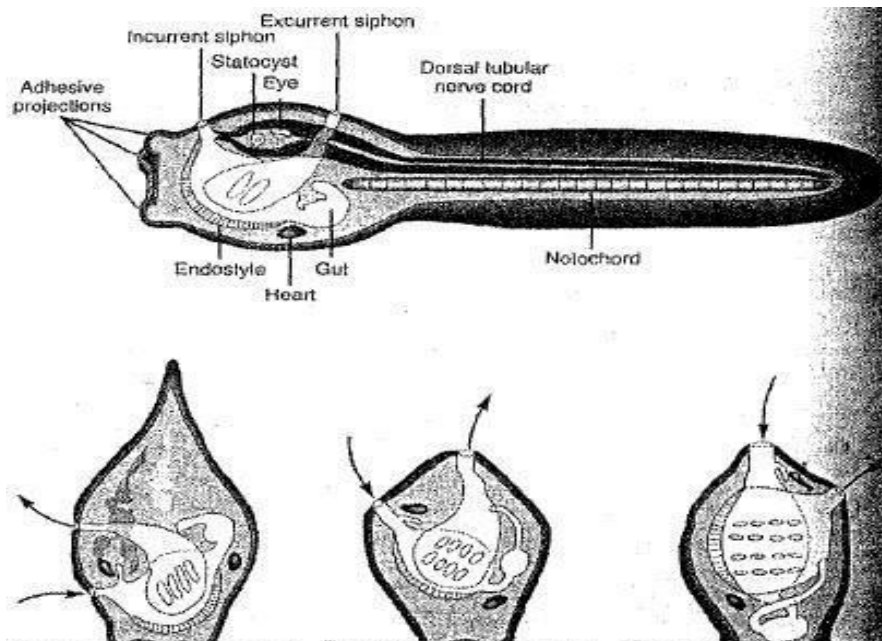


Fig 2.3 Stages in the metamorphosis of an ascidia larva

Class Thaliacea (salpians)

Salpians occur in the tropical and subtropical waters. Their body is barrel-shaped and the incurrent and excurrent siphons are on opposite ends (see Figure 2.4). They swim by contracting the ringlike muscles but 'some moved by water currents.

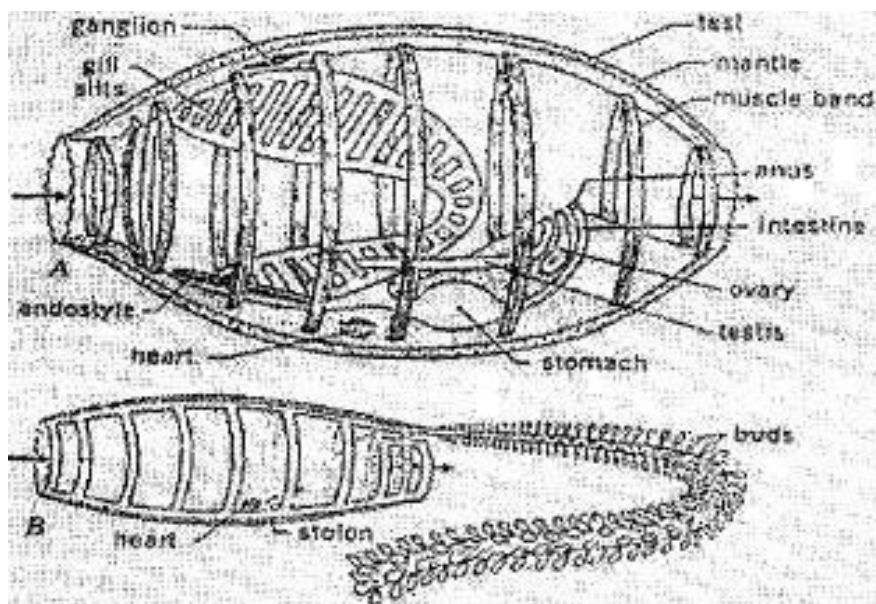


Figure 2.4 The thaliacean adult and larva

Class Larvacea (appendicularians)

They occur in the surface waters throughout the world. They are small, up to 5 mm long. The adults retain the larval tail. They lack a tunic but each secretes mucous housing in which it lives and through which it filters food.

2.5 Subphylum Cephalochordata (Lancelets)

These inhabit the tropical and temperate coasts. They usually burrow in the sand at the shallow shores leaving only the anterior end protruding. At times they emerge to swim. They are phylogenetically important because they have some features in common with larval urochordates and others in common with vertebrates.

The features they share with urochordates includes

- The notochord
- Nerve cord
- Pharyngeal gill slits.

Those they share with vertebrates includes

- The myomeres (block of segmental muscles)
- Ventral contracting blood vessel that is homologous to the vertebrate heart •
Separate dorsal roots of the nerve cord.
- Rudimentary olfactory and optic organs.

So lancelets seem to form a link between vertebrates and urochordates. They seem to be the closest relatives of vertebrates.

External features

The body is slender, long and •compressed from side to side. It is pointed on both ends. It has a median dorsal fin extending the whole body length. There is a tail fin and a pre-anal fin. The mouth is at the anterior end and is ventral. Study Figure 2.5 and identify the structures just listed above.

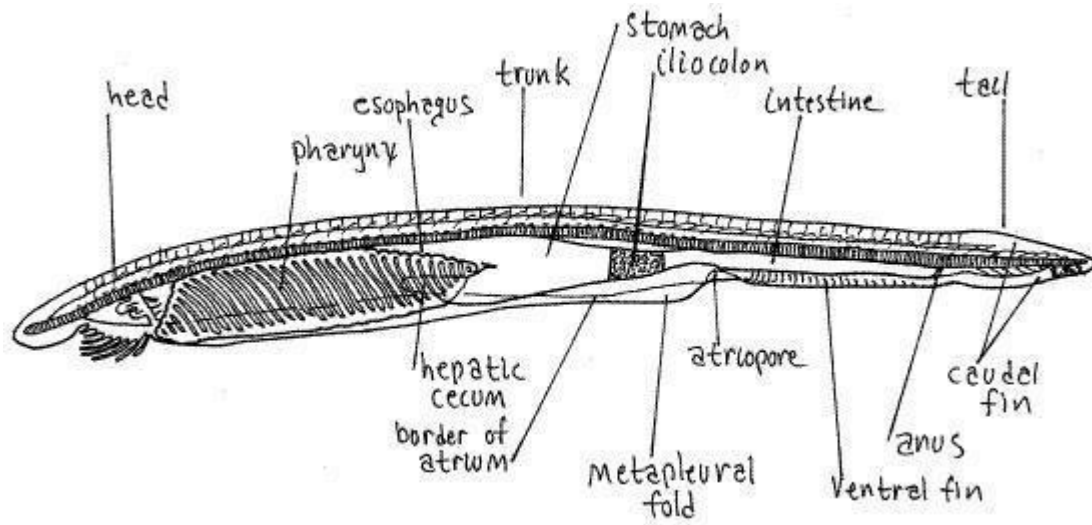


Figure 2.5 longitudinal section of amphioxus showing internal structures

Digestive system

Lancelets are filter feeders. Water containing food particles is drawn into the mouth through the oral hood by action of delicate projections known as cirri. The cirri prevent large particles from entering the mouth. Inside the oral hood is band of cilia known as the wheel organ whose beating action produces a current of water that enters the mouth. The mouth is a circular opening in a membranous velum. Behind the mouth is the pharynx with elongate gill slits at the sides. The pharynx hangs in the atrial cavity and leads to a narrow straight intestine that ends at the anus. There is a liver at the anterior end of the intestine. The atrial cavity opens to the outside through the atriopore.

As food passes through the pharynx, it is trapped by mucous in the endostyle while the water passes through the gill slits to the atrium and out through the atriopore.

Circulatory system

The circulatory system has somewhat a plan similar to that of higher chordates, with blood vessels (see Figure 2.7). However, there is no true heart. The blood from the digestive tract flows anteriorly in a sub intestinal vein to the hepatic portal vein entering the liver. From there it collects in a hepatic vein and together with other blood from the posterior part of the body passes forward in a ventral aorta below the endostyle and into branches at each primary gill bar. Each branch has a small pulsating bulb that functions as heart forcing the blood upwards into the gill bar where it is aerated. It then collects in the paired dorsal aorta. The two join behind the pharynx to form a single dorsal aorta in which the blood flows posteriorly to supply the body and intestine, and finally through the

capillaries to the venous side. Some oxygenated blood passes forward in the right dorsal aorta to the anterior end of the body.

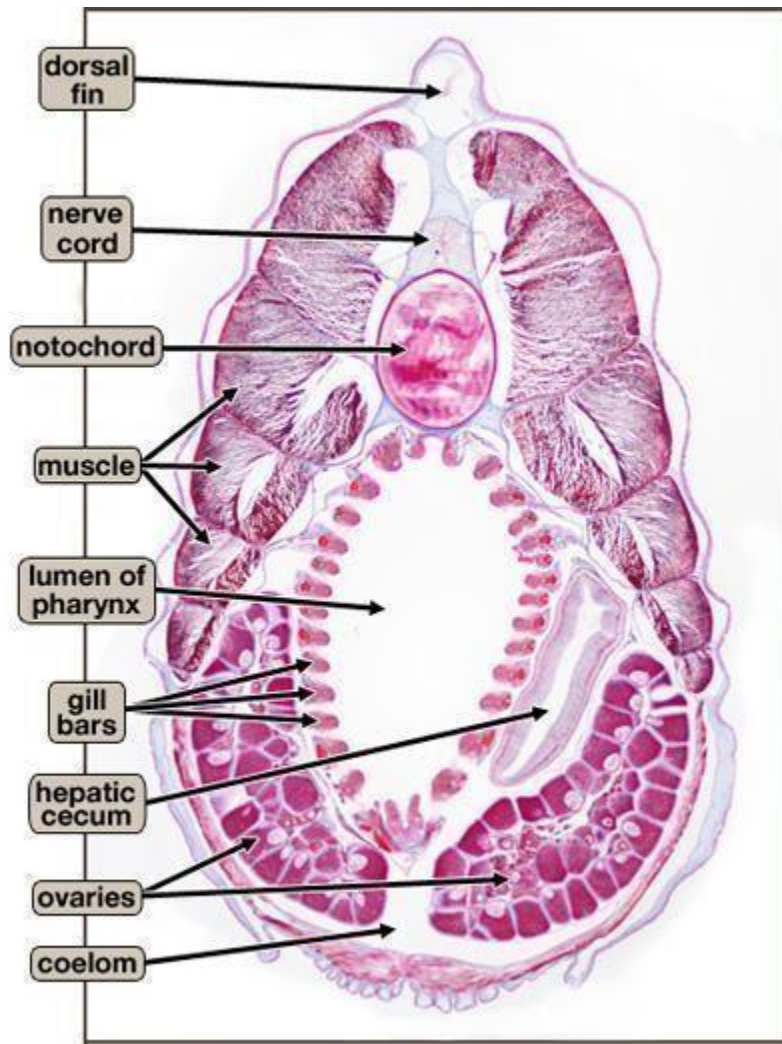


Figure 2.6 Amphioxus (Cross-section) and identify parts mentioned here

Excretory system

The excretory system comprises about 100 pairs of small ciliated nephridia in the coelomic cavity above the pharynx. They connect the coelom with the atrial cavity. They transfer excretory products from the blood in the dorsal coelomic canals to the atrial cavity.

The nervous system

The nervous system consists of tubular nerve cord located dorsal to the notochord. It is slightly enlarged at the anterior end but there is no brain. The cord sends off to each myomere a pair of processes. Amphioxus are sensitive to light, chemical and tactile stimuli although there is no elaborated sense organs.

Reproductive system

Sexes are separate. There are about 25 pairs of gonads. As the gametes ripen, they burst into the atrial cavity and pass out through the atriopore. Fertilization is external. The eggs are about 0.1 mm in diameter with little yolk. They hatch in about half a day into a ciliated free-swimming larva that resembles the adult. It grows for about three months within which it gradually assumes adult form. As it grows the ciliated body surface fails to keep pace with the body mass. So the larva eventually sinks to the bottom of the sea and adopts burrowing life. Lancelets are of little economic importance but in certain parts of China they are used as human food.



2.6 Subphylum Vertebrata

This is the largest chordate subphylum with about 40,000 species. They are placed in groups according to criteria of major evolutionary significance as we saw in lesson 1.

Origin of vertebrates

As we have seen in the preceding sections, vertebrates and cephalochordates have many similar features. Most likely they evolved from a common motile filter feeder ancestor. The two evolved differently because some of their adaptations such as those for motility are quite different. The myomeres in amphioxus are alternate while in vertebrates they are paired. Cephalochordate-like creatures may have evolved very early for an amphioxus-like creature, Pikaia, have been found in the mid-Cambrian deposits while vertebrates appear later in the fossil record.

The ancestor of cephalochordates and vertebrates is not known but theory puts it that they may have evolved from ascidia larva. The larva may have evolved into cephalochordate by retaining the larval features i.e. notochord and nerve cord, but failed to develop the tunic and other features that adapt adult ascidia to sessile life. Evolutionary retention of sexually mature adults of ancestry larval features is known as paedomorphosis. In evolution through paedomorphosis many of the

adult features that normally appear after the larval period fail to develop and new features can evolve.

Revision questions



- Explain why amphioxus is always studied in a vertebrate course.
- Why were hemichordates put in their own phylum while previously they used to be in the same phylum with chordates?

2.7 Summary



In this lecture we have learnt that it is possible that vertebrates evolved from invertebrates. A possible evolutionary sequence is: hemichordates - tunicate larva – amphioxus vertebrate. However, the fossil record does not provide links between the different groups.

Further reading



1. Harris C.L. (1996). Concepts in Zoology, Second edition. HarperCollins College Publishers,
2. Hildebrand M. (1995). Analysis of Vertebrate structures, fourth edition. John Wiley & Sons, New York.
3. Z. Young, The Life of Vertebrates.

LESSON 3: COMPARATIVE ANATOMY AND PHYSIOLOGY OF VERTEBRATES

3.1 The Vertebrate Teeth



Teeth are one of the most adaptive features of vertebrates. We can use teeth to infer the diet of a vertebrate. The great structural teeth variation makes them an important characteristic in the classification of vertebrates. Zoologists can identify the species and tell the age of an animal by examining its tooth. So they have been used to trace the general course of evolution especially because they preserve well as fossil. The tooth-like structures in agnathans are not true teeth and resemble the horny projections of a tadpole.

The teeth of cartilaginous fishes are attached to the skin by while those of most vertebrates are fixed on the bone of the jaws. Most vertebrates replace their teeth continuously throughout life and are said to be polyphyodont. Mammal and mammal like reptiles have two sets of generations of teeth, the milk and permanent teeth and are said to be diphyodont. Most fishes have teeth that are of the same size and are termed homodont. They attached to the margin of the jaws and in some on the roof of the mouth on gill arches and tongue. Most ray-fined fishes have teeth fused to the jaw bone and are termed acrodont. Acrodont teeth are not shed. Amphibians have small simple teeth. Reptiles have homodont teeth and mammals have heterodont teeth with different form and function. We will discuss mammalian teeth in lesson 11.



3.2 The Vertebrate Heart

The hearts of various groups of vertebrates are shown in figure 3.2. Refer to this fig as you read through this section.

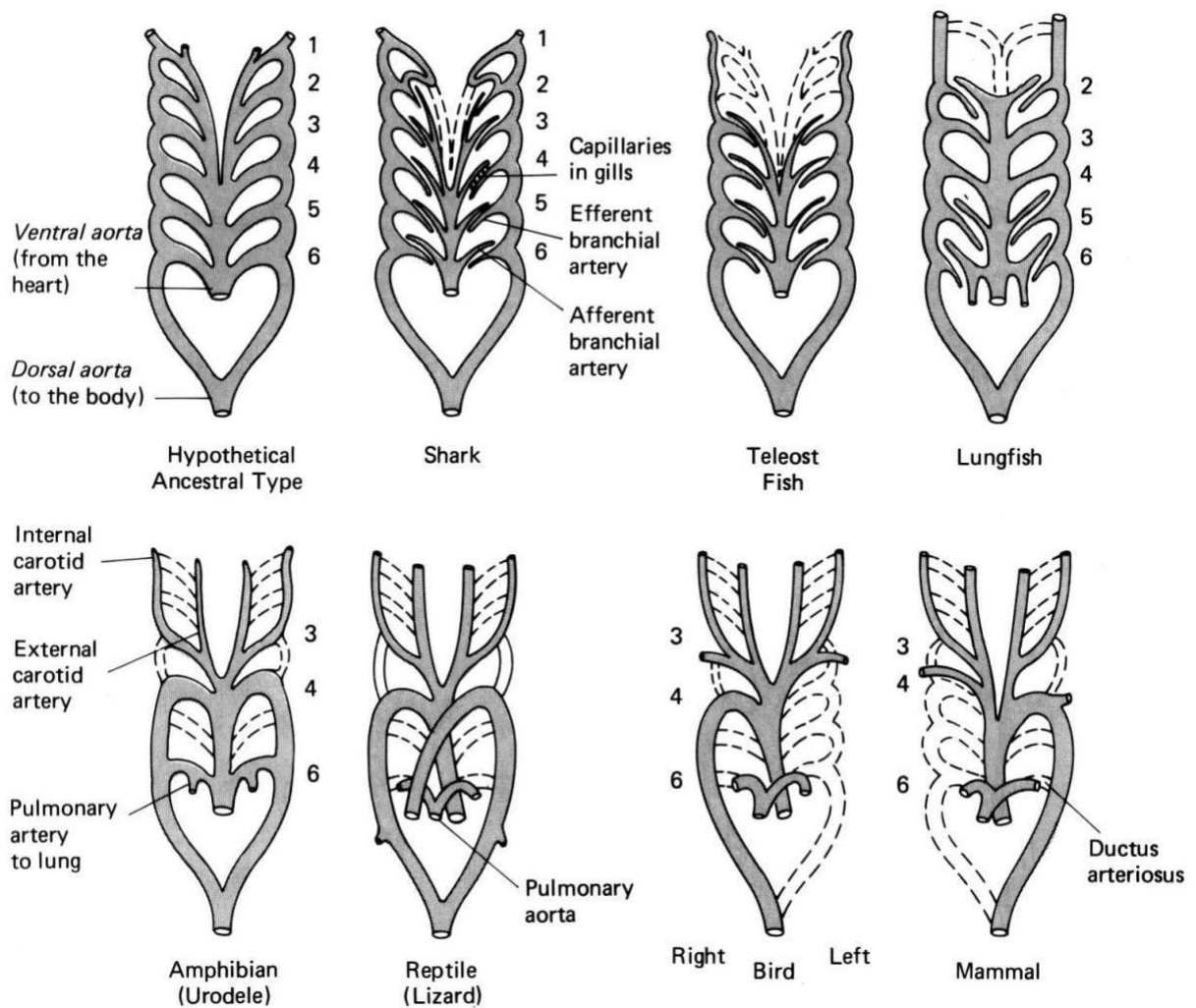


Figure 3.1 Evolution of the heart and arteries of vertebrates

In vertebrates the heart is divided into two major kinds of chambers, the atrium and ventricle. Atrium pumps blood from the veins to the ventricle while the ventricle pumps blood to the arteries. In addition to these two kinds of chambers, amphibians, birds and reptiles have another chamber, the sinus venosus that receive blood from veins and acts as reservoir for the atrium. In birds and mammals, sinus venosus occurs only in the embryo. Fish have another chamber, bulbus arteriosus that receives blood from ventricles.

Vertebrates vary in the number of major heart chambers, atria and ventricles, they have. The number of these chambers increases as you move from the _primitive to more advanced groups.

Fishes have two; amphibians and reptiles three while birds and mammals have four. Increase in the number of chambers is achieved by the atrium and ventricle being divided into two halves. This allows half of the chamber to pump oxygenated blood and the other deoxygenated blood.

In fish the chambers are arranged linearly. Deoxygenated blood enters the sinus venosus, then the atrium and finally the ventricle from where it is pumped into bulbus aterosus and to the gills. Amphibians have two atria and one ventricle. The two atria empty in-to the ventricle. Deoxygenated blood enters the sinus venosus from the major vein, the vena cava, and then goes into the right atrium. At the same time, oxygenated blood from the lungs or skin enters the left atrium from the pulmonary vein. The oxygenated and deoxygenated blood from the two atria enters the ventricle. Although the ventricle is one, there are structures that reduce the mixing of deoxygenated and oxygenated blood. Most of the oxygenated blood leave the ventricle through the dorsal aorta while deoxygenated blood go to the lungs and skin through the pulmocutaneous artery.

The hearts of most reptiles are similar to those of amphibians except that the sinus venosus is fused with the right atrium and the ventricle is more divided. In crocodiles and alligators, the ventricle is completely divided so they have four chambers. Birds and mammals have four chambers with complete separation of oxygenated and deoxygenated blood.



3.3 Vertebrate Kidney

The kidneys of vertebrates are paired organs on each side of the dorsal aorta. Vertebrate kidneys are composed of units called renal tubules or nephrons. As you will remember from the earlier unit on invertebrate zoology, nephrons end blindly and receive filtrate from the blood. The number and arrangement may differ among vertebrates. In ancestral vertebrates each kidney probably contained one nephron for each body segment. Each nephron opened at one end into the coelom by a nephrostome and the other end into a common archinephric duct. The nephrons drained into the archinephric duct that continued posteriorly to the cloaca. Such a kidney is known as holonephros (whole) because it extends the entire length of the coelom. Holonephros-like kidney is found in larvae of hagfishes but not in any adult vertebrate, amphibians and some reptiles. In most mammals the ureters connect directly to the bladder from which a median duct, ureter, discharges to the exterior, passing through the penis in males. The interrelated excretory and reproductive systems of the vertebrates are commonly termed urinogenital system.

In the embryonic development of higher vertebrates, reptiles, birds and mammals, there is a succession of three developmental stages of kidneys: pronephros, mesonephros and metanephros

(figure 3.4). The pronephros is the first to appear and is located anterior in the body. It becomes the persistent kidney in adult hagfishes. In all other vertebrates it degenerates during development and is replaced by a more centrally located and more structurally advanced kidney, the mesonephros.

The mesonephros became the persistent kidney in adult fish and amphibians. In the developing embryos of reptiles, birds and mammals, mesonephros is replaced by metanephros. The latter is structurally and functionally the most advanced of the three types of kidney. So, the embryonic nephrogenic mesoderm (kidney mesoderm) extends the whole length of the embryonic coelom on the dorsal part but only the most posterior part develops into adult metanephros. So the embryonic development of kidneys is similar to the evolutionary sequence of kidneys which gives a clue to the evolution of vertebrates.

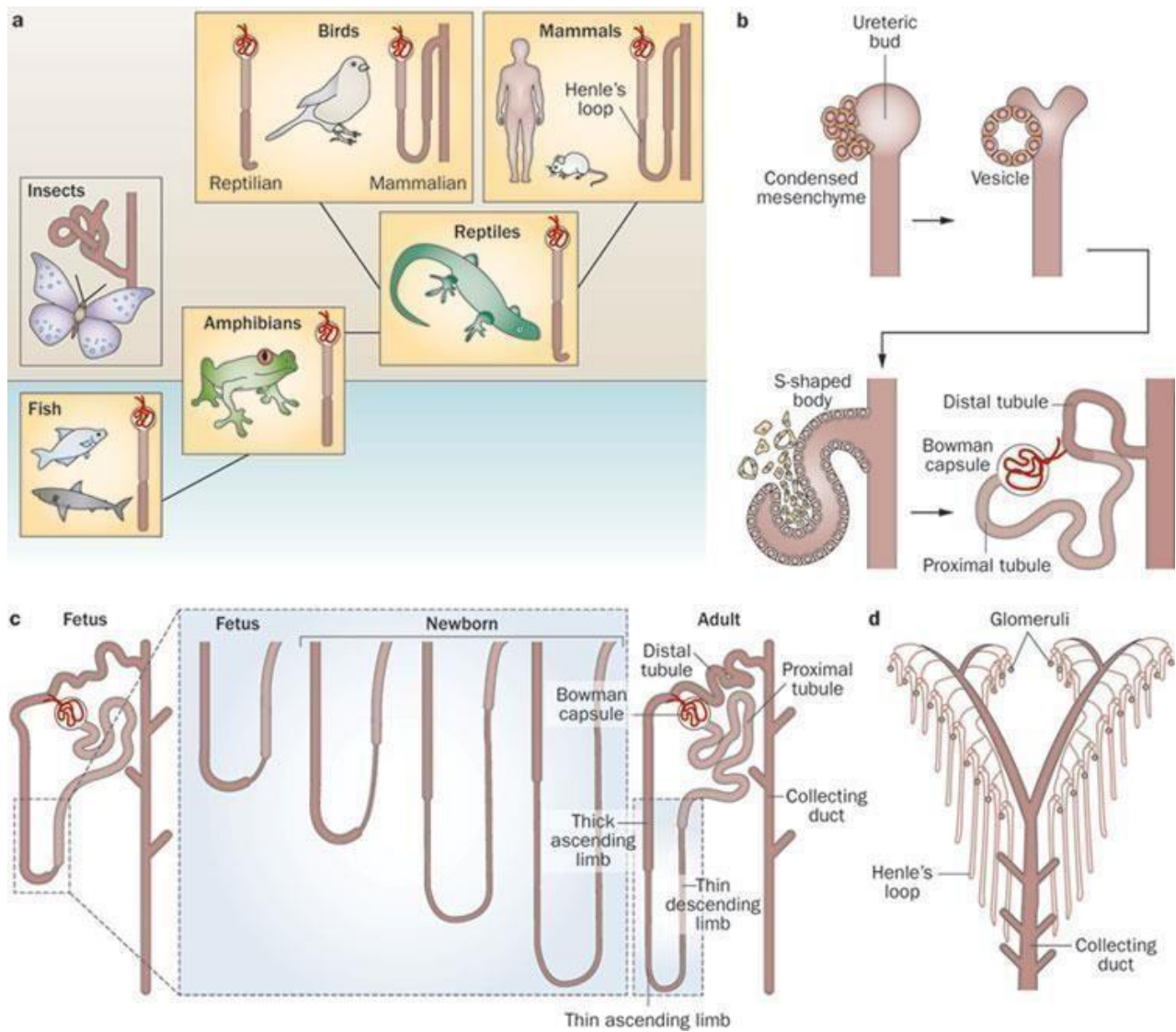


Figure 3.3 Evolutionary Sequence of the Vertebrate Kidney

LESSON 4: CLASS AGNATHA

4.1 Introduction



Welcome to lesson 4 where you will meet the first vertebrate class. In this lesson you will learn about vertebrates without jaws. They are all aquatic (live in water). Vertebrates in this group are of great evolutionary significance as they seem to link the other vertebrates and nonevertebrate chordates. They are of great economic importance as most parasitize other fish.

4.2 Objectives



At the end of this lesson you should be able to

1. List the characteristics of agnathans
2. Outline the evolutionary importance of agnathans
3. Explain why agnathans are regarded by some zoologists, as non vertebrates.



4.3 Characteristics of Class Agnatha

These are vertebrates without jaws. This group flourished in the middle Paleozoic era and was represented by several orders collectively known as Ostracodermns (meaning shell skins). Surviving members of this class are the hagfishes and lampreys. The main characteristics of these animals are;

- They lack the typical Vertebral column characteristic of other vertebrates. However, vertebrae are represented by cartilaginous elements on the dorsal surface of the notochord. The hagfishes have no vertebral column and lampreys have only nodule of cartilage along the notochord. So notochord persists in adults as the supporting structure.
- Their internal skeleton is made of cartilage.
- They differ from most fish in that their body is not covered with scales, having a single median nostril and light sensitive pineal eye present and absence of paired fins. For this reason they are usually considered not to be fishes.

We shall observe external characteristics of agnathans during practical 2. There are two groups of living agnathans, lamprey and hagfishes and many extinct species termed ostracoderms



4.4 Ostracoderms

These are extinct members of class Agnatha. Ostracoderms were small, fresh water and fishlike. They were dorsal ventrally flattened especially at the front of the body. They had a single median nostril and pineal eye on top of the head. The role of the pineal eye was to adjust the rate of activity to changing conditions of illumination. They had a median fin, dorsal, ventral, caudal, but no paired fins. The tail was heterocercal; the upper portion was larger and more ridged than the lower. Most of these characteristics are found in living agnathans so you will see them during practical 2. Fish that do not have lungs or swim bladder would sink in water but with a flattened head and pectoral fins raise the anterior off the bottom. The heterocercal tail helps raise the rear part of the fish.

The early Ostracoderms were covered with thick bony plates and scales all the skin. These structures were covered by dentine-like and enamel-like materials. The scales offered mechanical protection and also protection against an excess in flow of water from the fresh water environment in which these fishes lived. The absence of well developed fins, flattened body shape of most species and heavy bony scales and plates suggest that Ostracoderms were not good swimmers. They probably moved slowly at the bottom while the absence of jaws means that they were either filter feeders or scavengers. Ostracoderms became extinct by the end of the Devonian period but the lampreys and hagfishes survived to the present day.



4.5 Lampreys

Figure 4.1 shows external features of lampreys.

They have an elongate body that resembles that of eels. A sucker surrounds the mouth. The sucker is round when spread open for use but its lateral margins are drawn together when the lampreys swims. The edge of the sucker has tooth knobs of keratin that help to prevent the animal from slipping when attached to slippery surfaces of a fish. From the floor of the mouth a protrusible tongue projects into the mouth. A long central cartilage stiffens the tongue. Smaller cartilages at the tip of the tongue bear horny teeth that are used to rasp wounds in prey. Both sets of teeth are ectodermal, that is they develop from the out layer of the embryo and not homologous to the teeth of higher vertebrates. Since teeth are an important feature of vertebrates we will give a comparative outline of vertebrate teeth after we are thorough with lampreys. When a lamprey is feeding

blood is passed from the sucker into the gut without detaching the sucker. The sucker helps them attach to other since they are parasites the sucker enables them to attach to other fishes.

Behind the sucker are two eyes without eyelids. On the dorsal surface of the head is a single nostril that is immediately followed by a pineal eye. Posterior to each eye are seven small openings, the gill slits. Fish lenses are nearly always approximately spherical. They have a focal length of 2.5. radii not 4, the lens is not homogenous and has a refractive index is 1.53 at the center but less in the outer layers, When distant objects are focused the nearby one are not. Accommodations required into focus objects at different distance. Mammals accommodate by altering the lens and so the focal length, of the lens. Lampreys have muscles that flatten the cornea, and bringing distinct objects into focus. Other different vertebrates have different mechanisms accommodation.

The internal features of lamprey are shown in figure 4.2

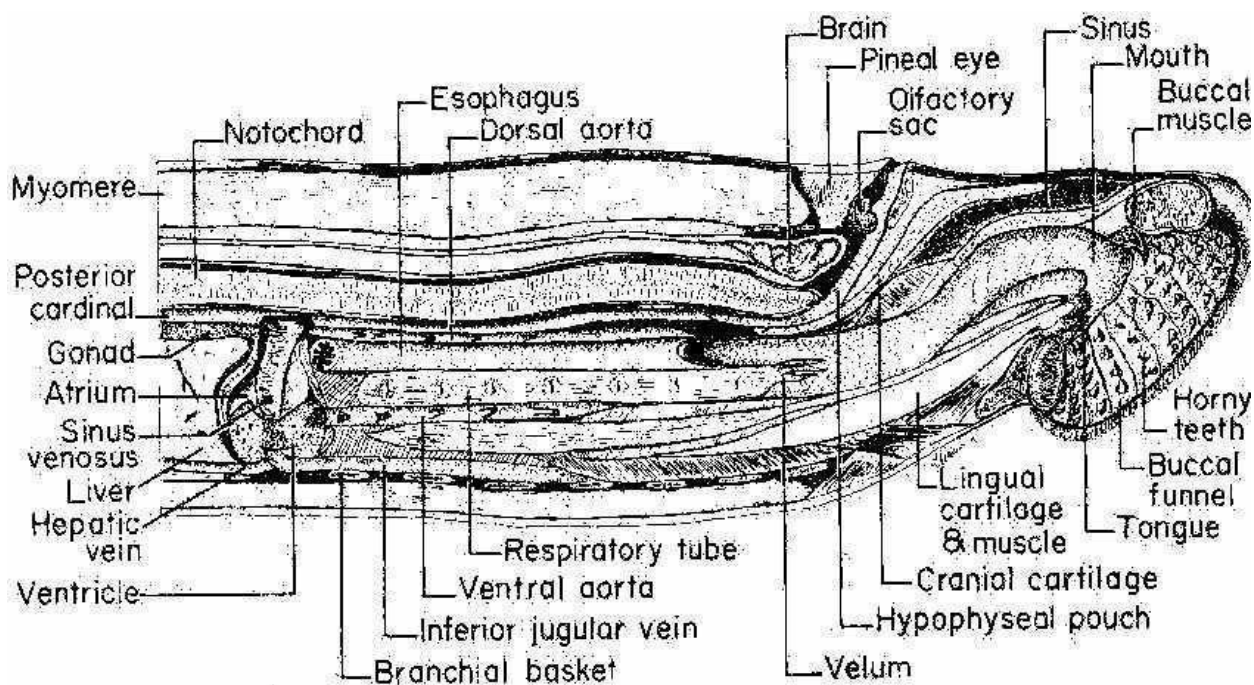


Figure 4.2 Internal features of lamprey

Activity 4.1



Study Figure 4.2 and note the position of each of the features mentioned above.

You will see and identify these features on preserved specimen during practical session.

Although Lampreys are marine dwellers, they reproduce in fresh water. Males make shallow nests in water by removing the larger stones with their sucker. During mating, the female attaches to stones at.

Study figure 3.2. Compare the arrangement of the basic excretory system of vertebrates with that of annelids. Are there any similarities between the two?

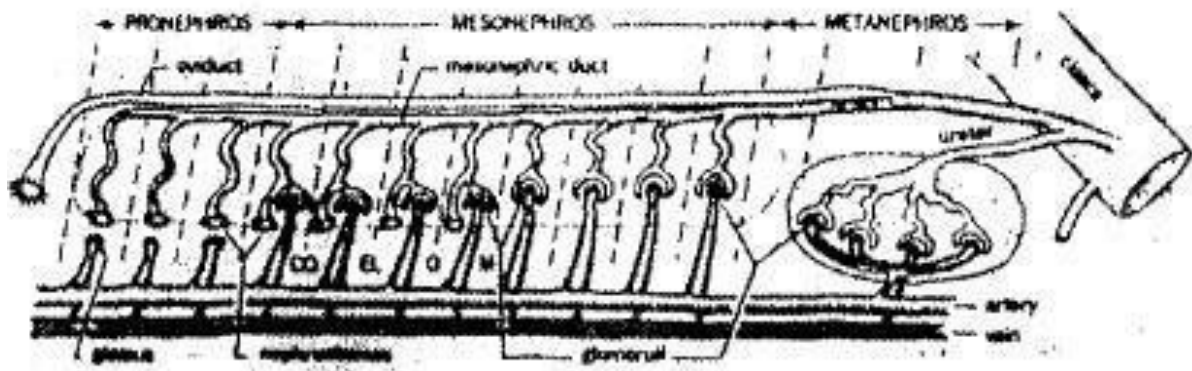


Figure 3.2 Holonephros kidney

In amphibians and adult fishes, the anterior tubules have been lost and some of the middle tubules are associated with the testis and there is a concentration of the tubules at the posterior side. Such a posterior kidney is called opisthonephros (behind). The original archinephric duct functions as excretory and sperm duct in males.

In reptiles, birds and mammals all of the middle tubules not associated with the testis have been lost and there is even a greater concentration of tubules at the posterior end. This is known as metanephros kidney. The tubules producing urine drain into a ureter, a tube that evolved as an outgrowth from the old archinephric duct. It carries the waste posteriorly. The archinephric duct has been taken over completely by the male genital system as the sperm duct. In amphibians, reptiles and birds, the two ureters discharge into the cloaca to which a urinary bladder connects in the bottom of the stream and the male attaches itself to the female. As the eggs are laid the male sheds sperms over them. Like in most fish, fertilization is external. Generally adults die soon after spawning although duration of adult stage varies from species to species. Fertilized egg hatch into a larva that is so different from the adult that for a long time it was thought to be a different species and given the name ammocoete. The name has been retained for the larva. The larval stage can last up to ten years but usually five.

Ammocoetes are small up to 20cm long but adults of the same species can be 1m long. Like the adults the larva has an eel-like body although much smaller than the adult's. Gill openings are small and round. There are only seven on each side of the body. Because the openings are few and small the gill apparatus is far less delicate than in amphioxus and is not protected in an atrium. There are flaps of skin over the openings that act as valves. Water can leave through them but is prevented from entering. Any inward movement of water through the openings make the flaps close over them. Similarly water can enter through the mouth but is prevented from leaving that way except in coughing, by the movement of the velum, a pair of flaps just inside the mouth.

Unlike the adults Ammocoetes are filter feeders and so they lack the sucker. The ammocoete feeds like the amphioxus. Since diatoms and detritus tend to settle on the surface of the sediments, and the water is drawn in from just above the sediments, the water that is filtered probably contains more diatom, and detritus than the bulk of the stream water. Diatoms make a large proportion of the diet but protozoa and occasionally other animals and detritus are eaten. The incoming water is strained through finger-like cirri as in amphioxus. The large particles are prevented from entering the mouth and are blown clear from time to time by a sort of coughing action,

Water is not propelled by cilia as in amphioxus but by muscular action but food is captured in a sheet of mucus in essentially the same way as in amphioxus and the tunicates. Some of the mucus is probably secreted by the endostyle which is much more complicated structure than in amphioxus and some elsewhere. Mucus with food trapped in it is drawn back into the intestine and digested.

The flow of water over the gills serves for respiration as well as well respiration. Gills have a complicated structure that gives them a large surface area. A transverse portion separates each gill slit from the next. From it project horizontal gill filaments in turn bear upper surfaces and down from the lower ones. The secondary lamellae are closely spaced so their total surface area is large and it is through their surfaces that most of the gaseous exchange occurs.



4.6 Hagfishes

These are found in temperate seas. Consequently we may not be able to see real specimens of these. Figure 4.3 below should give you an idea of how a hagfish looks like. Unlike lampreys they are not parasitic so they do not have a sucker. They feed on small invertebrates and scavenge dead fish. They have teeth on the tongue that they use to scrap tissues from carcasses.

Their body has small openings along the side of the body that lead to mucus secreting glands. The mucus is viscous and sticky and protects the hagfish from predators. Their body fluids have the same ion concentration as seawater and so do not need special osmoregulatory adaptations. They retain in adults two types of kidneys, pronephros and mesonephros as explained in lesson 3.

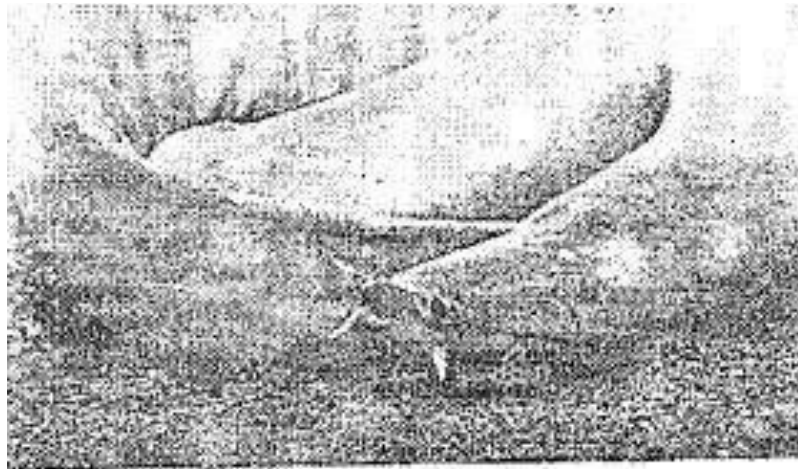
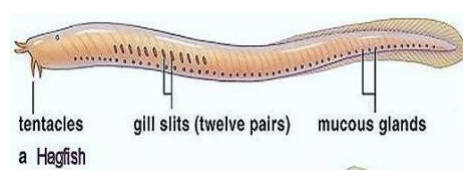
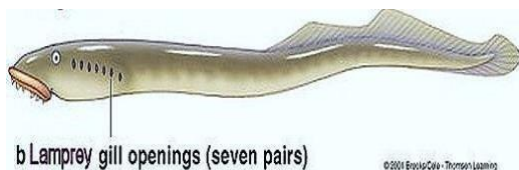


Figure 4.3 Hagfish

4.7 Distinguishing characters between Petromyzon and Myxine

Petromyzon	Myxine
1) It is commonly called lamprey	1) It is commonly called hag fish or slime eel
2) Internal ear has two semi-circular canals	2) Internal ear has one semi-circular canal
3) Cranial nerves are ten pairs	3) Cranial nerves are eight pairs
4) Seven pairs of gills are present	4) Six pairs of gills are present
5) It is a sanguivorous ectoparasite	5) It is a necrophagous animals
6) It does not produce slime	6) It produces enarmous amount of slime
7) Development is indirect (Ammocoetus larva)	7) Development is direct
8) Exhibits anadromous migration	8) Does not show anadramous migration



Activity 4.2



1. List the similarities and differences between adult and larval lampreys.
2. Distinguish between lamprey and hagfish

4.8 Summary



Agnathans lack certain characteristics that are typical of other vertebrates. They do not have jaws, true teeth and typical appendages. The gills are located in pouches.

Further reading



1. Harris C.L (1996), Concepts in Zoology, Harper Collins
2. Dorit R.L, W.F. Walker and R.D. Barnes (1991), Zoology, Saunders College Publishing.

LESSON 5: CLASS CHODRITCHTHYES CARTILAGINOUS FISHES

5.0 Introduction



Welcome to lesson 5. In this lesson, we will be dealing with animals that are more familiar than in the previous lessons. The lesson is about sharks and related groups. They are the first group of vertebrates with jaws (perhaps one of the reasons sharks are so predacious). The development of jaws enabled vertebrates to diversify their mode of life and perhaps one of the reasons they out- competed the ostracoderms.

5.1 Objectives



At the end of this unit you should be able to

1. List the characteristics of cartilaginous fishes
2. Explain the evolutionary advancement of this class over the agnathans
3. Draw and label the internal organ of a typical cartilaginous fish



5.2 Early Jawed Fishes.

Apart from agnathans all other vertebrates have jaws and are called gnathostomes (Gr gnathos, jaw + stoma, mouth). Jaws evolved as a modification of a gill arch. The gill arches of ostracoderms appear to have been firmly united with the bony plates covering the head. In other fishes each gill arch is movable. During the evolution of jaws certain of the anterior gill arches were lost but the most anterior remaining arch became enlarged and together with bone developed in the skin adjacent to it forming the jaw. This arch is known as the mandibular arch. Immediately posterior to this is the hyoid arch. And the others are the typical gill arches or branchial arches. In higher fishes the hyoid and mandibular arches are very close together and the hyoid helps to support the jaws. The gill slit one would expect to find between these arches is either absent or vestigial. The mandibular arch seems to have evolved independently and somewhat differently in agnathans and gnathostomes. It is possible that both groups diverged shortly after the origin of vertebrates although gnathostomes do not appear in the fossil records until the early Silurian period. Well after the first ostracoderms.

Activity 5.1



Study figure 5.1 and identify the position of mandibular, hyoid and gill arches in relation to each other.

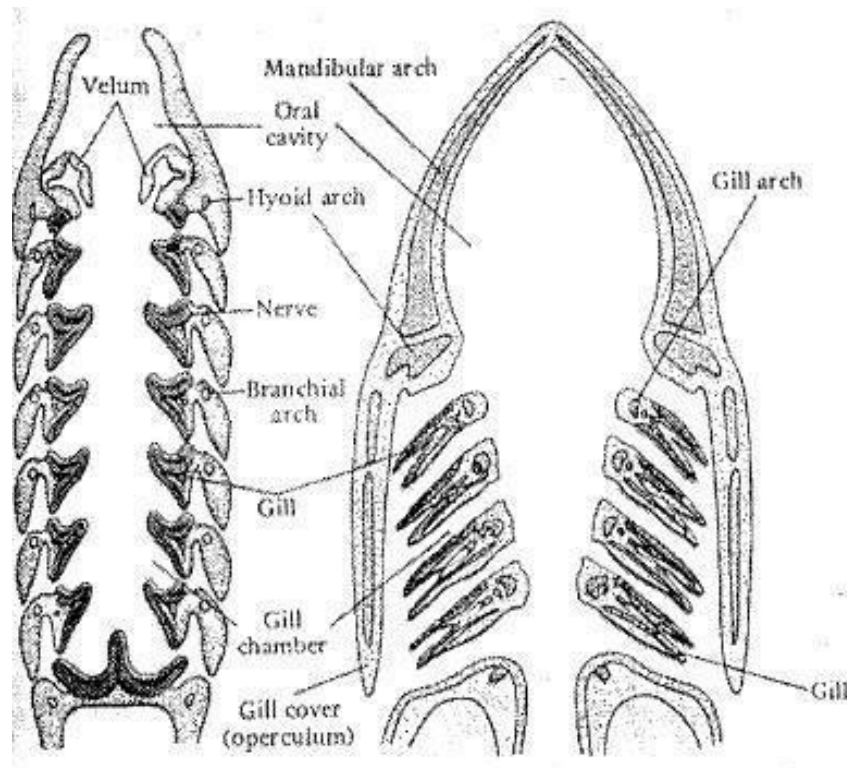


Figure 5.1 Section through the buccal cavity of (a) agnatha and (b) gnathostome

The second important advance common to all jawed vertebrates is the presence of paired appendages. In the earliest known jawed fishes appendages were in form of both pectoral and pelvic fins. The paired fins enabled the fishes to balance and propel in water. The presence of fins and streamlined body in early-jawed fishes indicate that they may have been active swimmers in contrast to the sluggish ostracoderms. Jaws enabled the fishes to feed upon a wide range of food sizes and types than could the jawless Ostracoderms. Consequently they diversified greatly. Earlyjawed fishes belong to two classes, Acanthodii and Placodermi.

I will now outline the characteristics of these two classes.



5.3 Class Acanthodii

These are primitive jawed fishes that existed and were abundant during the Devonian period and paleontologists do not agree as to how they should be classified. Some are related to contemporary sharks and rays, others to bony fishes and others are unique without obvious living relatives. Acanthodians had shark-like body shape and are called "spiny sharks" because of their bodies had numerous median and paired spines. However, these fishes were not sharks because sharks have cartilaginous skeleton but these had skeletons made of bone so they would appear to be a subclass of bony fishes. However, the absence of regular pattern of tooth replacement seen in bony fishes lead many paleontologists to place in their own class, Acanthodii. They had a series of paired spines protruding from the side of the trunk. Acanthodians became extinct during the Permian period. See Figure 5.2

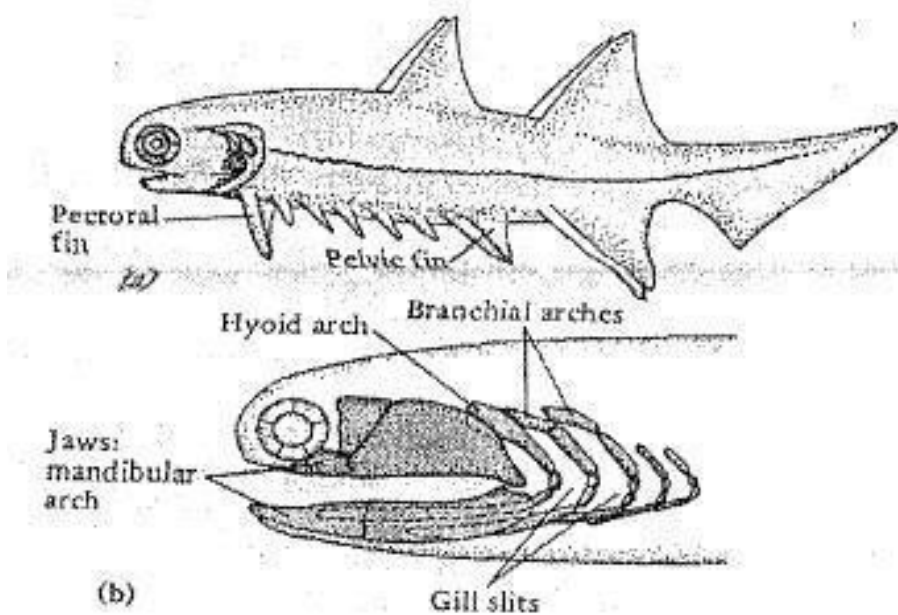


Figure 5.2 Spiny shark (representative of acanthodians) (a) External features (b) Jaw showing mandibular ark & gill slits



5.4 Class Placodermi

Most members of this class were heavily armored with bony scales or armor plate, the word "placoderm" means plate skin. They had a heterocercal tail and were slightly flattened dorsoventrally indicating that they could have been bottom dwellers. They had paired fins. Their gills were below rather than behind the brain case. They were most abundant at the time the amphibians evolved but became extinct before the other higher vertebrates had evolved.

5.5 Class Chondrichthyes

These are living fishes whose skeleton is cartilaginous. The earliest known species were marine inhabitants and the group has remained primarily marine ever since although few have secondarily adapted to fresh water. Examples of cartilaginous fishes are dogfishes, shark, skates and rays. We will use the dogfish as the representative animal of this class. We will study its external and internal features. Information presented in this lesson will be supplemented by what you will learn in practical

5.5.1 Characteristics of chondrichthyes

- They have jaws
- They have a well-developed electro-receptive system.
- Bony scales are either tiny placoid scales or no scales.
- Internal skeleton is entirely cartilaginous
- Tail is heterocercal
- They have five pairs of gill pouches that open to the outside individually on the body surface
- Males have a clasper on the pelvic fins with which sperms are transferred to the female. Fertilization is internal.

The body of cartilaginous fish is covered by minute placoid scales embedded in the skin. Each scale is a tooth-like structure containing a pulp cavity surrounded by dentine and capped by enamel-like material. The triangular teeth of sharks closely resemble an enlarged placoid scale and have undoubtedly evolved from them. The mammalian teeth have the same structures indicating that the two groups have a common origin.

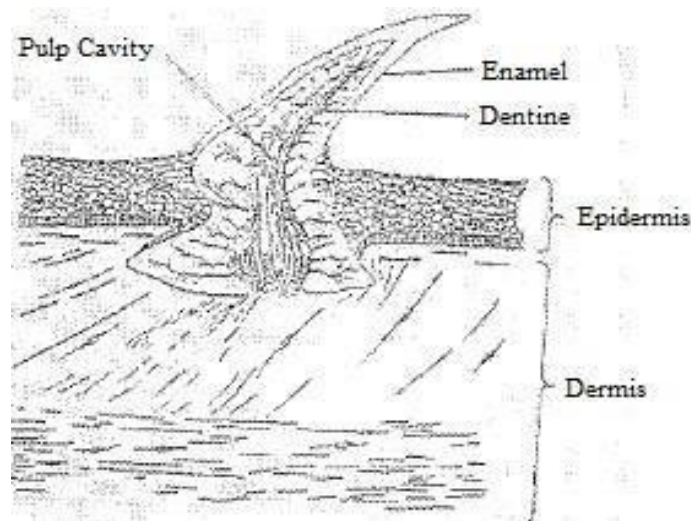


Figure 5.3 Placoid scale

Sense organs are well developed. Electro reception plays a prominent role in feeding behaviour. The electroreceptors are ampullary organs. They lie at the ends of long tubes filled with gelatinous material. These tubes are distributed all over the head in such a way that the fish can detect the source of weak electric currents generated by the muscle activities of prey organisms.

5.5.2 Digestive system

The mouth is on the ventral part of the head and the mouth cavity is continuous posteriorly with a pharynx. A spiracle containing a vestigial gill, and the gill pouches containing functional gills open from the pharynx to the body surface. An esophagus leads from the back of the pharynx to the stomach. A short straight intestine receives secretions from the liver and pancreas. It contains an elaborate helical fold known as the spiral valve. The spiral valve slows down the rate at which food moves down the intestine giving it time to be digested. It also increases the digestive and absorptive surface of the intestine.

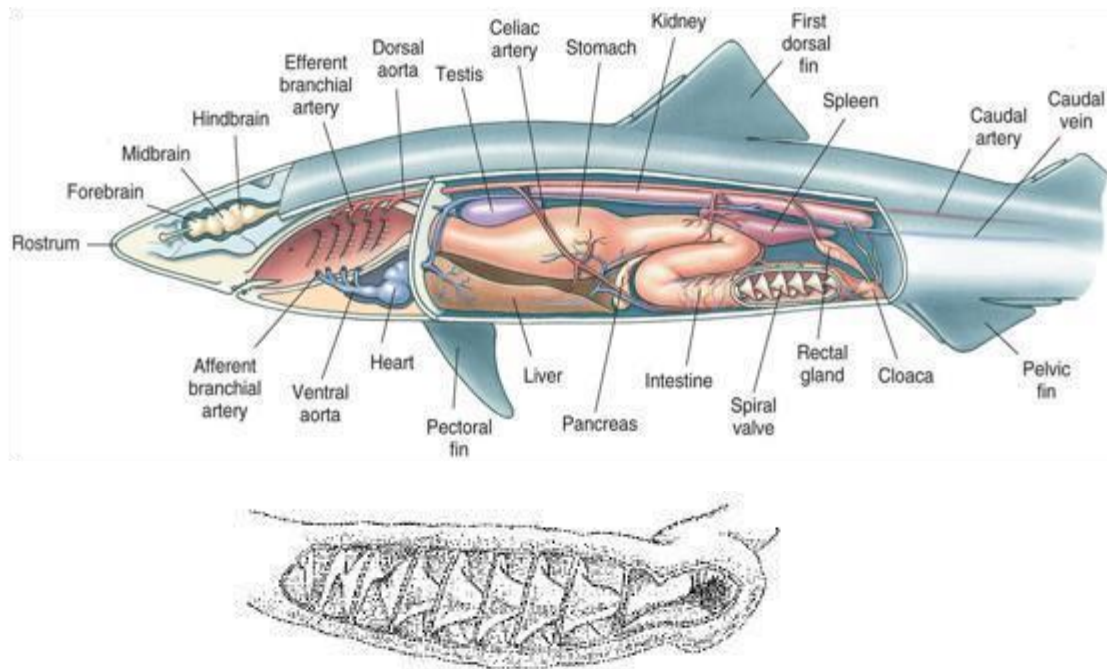


Figure 5.4 Digestive system of cartilaginous fish (a) fish open on one side (b) the spiral valve

5.5.3 Circulatory System

Circulation system is of the primitive type. Heart is undivided and pumps only oxygen poor blood to the gills where it is aerated, see lesson 3 for more details.

5.5.4 Excretory system

Kidneys are primitive, opisthonephric type. Remember we discussed the types of kidneys in lesson 3. The problem of conserve body water in marine environment is solved by retain considerable amount of salts so body fluids are slightly hyper osmotic. In sea water, some ions are eliminated by kidneys but most excess salts taken in with food is eliminated by salt- excreting rectal glands that empty into the end of the intestine and by salt excreting cells in the gills. Intestine and urinogenital ducts discharge into the cloaca that opens to the outside of the body.

5.5.5 Reproductive system

Chondrichthyes have internal fertilization. The fertilized egg develops in a uterus, a modified portion of the oviduct. Few species are viviparous and develop an intimate associating between

each embryo's yolk sac placentas. Some are ovoviviparous, they develop within a uterus but the young depends more upon stored yolk. Skates are oviparous and eggs have a lot of yolk and the embryo develops within it. Some sharks are oviparous but most brood their young internally.



5.6 Adaptive Radiation of Cartilaginous Fishes

Cartilaginous fishes are grouped into two subclasses.

- **Holocephali**, they have four gill slits that do not open directly to the outside but into a common chamber covered by a flap of skin. The spiracle is absent. Their upper jaw is fused to the cranium. The solid jaw allows them to feed on mollusks, crustaceans and other hard bodied animals. They have no scales. Males have club-shaped clasper organ in addition to the pelvic claspers.
- **Elasmobranchii**, have five to seven gill slits without covers. Usually they have a pair of gill openings modified to spiracles. The upper jaw is movable. Placoid scales usually present. These are divided into three groups: Squalomorpha, Galeomorpha, Batoidea. Squalomorph sharks are deep- water species. The dogfish belongs to this group.

Galeomorpha are subtropical and include the more familiar, predacious sharks "that are known to attack humans. Batoidea are the skates and rays. Skates and rays are bottom dwellers and are dorsal-ventrally flattened. Trunk and tail muscles are considerably reduced and they swim along the bottom by undulations of their enormous pectoral fins. Many rays and skates exhibit interesting offensive and protective adaptations among them snouts extended into "saws" in the saw-fishes, poison spines at the end of tail in sting rays and electric organs lateral to the eyes in torpedo rays or electric ray. **Activity 5.2**



We have many representatives of the groups of fishes mentioned here in the department and they will be displayed on the benches on the day you will be doing practical three. You make sure you identify as many as you can. You are strongly advised to visit any of the National Museums of Kenya where you will be able to see almost all the groups discussed in this lecture. The main museum in Nairobi is the best.

Revision questions



1. In what ways do members of Chondrichthyes differ from ostracoderms?
2. What do you mean by unicircuit heart?

5.7 Summary



We have seen in this lecture that Chondrichthyes were the first group to acquire true vertebrate structures. Acquisition of these features particularly the jaws enabled them to adapt too many modes of life.

Further reading



1. Dorit RL, W.F. walker and RD. Barnes (1991). Zoology. Saunders College Publishing
2. Harris C.L. (1996). Concepts in Zoology. Second Edition. HarperCollins.

LESSON6: CLASS OSTEICHTHYES- BONY FISHES

6.0 Introduction



Welcome to lesson 6. In this lesson we will be dealing with the most familiar of the fishes, perhaps the one you had for dinner yesterday. They comprise 95 % of all fishes and half of all vertebrate species. These are the most commercially exploited fishes and so fisheries officers, fishermen and even cooks would benefit from the contents of this lesson. Practical 4, that is supplementary to this lesson and activity 6.1 are mandatory if you are to understand this lesson.

6.2 Objectives



At the end of this lesson you should be able to

1. List the characteristics of bony fishes.
2. Outline the adaptive radiation of bony fishes
3. Explain the success of bony fishes



6.3 Difference between Bony Fishes and Cartilaginous Fishes.

Bony fishes are different from cartilaginous fishes of lesson 5 in the following ways

- They have a skeleton made of bone
- Their four gills on each side open into a common chamber covered by an operculum
- The tail is homocercal
- They have lungs and swim bladders except in few bottom-dwelling species.
- Intestinal spiral valve is absent except in very primitive species. Intestines long and coiled.
 - Most have external fertilization

As in lesson 5, we have one representative that we will study in details especially during practical 4 that is complementary to the lesson. The representative will tilapia preferably *Tilapia nilotica*.



6.4 Basic Features of Bony Fishes

The internal skeleton consists of cartilage replacement bone that has developed embryologically in association with cartilaginous rudiments that it gradually replaces. The bone in the scales and plates although histologically similar to the preceding type is dermal bone. It develops in the dermis of the skin and is not preceded by cartilage. The deeper portions of the dermal plates in the head and shoulder region is intimately associated with internal skeleton thus the skull and a portion of pectoral girdle of these fishes and terrestrial vertebrates that descended from them contain both types of bone.

The swim bladder is of particular interest. This is a gas-filled organ lying beneath the dorsal body wall. The bladder is filled with gases similar to those dissolved in water i.e. nitrogen, oxygen and carbon dioxide. It acts as a hydrostatic organ adjusting the specific gravity of the body so that the fish can stay at various depths with little effort. The bladder can also act as a reservoir for oxygen. Under conditions of oxygen deficiency, the fish can utilize the oxygen in the bladder. Gas, can be secreted into the bladder or absorbed from it as needed.

Another function of swim bladder is sound production.

In some species it is connected to the esophagus through the pneumatic duct and is known as physostomous fishes. In other fishes like the *Tilapia* the swim bladders has no connection with the esophagus and are known as physoclystic fishes.

In some fishes functional lungs are present instead of swim bladder. Some zoologists believed that lungs evolved from swim bladders but the current view is that swim bladders evolved from lungs.

It is believed that during the Devonian period fresh water fishes evolved lungs to cope with the unreliable fresh water lakes that would dry or become stagnant swamps with low oxygen content. Fishes with lungs would survive these conditions while others became extinct or migrated into the seas. Groups of bony fishes that have remained fresh water throughout their history tend to retain lung-like organs. Those that migrated back to sea no longer needed the lungs because ocean waters are rich in oxygen. Their lungs evolved into swim bladders. When conditions became more favorable in fresh water habitats, many migrated back into fresh water but retained their swim bladders. Hence the presence of swim bladders in fresh water bony fishes.

Digestive system

Jaws have many conical teeth to grasp food and further behind are pharyngeal and gill rakers' teeth helpful in holding and crushing it. Mucous glands are numerous but there is no salivary gland. The small tongue is attached to the floor of mouth cavity and may aid in respiratory movements. The pharynx has gills on both sides and lead to short esophagus followed by the stomach. A pyloric valve separates the latter from the intestine. There is a large liver anterior to the body cavity, gall bladder and a bile duct lead to the intestine, pancreas is usually diffuse.

We will be able to see these features when we dissect a bony fish.

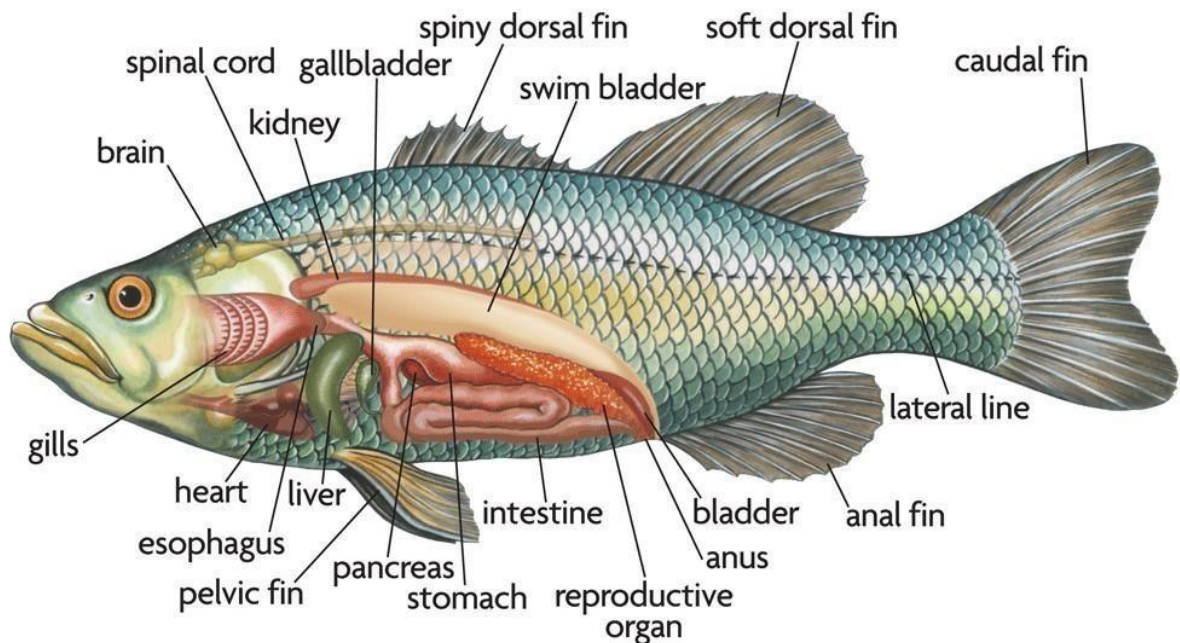


Figure 6.1 Digestive system of a bony fish.

Circulatory system

Bony fishes have a two-chambered heart that lies below the pharynx in the pericardial cavity. Venous blood passes into the sinuous venous to the thin walled atrium then to the muscular ventricle, all separated by valves that prevent reverse flow.

Rhythmic contraction of the ventricle forces the blood through the conus arteriosus and the short ventral aorta into four pairs of bronchial arteries distributing the blood into the gill filaments for oxygenation. It then collects in correspondingly paired different bronchial arteries leading to the dorsal aorta that distributes branches to all parts of the head and body. The principal veins are: the paired anterior cardinals and posterior cardinals and the unpaired hepatic portal circulation leading through the liver. Blood of fish is pale and scanty compared to that of terrestrial vertebrates.

Respiration system

Fish respire through gills. There are four gills in a common chamber on each side of the pharynx beneath the operation. A gill consists of a double row of slender gill filaments. Every filament bears many minute transverses plates covered with thin epithelium and containing capillaries. Each gill is supported by a cartilaginous gill arch. The bolder of the gill arch has gill rakes that protect against hard particles and keep food from passing out through the gill slits.

Bony fishes are grouped into two subclasses, Actinopterygii and Sarcopterygit.



6.5 Actinopterygii

These are referred to as ray-fined fishes. Their fins are reinforced with bony rays that are not attached to the rest of the skeleton. They include most fishes that have adapted to all imaginable aquatic habitats. Three infra classes are recognized, Chondrostei, Holostei and Teleostei.

One distinctive group is infraclass Chondrostei that include the Nile bichir and sturgeons. Chondrosteans are ancient and have many primitive features. Many have shark-like features such as ventral mouth, spiracles, spiral valves in the intestine, infused upper jaws, and skeleton that is largely cartilaginous. Unlike sharks adult chondrosteans have small teeth or none at all and their gills are covered by operculum. They suck invertebrates from the bottom. Chondrosteans are nearly extinct.

Scales

The remaining ray fishes belong to infraclass Neopterygii. Most neopterygeans belong to a large division known as teleosts.

Early Actinopterygian had thick bony scales overlain by layers of dentine-like (cosmine) and enamel-like material (ganoin), they are called ganoid scales. Few actinopterygians have these types

of scale. In most living Actinopterygians the scales have become thinner and the ganoin and cosmine layers have been lost and the underlying bone reduced to a thin disc that develops in the dermis of the skin. Such a scale is called cycloid scale when its surface is smooth and ctenoid scale when the exposed portion bears minute spiny processes (ctenii). Such are the scales found on Tilapia

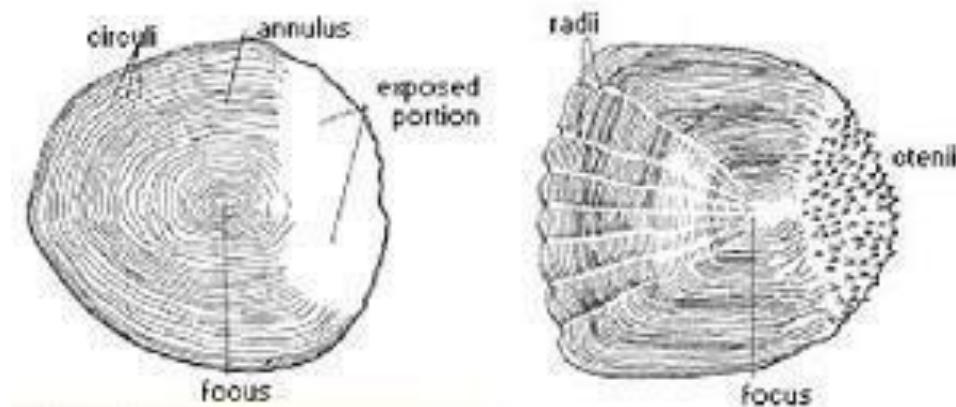


Figure 6.2 Scales of bony fishes (a) cycloid scale (b) ctenoid scale

Buoyancy

As the functional lungs of early actinopterygians gave way to swim bladder and control over buoyancy improved, and the body became more streamlined, the primitive heterocercal tail of most Chondrosteans became superficially symmetrical or homocercal, in teleosts. Also increased buoyancy relieved the paired fins of their primitive hydroplaning function. The point of attachment of pectoral fins shifted dorsally so that they were closer to the horizontal plan passing through the center of gravity and the pelvic fins shifted anteriorly. These changes enhanced the balancing and turning function of the fins.

Adaptive radiation of teleosts.

Are found in all aquatic habitats and have a wide range of feeding methods. Some are filter feeders and many are carnivores and herbivores. The resources of the environment can be utilized more fully if subgroups became specialized for certain parts of the environment than if they try to compete with each other in the entire environment. The following are examples of adaptation of these fish.

Fish that live at bottom such as halibut, soles and flounders tend to lose their swim bladder and are frequently flattened and glide along the bottom using up and down undulatory movements. Unlike the skates and rays that are dorsal ventrally flattened, they are greatly compressed from side

to side and swim turned over on the side. During development the eyes that would be on the side that becomes the bottom migrate to the topside but the mouth does not change position. Skates and flounders are a good example of convergent evolution, by which animals that are not closely related evolutionary adapt to similar modes of life and acquire similar features. Teleosts have developed flattened bodies in two different ways.

- Dorsal ventrally - anguilliform.
- Side to side- flounders, sole and halibut.

Other teleosts have adapted for life among seaweeds and in coral reefs. The sea hares with their monkey-like tail and the elongate moray eels are examples.

Some are adapted for life at the bottom of the sea where it is completely dark. Such fish have light-producing luminescent organs probably for species recognition, large mouths and extensible stomachs to take advantage of occasional meal that may come their way.

Few teleosts can exist out of water for short periods. Mudskipper is a good example. It hops about on the mud flats of mangrove swamps at low tide in search of food. It has muscular pectoral fins to help pull itself along the land. It can close its operculum so that they maintain high humidity in the opercular chamber and extract oxygen from the air with gills.

Some teleosts live in close association with other fishes. For example, the remora has an anterior dorsal fin modified as suction cup and is used to attach to sharks. It feeds upon crumbs of the larger fish's meals or obtains free rides to favorable feeding grounds. This type of relationship is known as commensalism.



6.6 Sarcopterygii

These are referred to as lobe-finned fishes because their lateral fins are elongate and have fleshy lobes supported by a fleshy central axis of bone that articulates with the rest of the skeleton and are homologous to the limbs of terrestrial vertebrates. In many species their olfactory sacs connect to the body surface through an external nostril and to the front part of the roof of the mouth cavity through an internal nostril. The caudal fin tapers posteriorly to a point. The shape is known as diphyercal. Examples of lobe-finned fish are the lungfishes.

Primitive sarcopterygians had lungs, internal nostrils, strong fins and so they are thought to be, the ones that gave rise to terrestrial vertebrates. These primitive sarcopterygians had cosmoid scales, two dorsal fins and heterocercal tail. Sarcopterygian's evolution diverged early into three lines: Orders Crossopterygii, Brachiopterygii and Dipnoi.

Crossopterygian

Crossopterygians have well-ossified internal skeleton, a unique jointed braincase and small conical teeth suited for grasping prey. Those belonging to suborder Rhipidistia were the largest and most •predatory freshwater fishes of Paleozoic era and probably it is from these that amphibians evolved although some zoologists believe that they evolved from lungfishes. They were thought to have been extinct 60 million years ago until in 1938 when one was caught in Cornaro Islands and named *Latimeria chalumnae*. Since then more specimens have been caught. One was caught at the Kenya coast a few years ago and is preserved at the National Museums of Kenya. *Latimeria chalumnae* has spiral valve in the intestine and a vestigial lung filled with low-density fats that help it maintain buoyancy. The female gives birth to live young from eggs that are the size of a tennis ball. Males have no copulatory organs and so it is still a mystery how the eggs get fertilized.

Brachiopterygii

These are the bichirs and reed-fishes. They inhabit stagnant ponds and streams in Africa. They get most of their oxygen by gulping air into their lungs rather than by using gills. They use their lobed fins to propel themselves along the bottom and through the vegetation. They can also feed on land. **Dipnoi**

The vertebral column of Dipnoi (lungfishes) is not fully ossified and paired appendages tend to be reduced. They have crushing tooth plates that enable them to feed on crustaceans and mollusks. Three genera of lungfish survive in tropical rivers and lakes, one each in Africa, America and Australia. Their lungs allow them to live in unsuitable waters that are subject to drought and stagnation. These fishes give us a clue as to how vertebrates overcome their respiratory and circulatory problems associated with transitions from water to land. However, their skeleton is too weak and dentations too specialized for them, to have been the ancestors of terrestrial vertebrates.

Activity 6.1 (Optional)



Make a trip to the National Museums of Kenya, Nairobi. Go to the ichthyology section where you will be able to see most of the fish discussed in lesson 5 and 6. There is also an aquarium next to the snake park in Nairobi where you can see live fish including the marine ones.

Revision questions



1. Describe the different types of scales found in fishes.
2. Explain how buoyancy is maintained in bony fishes.
3. List the differences between crossopterygians and sarcopterygians
4. Under what conditions did lungs evolve in bony the fishes.

6.7 Summary



Bony fishes are so diversified that we cannot study all the specialized adaptations within a general course like "Vertebrate Zoology". The lecture gives you an overview of the different

groups of bony fishes. The fishes discussed in this lecture are the most advanced and have evolved features that adapt them well to their aquatic life. However, they retain a few primitive features such as the bony scales that remnants of the armors of their ancestors.

LESSON 7: AMPHIBIANS

7.1 Introduction



Welcome to lesson 7 in which you will meet terrestrial vertebrates for the first time in this course. In this lesson we encounter the first vertebrates able to breath and move on dry land. So in this lesson we will look at features that enabled the transition of vertebrates from aquatic to terrestrial animals.

7.2 Objectives



At the end of this lesson you should be able to

1. List the anatomical and physiological transformations that enabled aquatic vertebrates to adapt to life on land.
2. Explain the differences between different groups of amphibians



7.3 Characteristics of Amphibians

We saw in lesson 6 that crossopterygians are most likely the ones that give rise to terrestrial vertebrates. Life on land presented the following challenges

- Locomotion without the support of water is difficult.
- Desiccation. The skin and excretory system had to be modified to prevent desiccation.

- Respiration. Since there was no water to bathe the gills all the time, the respiratory system had to change to be able to use gaseous oxygen. The change also affected the circulatory system.
- Signal receptors, Lateral line, the main sensory organ for fishes, would not be effective on land. So new receptors had to evolve. Sense organs near body surface had to be protected and kept moist. Sound waves in air could not reach the inner ear unless they were amplified and transported through the dense tissues of the head.
- Air is less dense than water so it gives less support and offers less resistance to movement. So strong skeleton and muscles were needed to support the animal.
- Delicate sperm and egg if deposited on land would die from desiccation and larval stages were very vulnerable. So this had to be dealt with.

Digestive system

The wide mouth leads into a capacious buccal cavity which has in its roof the internal nostrils. On its floor is the tongue, attached in front and free posteriorly where it ends in a double point. The tongue can be suddenly projected, to capture and engulf insects. Immediately behind the tongue is a slit-like glottis. Teeth are arranged in a single series round the edge of the upper jaw. There is also a small patch of teeth just internal to the posterior nostril (vomerine teeth). The buccal cavity narrows towards the pharynx, Which leads by a short esophagus into a stomach consisting of a wide cardiac and a short pyloric division. The stomach epithelium produces pepsin and HCl. A pyloric sphincter guards the entrance to the duodenum. The rest of the small intestine (ileum) is coiled. The large intestine (rectum) is very wide and short, and passes into the cloaca, which possesses a cloacal aperture. The liver is composed of two large lobes, the left lobe partly divided. Between the two lobes lies a large gall bladder. The bile constantly produced by the liver passes into the gall bladder via cystic ducts as well as directly into the bile duct by way of minute hepatic ducts. The bile duct runs from the gall bladder through the pancreas (between stomach and duodenum). Digestive juices flow down the pancreatic duct which is bound to the bile duct within a common sheath and they empty into the duodenum- through a common bile and pancreatic duct. The spleen is morphologically associated within alimentary canal, a small red globular blood reservoir.

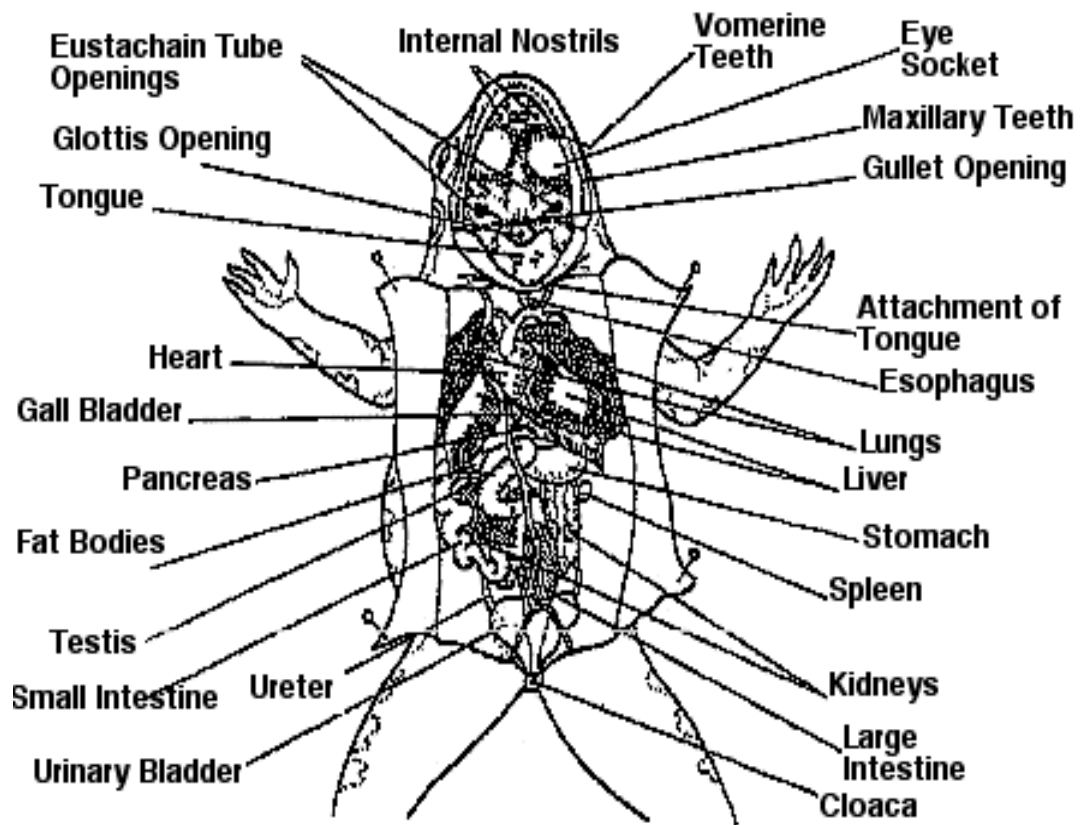


Figure 7.1 Internal Structure of the Frog

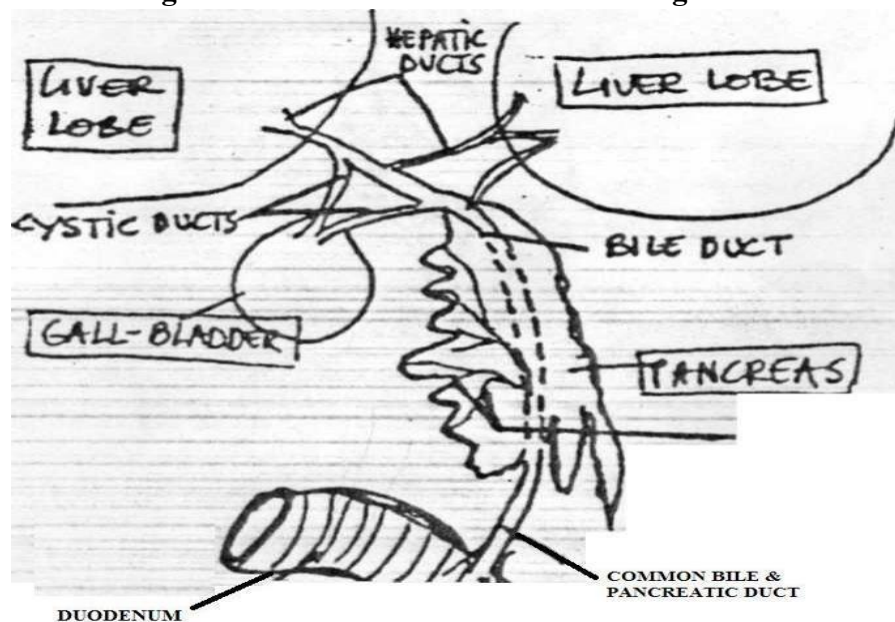
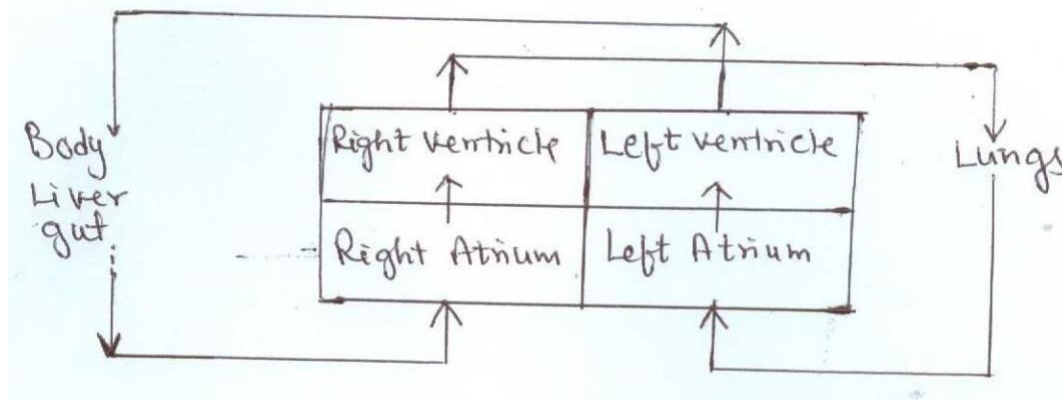


Figure 7.2 Diagram showing digestive glands

Circulatory system

In the typical fish heart is a simple structure with simple structures: in it is received, in a single stream the blood from all the organs. This is 'spent' blood deprives of its oxygen in its passage through the body, and thus all of it is ready to pass to the gills to receive a new supply of oxygen. We find thus the fish heart is basically a single tube structure pumping all the blood it receives forward in a single stream to the gills.

In lung breathers however, a major complication arises: Two blood streams are present. Fresh bloods from the lungs should pass to the major body structures while spent blood should be pumped to the lung. In reptiles, birds, and mammals this problem has been adequately solved. Both major parts of the heart (atrium and ventricle) have been divided down the middle so that a double-barreled pump has been created and the two blood streams are completely separated. The double type of circulatory system (reptiles, bird's mammals) scheme is:



Schematic diagram of double type of circulatory system (reptiles, birds & mammals)

In the frog, the heart is incompletely divided: it has a separate left and right atria but an undivided ventricle (three-chambered heart)

The two blood streams enter the heart separately but then become mixed to a certain degree in the undivided ventricle. This is not a disadvantage for the frog, since the blood can be oxygenated by two different ways (lungs, and skin + lining of the mouth). It has been shown that in the aorta a spiral valve is present which can co-ordinate the co-operation between lungs and skin if blood contains enough oxygen, the main part of the blood will be sent toward body, liver, gut, kidney (skin off course). If the blood does not contain enough oxygen which means that the skin was not able to enrich the blood.

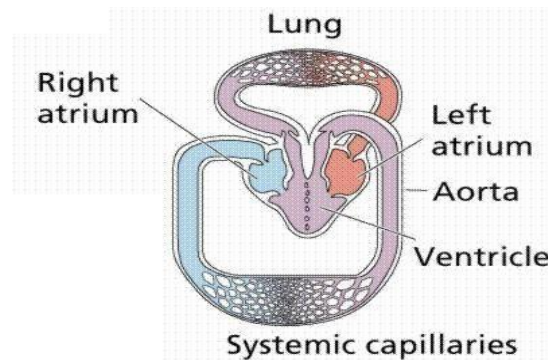


Figure 7.3 Amphibian Heart

With **enough** oxygen (for example during the hot season when oxygen requirements are higher) a bigger part of the blood is sent towards the lungs. (Spiral valve switches more blood towards the lung circuit). In other words, thanks to the fact that the two blood streams are not yet separated completely into two circuits, the spiral valve can co-ordinate the co-operation between lung and skin. In summary:

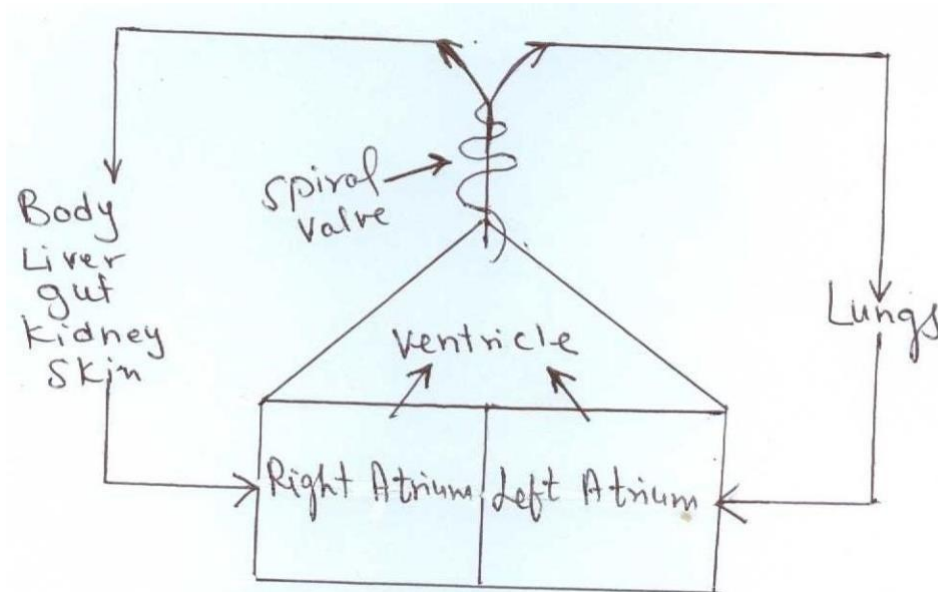


Figure 7.4 Schemetic diagram of an Amphibian Circulatory System

This system makes it possible for a frog to aestivate in the mud during the cold winters, using the skin only to obtain its necessary oxygen, and to switch (gradually) to the 'lung circuit' as the skin becomes less efficient to fulfill the increasing oxygen demands during the warmer periods where the frog is more active.

Urogenital system

Ophisthonephric kidney in adult amphibians the larvae have the same type of kidney as larvae the acnatha (holonephric kidney). There is a urinary bladder (adults) which opens directly into the cloaca.

The testes are white ovoid bodies and are attached to the anterior ends of the kidneys. Sperm is passed, via the vasa efferentia, into the urinary tubules and carried off by the Wolffian duct.

Attached to each testis are the fat bodies which are food reserves during aestivation.

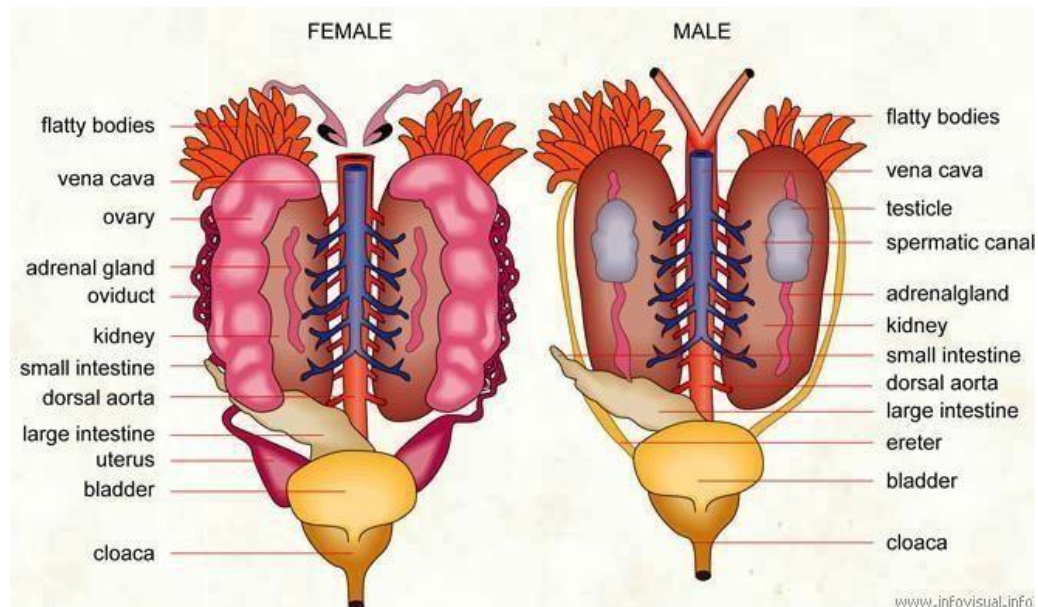


Figure 7.5 Urogenital Organs of the Frog

Amphibians ovaries are secular with fat bodies associated anterior to the ovary following release for the ovaries, the eggs are wafted towards the ostia (funnels) of the mullerian ducts by cilia on the peritoneum, liver, etc. for most of their length; these paired ducts have thickened glandular walls that secrete the gelatinous covering of the eggs. A posterior swelling forms parasite of temporarily storage for the eggs prior to laying. The black colour of the eggs enables them to absorb solar radiation in an efficient way.



7.4 Adaptations of Amphibians to Terrestrial Life

Amphibians were the first vertebrates to overcome some of these difficulties to become the first terrestrial vertebrates. They did so by retaining the lungs and sturdy paired appendages of their crossopterygian ancestors and also evolved many additional features that adapted them to life on land. They have also retained many fish-like characteristics and so they are not completely independent of water. Hence they are intermediate between aquatic and fully terrestrial vertebrates.

Skin

The skin is adapted for respiration. It is without scales that are a feature of fishes and reptiles. It is usually thin, and covered with mucous that help to keep it moist all the time. The epidermis is few layers thick so gaseous exchange is possible. The skin is also highly vascular (has many blood vessels) and aids in respiration. This combination also allows loss of considerable amount of water through the skin, a factor that limits the amphibians to habitats with high humidity. There are numerous glands and when the animal is stressed they secrete a frothy substance that contains poison. Many toxic amphibians have warning coloration.

Some of them are camouflaged. The colour is produced due to chromatophores, pigment cells, in the skin. In the leopard frog and others there are three layers of chromatophores that form a system responsible for changes in colour. Just beneath the epidermis are xanthophores that impart yellow colour. Next are the iridophores with light reflecting organelles that produce light scattering resulting in blue colour. The blue filters through the overlying yellow pigment resulting in green colour. The black melanophores are extensively branched cells under the iridophores and surround them with finger like process that extends between them and the xanthophores. The skin darkens as the melanin granules move outward from the cell body into the branches and obscure the iridophores. Pituitary hormone, intermediodin, causes the dispersal of melanin granules and contraction of iridophores. A reverse process of both produces a paler skin

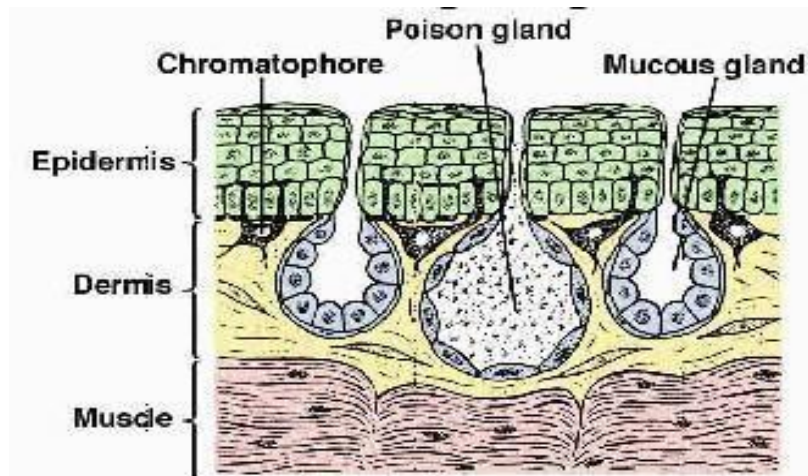


Figure 7.6 Section through the skin of a frog



7.5 Skeleton and Muscles

Amphibians have evolved a strong skeleton and associated musculature to support the weight of the body on land, raise the head during feeding. The vertebra centra are thoroughly ossified and

successful vertebrate interlock secured by means of overlapping processes on the vertical arches. The lobe-shaped paired fins of crossopterygian have been transformed into limbs supported by strong girdles, the hip or pelvic and the shoulder or pectoral girdle. Muscles of the trunk are powerful and lateral adulations of the trunk and tail help to advance and retract the feet

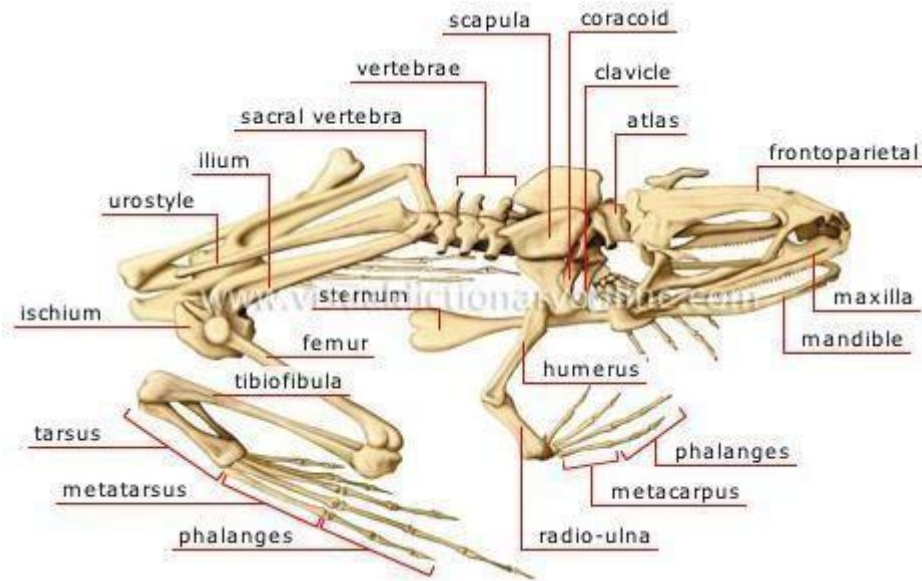


Figure 7.8 Skelton of frog

Sense organs and nervous system

Many changes occurred in the sensory organs during evolution. In the Larvae and aquatic adults, there are lateral line systems with mechanical and electrol receptors that are absent in terrestrial adults. Eye is clear and protected by tear glands and eyelids, respectively. The lens is flattened and refraction is by the cornea. The inner car can detect position and acceleration of the head, and auditory vibrations. When in water they hear in the same way that fish do, by detecting vibrations that pass through the body tissue to the inner car.

Respiration

Amphibians have more means of respiration than any other animal group reflecting transition from aquatic to terrestrial life. In different species the gills, lungs, skin and buccal-pharynx serve separately or in combination as respiratory organ.

- **Gill.** Three pairs of external gills occur in most embryos and larvae and persist in adults of some aquatic salamanders. In tadpoles water is drawn in through the mouth and nostrils then forced over the gills and out through the spiracles.
- **Lungs.** Lungs are primary sites for gaseous exchange in adults of most species. The lungs are merely sacs although there may be internal walls that increase surface area for gaseous exchange. They are not suction lungs but force pump lungs similar to those of lungfishes. Air is pumped into them by lowering the floor of the mouth and pharynx. The amount of air pumped is proportional to the size of the mouth cavity and pharynx. Lungs also serve as hydrostatic organs being inflated when the animals are floating.
- **Skin.** The skin contains many blood vessels that aid in aeration of blood. Cutaneous respiration is very important in eliminating CO₂ because the lungs are not ventilated frequently enough to flush this gas out as fast as it accumulates. Cutaneous respiration also helps the animal to remain submerged in water for long periods and to hibernate in ponds.
- **Mouth cavity.** Some species have buccal-pharyngeal respiration in which gases diffuse across the vascular lining of the mouth cavity and pharynx.

Circulation

The heart of the larvae is like that of a fish with one atrium and a ventricle. It receives only deoxygenated blood that is pumped directly to the gills. Adults have two atria and one ventricle. The left atrium receives oxygenated blood from the skin and lungs, the right deoxygenated blood from the general circulation. Despite the single ventricle there is some separation of a trail blood that sent to respiratory surface is almost all deoxygenated (see fig).

Excretion and water balance

Most species do not drink water so it passes through the skin to enter or leave the body. The kidney eliminates some water and reabsorbs ions. The kidney compensates for water passing through the skin by either reabsorbing water from the urine or producing copious amounts of dilute urine. The urine passes from the kidneys through the ureters into the cloaca and is excreted or stored in the bladders. Larvae of most species resemble fresh water fishes in that they excrete a large part of their nitrogenous waste as ammonia. Terrestrial species eliminate most nitrogen as urea while several tree species conserve water by eliminating up to 80% of their nitrogen waste as uric acid.

Reproduction and development

The reproductive organs of Amphibians are closely associated with the excretory organs. Sperms pass from the testes to the kidneys and ureters where they are stored. They are released through the

cloaca during copulation. The eggs enter the body cavity and are taken into the oviducts. The oviducts surround the eggs with jelly coat. The jelly swells when in water.

Most species have to return to water to breed. They mate in water where their eggs are deposited and hatch and the larvae live and grow until they metamorphose to adults. Each species has its own characteristic type of breeding place such as a pond, stream, pool and so on. Few breed on land.

The gelatinous coatings protect the eggs against shock, desiccation and predators. The form of egg masses and jelly coats are distinctive for each species. Toads' eggs have long strings, those of frogs in masses and for salamanders in clumps. In few species like the land salamander, eggs are deposited individually.

Evolution of Amphibian

The earliest amphibian belongs to subclass Labyrinthodontia and can be traced back to Devonian period. These were large creatures that could reach up to 1 meter in length. They were abundant in Paleozoic and Mesozoic eras but none survive today. They had lungs but retained fishlike tails and lateral line canals in the brain bones. This indicates that they spent most of their time in water.

Many early labyrinthodonts also retained small bony scales and notochord persisted in adults. Presence of scales and the large size of most species mean that cutaneous respiration, if used, was less efficient. The enamel and dentine of their teeth was folded to form a complicated pattern as in figure 7.3. This structure is termed labyrinthodont. Hence they were given the name labyrinthodont.

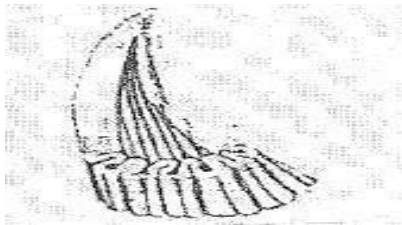


Figure 7.9 Labyrinthodont tooth characteristic of crossopterygians

Many people believe that Labyrinthodonts evolved from fresh water crossopterygian fishes. However, Labyrinthodonts probably gave rise to another extinct subclass the Lepospondyli that became extinct during the carboniferous period without leaving descendants. Living amphibians belong to the following orders

- **Urodela**- Salamanders
- **Anura** - frogs and toads • **Gymnophiona**- Caecilians.

Herpetologists group them into one subclass, Lissamphibia, because they share several unique characteristics. It is probable that Lissamphibians evolved from labyrinthodonts but fossil records of modern groups is very poor.

Salamanders: Order Caudate (Urodela).

They are closer in structure to ancestral amphibians than other groups of amphibian. Though smaller than labyrinthodonts, they retain the primitive body form with a long trunk and tail and legs that are approximately equal in size. They have a distinct head, trunk and tail. Salamanders live beneath stones and logs in damp areas. Some species are entirely aquatic. Most are small ranging in size between 5 and 15cm and primarily found in temperate regions.

Although adults have lungs, one family, Plethodontidae (mainly North America) have neither lung nor gills and rely totally on the skin and membranes lining the mouth and pharynx for gaseous exchange. The skin is exceptionally thin and capillaries are close to the body surface making it very efficient as a respiratory organ. Loss of lungs is curious adaptation for a terrestrial animal. It is not known why this happened.

Several families of salamanders have become paedomorphic; they retain many larval features as adults, meaning that they do not complete their metamorphosis. Paedomorphosis has evolved by neoteny, a process in which somatic development slows down relative to sexual maturation so that the animal becomes sexually mature before it has completed its metamorphosis. Neoteny is an adaptation that helps these animals to remain aquatic and so escape the difficulties associated with terrestrial life.

Salamanders are prey to many animals including snakes, birds and mammals. So many species have evolved adaptation that discourage predation such as secretions produced by glands on skins and are noxious this is coupled in some species with warning colouration. For example the terrestrial stage of red spotted newt, *Notophthalmus viridescens* is brilliant red and has particularly toxic skin. Birds soon learn to recognize this colour and avoid eating them. Some species can discard their tails, autotomy, when caught by a predator so that the predator is left with the tail as the animal moves away. The discarded tail thrashes about attracting the attention of the predator while the rest of the animal moves away. It grows another tail.

Reproduction takes place in water. Males of most species package their sperms in small capsules called spermatophores. They engage in an elaborate courtship ritual as they deposit them. The females take the spermatophores into their cloacas. Fertilization is internal. Eggs hatch into a larva that resembles the adult more than tadpoles do to frogs. The larva resembles adults in general body form and have teeth in both jaws. It differs from the adult in having pharyngeal gills slits and

external gills. They are carnivorous and retain external gills until metamorphosis. Limb girdles are largely cartilaginous.

Many of the terrestrial species reproduce on land by laying small clutches or eggs in damp sites under stones and logs. Parents often brood, protect and keep them moist. The eggs are provided with a large store of yolk and hatch as miniature adults.



Figure 7.10 Salamander

Order Anura

These are the toads and frogs. The terms frog and toad have no taxonomic meaning but refer loosely to adaptations for aquatic and terrestrial habitats respectively. Frogs are usually semiaquatic their skin is thin and moist and the hind legs are long and webbed to aid in jumping and swimming respectively. Toads spend more time on land and their hind limbs are shorter and the web is reduced.

The African clawed frog *Xenopus laevis* has adapted to aquatic life by becoming neotenic and retaining many larval characteristics such as lateral line in adults.

Anurans are more widely distributed than the other orders of amphibians. They occur on all continents except Antarctica but are absent from many oceanic islands because most species cannot tolerate salt water so they could not cross to the islands on floating vegetation or natural rafts. The number of known species is about 3500. Most measure 2 -12 cm although one species from West Africa, *Gigantorana goliath*, attains a length of 1 meter.

Anurans are specialized for terrestrial jumping. Thus they have no tail, hunk is short and compact, limbs are fused to the vertebrae and their hind legs are long and strong. They can see and breathe even when the rest of body is submerged in water because their eyes and nostril are located on top of the head. Frogs and toads get more exposed than salamanders and this cause high evaporative loss of body water. They make up for this by absorbing water and salts through the skin from their

surrounding. Anurans are prey to other animals and many have poison glands on their skin to keep off predators. Some frogs combine poison with warning colouration.



Figure 7.11 Frog

Order Gymnophiona

Caecilians are the least familiar group of living amphibians. They occur only in the tropics, Africa, Central and South America, India and the Seychelles. They spend most of their life burrowing through damp soil with their trowel-shaped heads. They have elongate bodies, ranging from 10cm to 1.5m, without legs or girdle. Early caecilians had limbs but have been lost as an adaptation for burrowing. They have 60 to 285 vertebrae.

At a glance they can be confused with a large earthworm. They are about the same size and annular folds on their skin give the appearance of segments. However caecilians, unlike earthworms, have a head with small jaws, nostrils and a pair of vestigial eyes partly hidden beneath the skin.

They have a pair of protractile tentacles lying in a pit between each nostril and eye these are important sense organs. The Cloaca is not quite at the end of the body so there is a short tail. Some species have small bony scales embedded in the skin and protect from abrasion. The lungs are long and thin and the left one is often smaller than the other. This is an adaptation to accommodate internal organs in a narrow space. We will see similar arrangement of organs in snakes.

They feed on invertebrates that they locate by the tentacles, fertilization is internal. During copulation, the male cloaca is averted to form a penis-like organ that transfers sperms to the female's cloaca. Most species are oviparous and lay their eggs in damp soil. The eggs hatch and the larvae complete development in nearby bodies of water. Others brood their eggs externally and the embryos develop directly into miniature adults. A few species are ovoviviparous, nourishing the embryos with uterine secretions but there is no placenta.

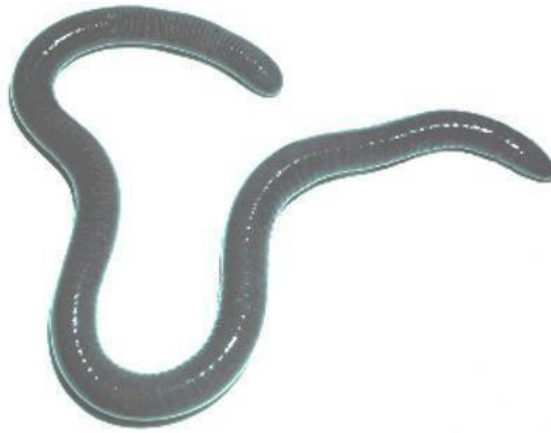


Figure 7.12 *Caecilian*

Activity 7.1



Visit a pond, temporary pool or a stream in your area and search at the bottom for amphibian eggs and larvae. Collect some and observe them using the hand lens in your dissecting kit. Note and draw the external features of the larvae. Try to differentiate frog and toad eggs.

Revision questions



1. Outline the differences between the three orders of amphibians
2. List the respiratory surfaces of amphibians.
3. Why do zoologists believe that salamanders are the most primitive amphibians?

7.6 Summary



We have seen that amphibians are adapted to terrestrial life by having limbs, lungs for breathing air, denser and stronger bones. However, few are completely independent of water because the immature are almost always aquatic and the skin of adults is highly permeable to water.

7.7 Further Reading



1. Harris C. L. 1996. Concepts in Zoology, Second edition. HarperCollins College Publishers
2. Hildebrand M. 1995. Analysis of Vertebrate structure, Fourth edition. John Wiley & Sons, New York.

LESSON 8: REPTILES

8.1 Introduction



Welcome to lesson 8. In lesson 7 we saw that amphibians adapted to life on land but their adaptation was deficient in that they had to return to water to breed and could only exist in moist habitats very much the same way present day amphibians do. However, one lineage of ancestral amphibians gave rise to vertebrates that were completely independent of water, the reptiles. In this lesson we will learn about the reptiles and the adaptations that made them independent of water.

8.2 Objectives



- List the characteristics of reptiles
- List the adaptations to fully terrestrial life
- Outline the adaptive radiation of reptiles



8.3 Features That Enabled Reptiles to Be Fully Terrestrial

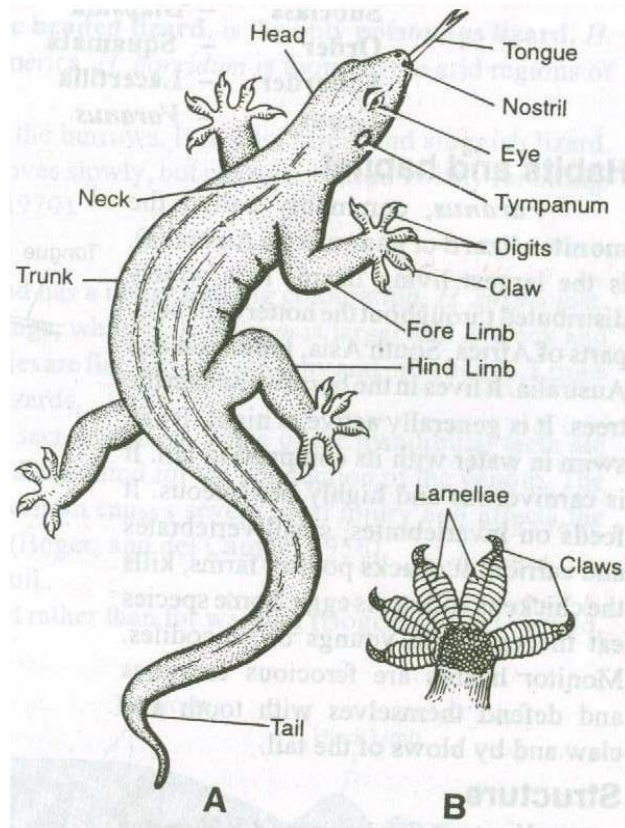
In Africa reptiles are surrounded with myths. You have heard of the snake that runs faster than a car on the highway, the woman who carries a log of wood home but the log turns into a python when put on fire, the "husband" snake that revenges the death of his "wife" three days later and so on.

These myths indicate how little we know about reptiles and the fear many of us have for them. But what is a reptile?

There is no simple definition for this group of animals because there is no single characteristic that is unique to reptiles. Generally, they are vertebrates that have the following features

- Body covered by horny scales.

- Heart usually three chambered but the with further separation of oxygenated and deoxygenated blood than in amphibians
- Skeleton fully ossified
- Eggs suited for development on land with membranes and shell to protect the developing embryo.



A – Dorsal View B – Ventral Side of the foot

Figure 8.1 Hemidactylus (House or Wall Lizard)

All these features are adaptations for terrestrial life and their acquisition made reptiles completely independent of water. The structure of the reptilian heart was discussed in lesson 3. Here I will only explain the first and last point in detail.

Integument (body covering)

The epidermis is dry and covered with horny scales with large amount of keratin, a water-insoluble protein deposited by epidermal cells. One difference between the scales of reptiles and those of fishes and amphibians is that they arise from the dermis rather than from the epidermis. The scaly skin protects the animal from desiccation, mechanical damage and harmful radiation. There are few glands on the skin although some have scent glands that produce secretions used in species and sexual recognition. In some snakes and lizards glands secrete irritating or sticky material that protects them from predators. Bone may develop in the dermis beneath the bony scales, as is the case with the bony plates in a turtle shell and small nodules of bone in some lizards and crocodiles.

The amniote egg

One of the most important achievements of reptiles was the evolution of the cleidoic egg, a self-contained system (Gr kleidoun = to lock up). The egg is provided with all the necessary food material and with fats that when metabolized yield water for the developing embryo's needs. The eggs obtain additional moisture through the shell and so most reptiles deposit their eggs in moist places. This egg is enclosed within a shell that provides protection and reduces the chances of desiccation and yet permits exchange of gases. Most reptilian eggs have leathery shells that are more permeable to carbon dioxide and oxygen than the hard shells of birds' eggs. Embryonic waste materials are stored in the eggs. Well found cleidoic eggs are found in reptiles, birds and insects and egg-laying mammals. Figure 8.1 shows the basic structure of a cleidoic egg.

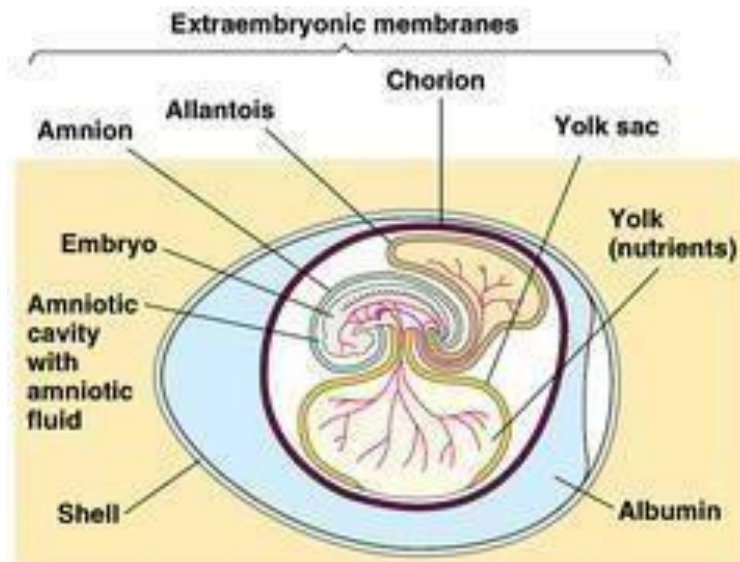


Figure 8.2 The cleidoic egg

In viviparous reptiles and mammals that give birth to live young, the eggs have little or no shell but the egg extra-embryonic membranes, are still present. Four extra-embryonic membranes are present. They lie outside of the embryo and play an important role in protecting and maintaining the embryo. They are;

- Yolk sac that surrounds the yolk mass.
- Amnion encloses the embryo and is filled with fluid which protects the developing embryo.
- Reptiles, birds and mammals are called amniotes because they all have this membrane.
- Chorion lies beneath the shell and surrounds both the yolk sac and amnion.
- Allantois grows out of the posterior end of the gut and comes to lie against the inner side of the chorion). Blood from the embryo is circulated through the wall of the allantois by allantoic vessels (called umbilical vessels in mammals). Gaseous exchange occurs here. Nitrogenous wastes in form of uric acid crystals are deposited in the cavity of allantois.

Circulatory System

The blood leaves the heart through 3 separate arteries that are twisted around each other, not through a conus and spiral valve as in the amphibians.

The pulmonary artery goes to the lungs. The left and right systemic arteries serve the rest of the body. As in amphibians, the systemic arteries join together to form a median dorsal aorta. The carotid arteries, which serve the head, branch from the right systemic artery.

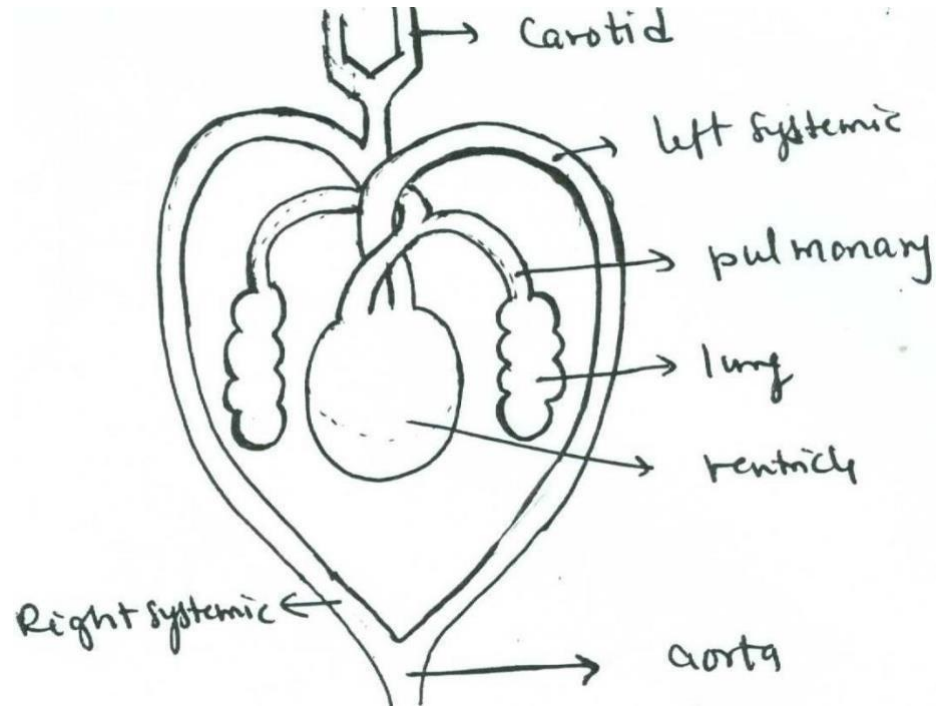


Figure 8.3 Diagram of a Reptilian (Crocodile Heart)

The heart of crocodile is different from those of other reptiles discussed because this is more easily understood

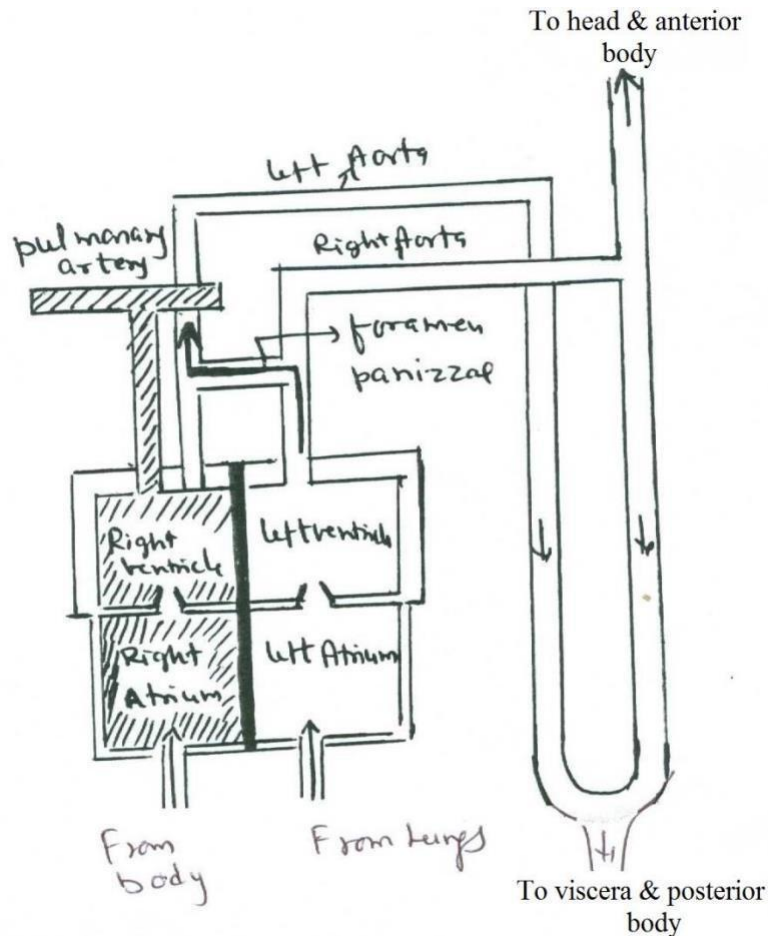


Figure 8.4 Schematic diagram of circulatory system of a crocodile

They have separate left and right atria as in frogs but they also have a partition separating left and right ventricles. The pulmonary and left systemic arteries open out of the right ventricle, and the right systemic from the left ventricle. (The twisting together of the arteries brings about this rather confusing situation in which the one that goes to the left arises on the right, and vice versa)

Where the left and right systemic arteries are in contact as they leave the ventricle, there is an opening between them known as the foramen panizzae. Small samples of blood from the main blood vessels

of crocodiles show that the pulmonary artery carries deoxygenated blood, with about the same oxygen content as the venous blood returned from the body to the sinus venosus. Also, the right systemic-artery carries oxygenated blood with about the same oxygen content as the blood returning from the lungs in the pulmonary vein. Less predictably, the left systemic artery normally

carries oxygenated blood of similar oxygen content. It apparently does not receive its blood from the right ventricle, but mainly or entirely through the foramen Panizzae from the left ventricle.

An explanation for this has been found by using pressure transducers to record pressures in the heart and major arteries. Pressures in both systemic arteries are generally considerably higher than those in the right ventricle and pulmonary artery

This pressure difference must prevent the valve from the right ventricles to the left systemic from opening and ensure that this artery receives blood only through the foramen.panizzae.

It thus seems that all the oxygen blood from the left atrium goes to the system circulation and all the deoxygenated blood goes to the right atrium goes to the lung. The difference in pressure between the two sides of heart must be due to the pulmonary circulation offering less resistance to flow than the systemic circulation. However, its resistance can apparently be increased by constricting the base of the pulmonary artery so that this happening experiments in which diving was stimulated. If it happens in normal diving it must help to conserve the oxygen in the lungs (by reducing blood flow to the lung while ensuring that the blood supplied through the carotid arteries to the brain is oxygenated blood which is thus a mixture from both sides of the heart.

Reptiles other than crocodiles have only one ventricle ,and no foramen panizzae the single ventricle has a rather complicated arrangement of incomplete partitions which have more or less the same effect as the separate ventricles and foramen penizzae of crocodiles. Both systemamic arteries normally carry oxygenated blood but if the pulmonary vessels are constricted (as in turtles when they dive) the left systemic receives deoxygenated blood.

Reproduction

Sex determination is by genes. In many lizards females have two x chromosomes and males one. In others it is the reverse. In many turtles, some lizards and all crocodiles, sex determination depends on the temperature during embryonic development. (Harris 1996). Fertilization is internal. Male reptiles have penis that introduce sperms into the female's reproductive tract. Snakes have two such structures known as hemipenes but the mate uses only one at a time. Yolk is stored within the egg while still in the ovary. The egg is fertilized in the oviduct and after fertilization the shell is added. Reptiles lay fewer eggs than fishes and amphibians possibly because mortality of embryos is low. Most are oviparous, some are ovoviviparous but few have developed a placental relationship with the embryos. Most do not care for their young but crocodiles do. They make a nest in the sand and when the young hatch parents lead them into the water and teach them to swim.



8.5 Evolution and Adaptive Radiation of Reptiles

Since they were the first vertebrates to be fully terrestrial, they had few competitors and they multiplied rapidly and occupied all available niches and specialized accordingly. Their adaptive radiation involved different methods of locomotion and feeding. Different feeding methods resulted in modification of jaw muscles and the temporal regions of the skull and so reptiles are grouped according to the morphology of the skull.

- In the most primitive reptiles, the roof of the skull has openings for the major sense organs only, nostril, eyes and pineal eye. Such a condition is known as anapsid and reptiles with that kind of skull are placed in subclass Anapsida. Example of anapsid reptiles are the turtles.
- In other reptiles there are other openings in the temporal region known as temporal fenestrae. Those with a single opening high on each side of the cheek are put in subclass Euryapsida.
- Those with a single opening on the side but more ventrally placed than in euryapsids belong to subclass Synapsida, the mammal like reptiles that are all extinct. They give rise to the mammals.
- In diapsid condition, both pairs of fenestrae occur. There two subclasses with this type of condition, Lepidosauria (for example snakes) and Archosauria (example crocodiles). They give rise to birds and other reptiles like snakes and crocodiles.

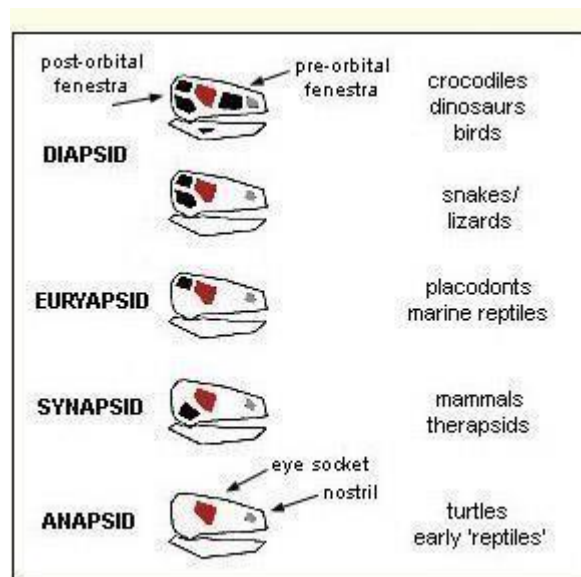


Figure 8.5. Temporal openings of different evolutionary lines of Vertebrates

Dinosaurs and Their Allies

During the Mesozoic era, reptiles dominated the land. They belonged to several orders that had many features in common. They had reduced pectoral limbs, enlarged pelvic appendages and heavy tails.

Saurischians evolved from small ancestors to become giants inhabiting swamps. Tyrannosaurus was the largest carnivore in the history of the world. It had large jaws with 1.5cm long teeth. It was 20 about 6 metres high. Other saurischians were herbivores.

Ornithischians were terrestrial, walked in twos just like saurischians. They had many protective devices such as bony plates on the body and horned skull.

The large size of the dinosaurs may have enabled them to achieve a more constant temperature but the real reason is not known. In one group, Pterosaurians, the front limbs, became modified into wing giving rise to the birds. The reptiles declined suddenly but the reason is not known. Today only four orders survive of the 16 that existed during the Mesozoic era. They are listed below.

Turtles (order chelonia)

These are the only surviving anapsids. They have no teeth and have hard horny jaws. Have a tympanic ear. Body encased in a protective shell made of pieces of dermal bone that is fused to each other and to the ribs and vertebrae. The bony plates are covered by horny material similar to the scales of other reptiles. The portion of shell covering the back is known as carapace and the ventral part is called plastron.

The shell does not allow lung ventilation by a change in diameter of the rib cage. Instead, muscles that line the skin pocket into which limbs can be withdrawn act as diaphragms. Aquatic species can also exchange gases under water through the vascular linings of their mouth and pharynx and through sacs that bud off the cloaca.

Most species are semi-aquatic and tend to have flattened shells that offer less resistance to swimming. Several groups are entirely aquatic and their front legs are modified as flippers. Tortoises are completely terrestrial and have dome-shaped shells that predators find difficult to bite, crush or swallow. **Diapsids**

All other reptiles are diapsids. They belong to three orders, squamata, Crocodilia and Sphenodonta.

Sphenodontids or tuatara

These were abundant in the early Mesozoic era but today only two species survive; *Sphenodon punctatus* and *S. guntheri*. Until recently it was thought that only the former species existed. They are confined in distribution to New Zealand. They look like lizard but have a more primitive skull structure than any true lizard. They move using legs, lateral undulation of the trunk and tail. They have a pineal eye on top of the head. Males have no distinct copulatory organ and transfer of sperms is by pressing the cloacas together. They are referred to as the "living fossils" because they have not changed much from the species that exist 150 million years ago.

Lizards

Lizards and snakes belong to order Squamata; lizards are the oldest and most primitive squamates and, probably evolved from sphenodontid-like ancestor early in the Mesozoic era. They are placed in suborder Lacertilia (or Sauria). Many species are slender, some compressed laterally and flattened dorsal ventrally. Limbs may be long or short, are reduced on some and entirely lacking in a number of limbless lizards. All limbless lizards live in the soil. The tail vertebrae in many species are incompletely ossified. If the tail is seized the vertebrae separate and animal escapes and in time regenerates another tail.

Snakes

They belong in order Squamata, suborder Ophidia. They have elongate body. They have no limbs or girdles. The sternum, eyelids, external ear openings and bladder are also lacking. The eyes are covered by a nictitating membrane. Their jaws are extensively flexible and can swallow prey much bigger than a mammal of the same size.

They can detect low frequency vibrations through their skull bones. They compensate for lack of senses and sight and hearing with a very sensitive tongue. Particles adhere to it, it is withdrawn into the mouth and directed towards a pair of openings in the roof of the mouth that lead into a specialized parts of the nasal cavity, Jacobson's organ, where the odor of the particles is detected. Although this organ is present in many other terrestrial vertebrates it is very highly developed in snakes. Some species including pythons, pit vipers and boas that feed on warm-blooded animals have sensory pits that detect infrared rays. This enables the snakes to detect objects warmer than their surrounding.

The Skull is delicate and most of the bones can move upon one another. Teeth slant backwards and are fixed on jaws and on bones on the roof of the mouth. The used to hold food while being swallowed poisons snakes have a pair of specialized teeth or fangs on the lower maxillary bone.

Snakes are feared because of their venom or poison. Venom is saliva containing a mixture of digestive enzymes such as proteases, collagenases and phospholipases. Venom kills and starts the digestion. In some snakes the fangs are at the front part of the upper jaw and they inject venom into the prey before it is swallowed. In others, the fangs are towards the back of the mouth and venom is injected as the prey is swallowed. Some of the venoms work by affecting the nervous system, neural toxic venom, and kill very fast by paralyzing the prey. Others have poison that affects the blood and tissues, hemotoxic venom, causing hemolysis, clotting and so on. Many snakes (about $\frac{3}{4}$), however, are not poisonous but suffer because most of us in Africa believe that a snake once spotted should not be left alive. There are two species of lizards that are poisonous.

Crocodiles and alligators, order Crocodilia

These are the most advanced reptiles. The heart is completely divided into four chambers. Though they are poikilotherms, their body temperature is closer to constant than in other reptiles.



8.6 Interaction of Reptiles and Humans

The deliberate killing of reptiles by humans is a threat to continued existence of these animals. Many people do not think they are of any benefit to man. However, reptiles can be a source of food. Even snakes can and are eaten by certain people. They can also control the population of rodents.

Activity 8.2



- Find out which species is the most poisonous, colourful and largest snakes.
- List five examples of snakes that have neurotoxic and live that have haemotoxic venom.

Revision Questions



- Why is it difficult to give a precise definition of a reptile?
- How did the cleidoic egg enable reptiles to become independent of water?
- What characteristics distinguish a snake from a lizard?

8.7 Summary



We have seen in this lecture that the most important evolutionary achievement of the reptiles was to become adapted for terrestrial life apart from water. The acquisition of dry cornfield skin that reduce loss of water from the body and the production of eggs capable of developing on

land made this possible. These achievements resulted in ancestral reptiles to diversify and occupy all available niches. Their decline that was rather sudden cannot be explained with the current knowledge.

LESSON 9: BIRDS

9.1 Introduction



Welcome to lesson 9. In this lesson you will learn about one of the most advanced classes of vertebrates; the birds. They are the most colourful, vocal and familiar of all vertebrates. The distinctive colouration and voices of birds appeal to the human eye and ear thus they are some of the well-known animals. They appeal even to non-zoologists and we have many groups of organized bird watchers in Kenya with participants of varying backgrounds: lawyers, botanists, accountants and so on. You will no doubt enjoy this lesson.

9.2 Objectives



At the end of this lesson you should be able to

- State the characteristics that distinguish birds from other vertebrates
- List adaptations of birds for flight.
- Draw and outline the structures of feathers
- Describe the digestive, circulatory, respiratory, excretory, nervous and reproductive systems of birds.



9.3 Characteristics of Class Aves

Birds are a very advanced group of vertebrates only comparable with the mammals from which they notably differ in the type of body covering and in their reproduction. At first glance they appear to be quite variable in size, colour, behaviour etc. However, a closer look shows that they exhibit less morphological, anatomical and physiological diversity than do species in other vertebrate classes. This is because they are all adapted for flight. The class adapted for flight early in their evolution and so even the few species that have reverted to completely terrestrial life show anatomical and physiological features that reflect their evolution from flying ancestors.

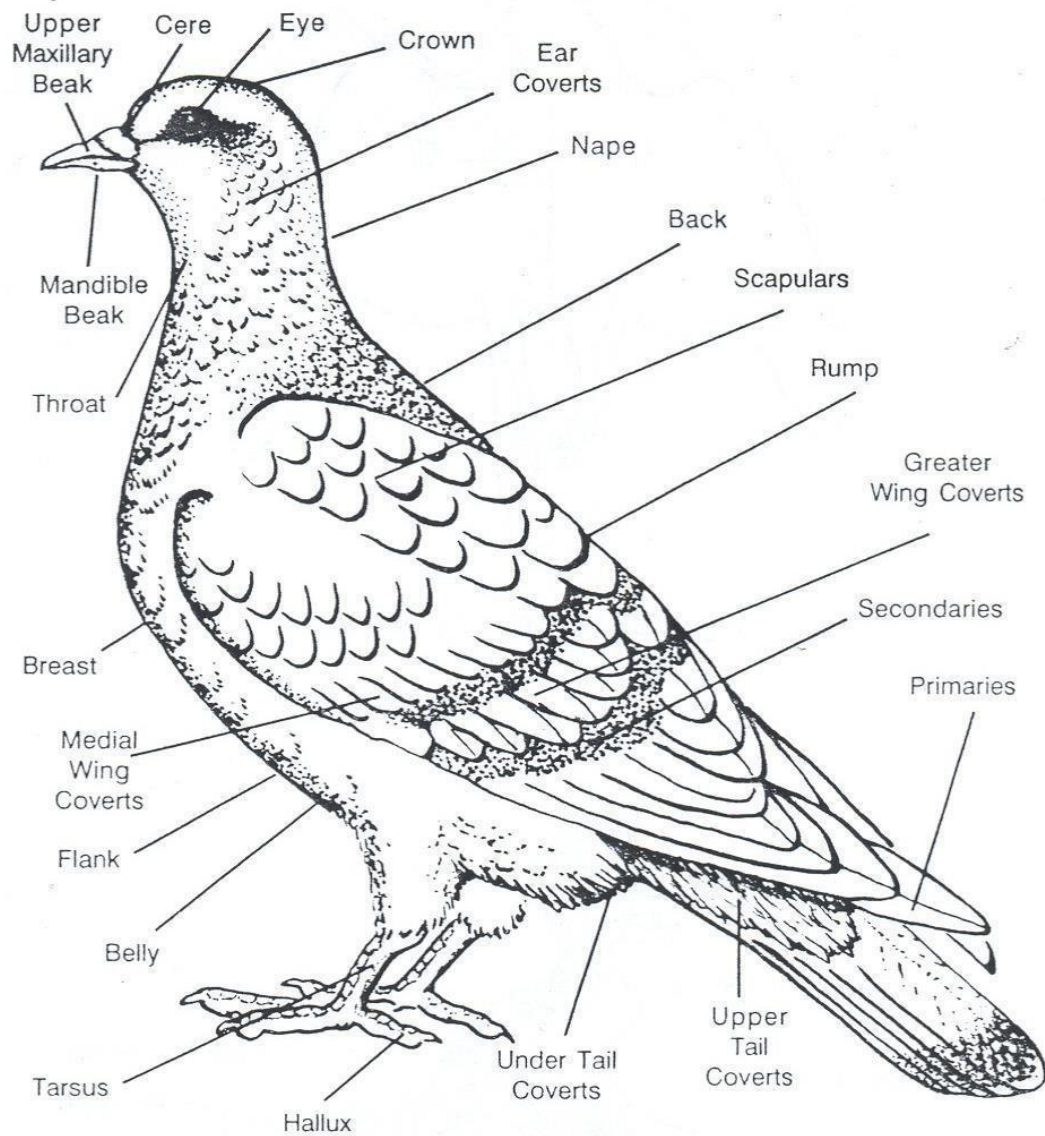


Figure 9.1 Diagram of a Pigeon

The single most important diagnostic feature of birds is the feathers covering their bodies. No other animal possesses feathers. Birds share many reptilian features such as scales on their legs, large yolky eggs and the arrangement of many internal structures. The only fundamental difference between birds and reptiles is the feathers. The feathers in turn enable the bird to maintain constant body temperature and fly. Of course flight required anatomical modifications that that lack in reptiles. Many adaptations for flight boil down to meeting two requirements for flight: high power and low weight. The ability to fly enables birds to occupy habitats inaccessible to other animals. Thus birds occur in all continents and all habitats.

The following are features that distinguish birds from other animals.

- Body covered by feathers
- Two pairs of limbs. The fore limbs modified into wings.
- Light skeleton though strong and fully ossified.
- Mouth modified into a beak (bill) with horny sheath. No teeth.
- Heart four-chambered so deoxygenated and oxygenated bloods are completely separately
- Body temperature internally regulated, endothermic.
- Fertilization internal
- Eggs large and yolky
- No urinary bladder. Excretion semisolid
- Female usually with only the left ovary
- Eyes have upper and lower eyelids beneath which is the nictitating membrane- which can be drawn independently across.
- Specialized care for the young
- Highly developed voices, hearing and sight

Birds are generally small, the largest being the African Ostrich, *Struthio camelus*, that is about 2 m tall and about 150 kg. The smallest bird is Helena's hummingbird of Cuba that is 5cm long and weighs less than 3 g. No bird attains as large size as mammals or fish. The largest mammal is the Blue Whale, *Balaenoptera musculus*, and weighs 136,200 kg



9.4 Structure of a Bird

In this section, we will look at the architecture of the bird and how it is modified to meet the requirements or flight.

9.4.1 Skeleton

The skeleton is designed to reduce weight. So

- The bones are hollow but strong and flexible.

- Bones separate in other vertebrates are fused to increase their strength. For example skull bones are fused except for the nasofrontal hinge that permits slight movement of the upper jaw in many species.
- The cranium is rounded and paper-thin. The cranium has large orbits sheltering the eyes.
- They have no teeth so their jawbones and jaw muscles need not be heavy.

The neck contains 16 cervical vertebrae that are articulated in such a way that the head and neck are very mobile. This enables the bird to use the beak for feeding, building, preening etc. The trunk vertebrae are closely fitted together. Those of the thorax have ribs articulating laterally and the rest are fused into a solid synsacrum to which the pelvis attaches. The ribs are long, flat, thin and jointed to allow extensive movement for breathing and flying. Each rib has a distinct vertebral and sternal part, the two joining nearly at right angle. Each ribs overlaps with its neighbour to strengthen the thorax. The sternum or breastbone in flying birds is modified by addition of a thin flat keel for attachment of wing muscles. The sacral and hip girdle bones are fused and modified to form a tube-like structure. Except in ostriches, the pubic bones are joined mid ventrally but leave a wide open passage for the large eggs.

The forelimb has the same parts as in other vertebrates. It consists of the humerus, radius and ulna but other bones are modified to provide stable support for light feathers. Only two carpals and three digits are present. The other carpal bones are fused to the three metacarpals to form carpometacarpus.

The hind limb consists of femur, tibiotarsus paralleled by incomplete fibula, the fused tarsometatarsus, knee bone and four toes three in front and one behind.

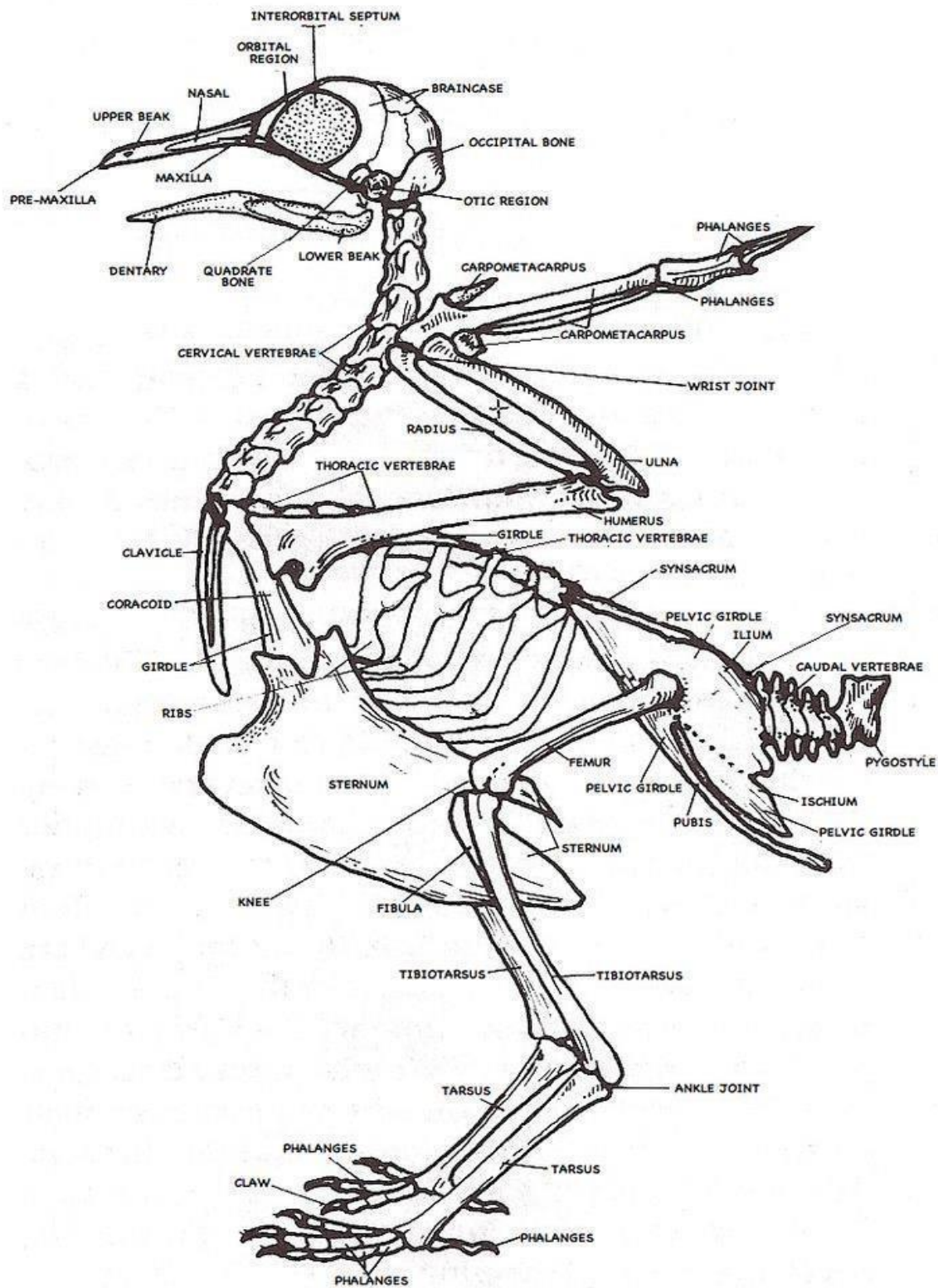


Figure 9.2a, Skeleton of a bird

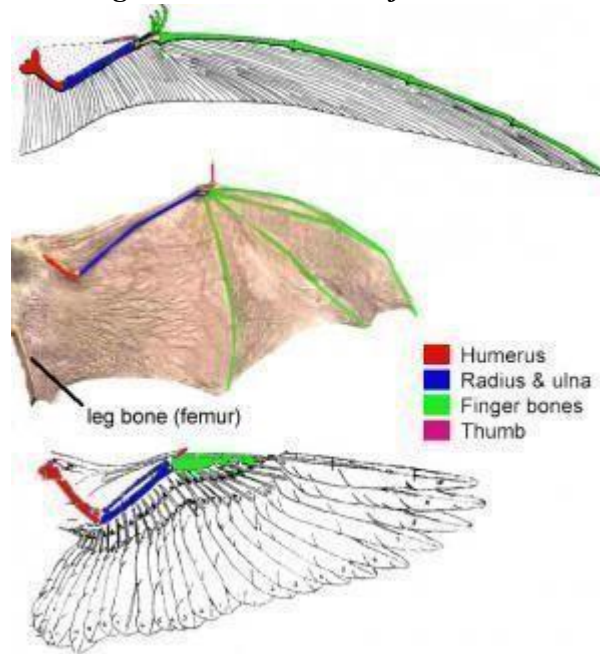


Figure 9.2 b, Structure of bats, birds compared to the ancestral bird

9.4.2 Body covering

The skin is soft, flexible and loosely attached to the muscles beneath it. It has no glands except the oil gland (also known as uropygial gland or preen gland) at the base of the tail. The gland secretes oil that is used to clean feathers and prevent the bill from becoming brittle. The body is covered with feathers that grow from follicles in the skin. These epidermal outgrowths are made of keratin, a flexible protein. Feathers insulate, streamline and reduce weight of the body of a bird. They also provide the wings with large surface area needed for flight. The varied patterns, colour, texture and shape of feathers help birds to signal their age, sex, social status and species identify to one another. Some birds have plumage (assembly of feathers on a bird at any one time) that blends in with their surrounding to provide camouflage so they are not easily seen by their predators.

9.4.3 Feathers

Feathers are homologous with the reptilian scales. The growth of a feather begins just like the scale of a reptile. It begins with a dermal papilla that thrust up the overlying layer of epidermis. The base then sinks into a circular depression, the future follicle that will hold the feather into the skin. The outermost epidermal cells on the bud form a smooth cornified sheath within which other epidermal layers are arranged in parallel ribs branching at an angle from a larger median rib that becomes the future shaft. The others become barbs. Blood vessels and nerves from the dermis enter the papilla,

forming a pulp. The pulp supplies nutrients to the surrounding epidermis so that it grows and produces keratin that forms the feather. When development is complete, the pulp dries up leaving a hollow quill to the feather is purely an epidennal structure. When growth is complete the sheath breaks and crumbles or is removed by preening, then the feather spreads to its completed shape. See

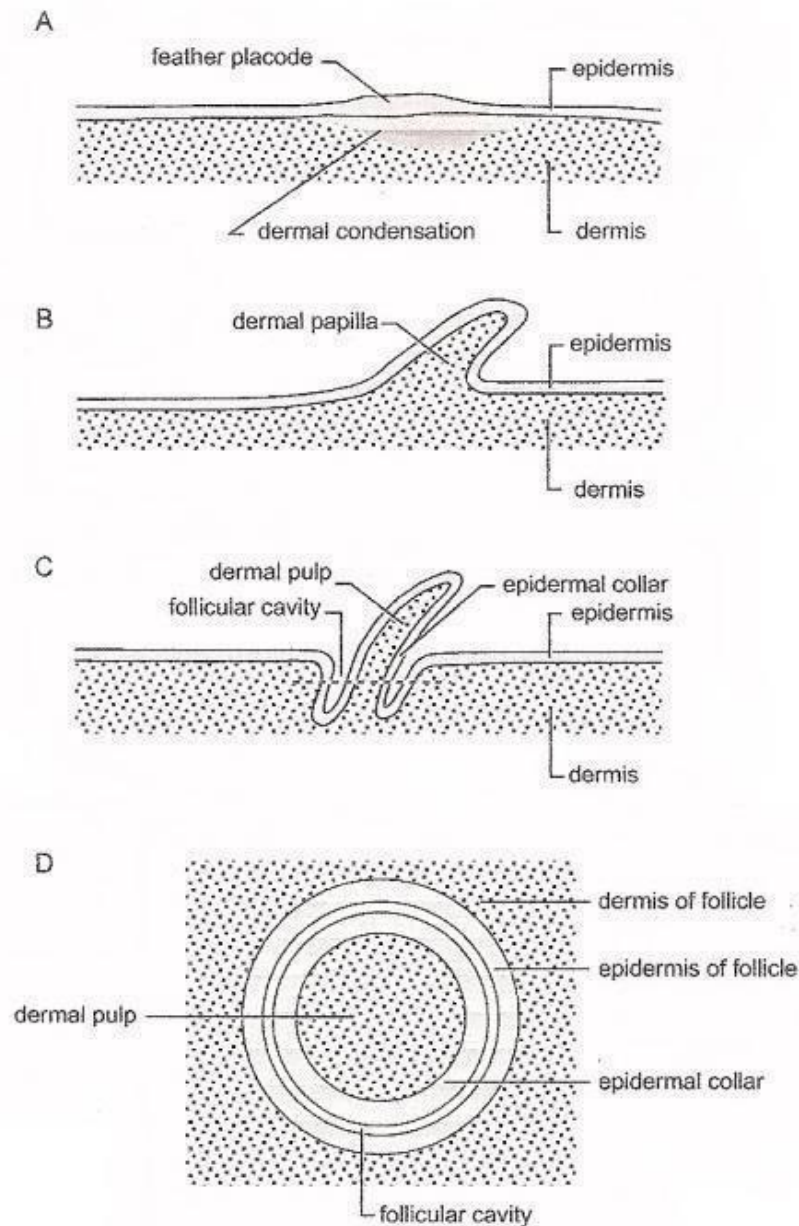


Figure 9.3 Development of a feather.

(C) The shaft and attached barbs develop between protective sheaths. (D) When mature, the sheath bursts and the feather expands to full size.

Pigment for coloration is deposited in the epidermal cells during growth in the follicle. The black, brown and dark red colours are due to melanin pigments. Bright yellow, orange and red are due to carotenoids, while bright red may be due to porphyrin pigment. Other colours may be due to interaction of light with the feather. For example the metallic blue of sunbirds and starlings is due to scattering of light by particles or overlying keratin layer.

Types of feathers

Contour Feathers

These provide the external covering and establish the colour of the bird's body. Each consists of a flattish vane supported by the central shaft that is an extension of the quill attaching to the follicle. Each half of the vane is composed of many narrow parallel and closely spaced barbs joining the sides of the shaft. On the proximal and distal side of each barb are numerous smaller parallel barbules with minute barbicels or booklets that interlock those of the adjacent barbules. If the barbules separate the birds preens the feather until they are together again. In birds that do not fly, for example the ostrich, the hooks are lost and the feather is very fluffy i.e. for insulation only. The contour feather at the posterior border of the arm, hand and tail form the flying surfaces and are called flight feathers. **Down feathers**

This is type of feather that covers young birds. Also found under contour feather in some adults. Their barbules lack hooks so they are fluffy and provide excellent insulation.

Filoplume

These are minute hair-like feathers with a slender whip-like rachis bearing few barbs and barbules at its tip. These are interspaced among the flight feathers. Their follicles have many nerve endings suggesting that they could be sense organs may be controlling movement of flight feathers.

Bristles

Some birds have hair-like growths that are modified feathers each with a short quilt and slender shaft with a few vestigial barbs at the base. They are commonly found around the eye where they

- Protect the eyelashes as in ostriches. They are also found around the nose where they filter dust and round the mouth of flycatchers where they are believed to act as nets to catch insects.

Since feathers are not living and they get damaged most birds molt them at least once a year. Molt is the process of feather replacement. As feathers become worn out, they loosen on their follicles and drop out, pushed by the already growing feather underneath them. This is an orderly and gradual process so that portions of the body are not left bare. Large feathers of wing and tail are shed in symmetrical pairs so that flight is not hindered. Although the main function of molting is to replace worn out feathers, it also serves as a means of acquiring different plumage especially prior to breeding or it may serve hygienic purposes.

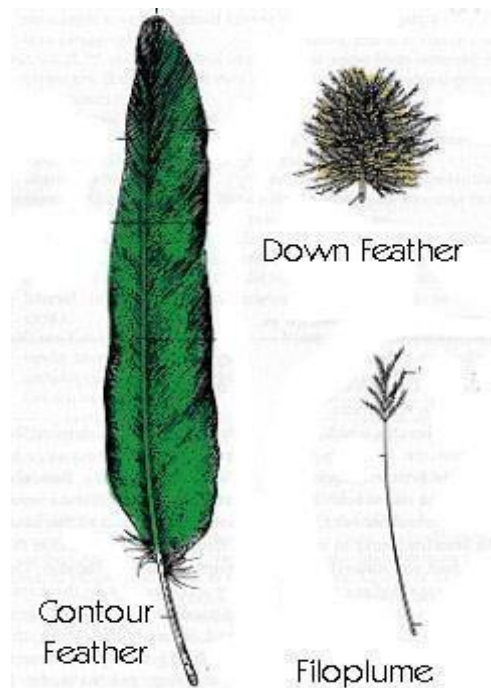


Figure 9.4 Types of feathers of a bird

Activity 9.1



1. Pluck feathers from the tail of a chicken, the breast and around the beak. Which of the different types of feathers discussed above can you identify?

9.4.4 Muscular system

As a result of adaptation for flight, the muscles of birds have become altered in structure and distribution. The main change has been in locomotor muscles of the wing and legs. There is great reduction in the muscles on the back. The rigidity of the sacrum and dorsal bone eliminates the need for strong dorsal muscles. Limb muscles enlarged to enable rapid activity. Movement of the wings is achieved by contraction of large pectoral muscles on the breast, pectoralis, which originates from either side of the keel and inserts on the ventrolateral head of the humerus. Its contraction moves the wing down and lifts the body in flight. The strong muscles of the femur and tibiotarsus are used for running and perching. The shanks and feet have little muscles, an adaptation to reduce heat loss, as these parts are not covered by feathers. The toes are moved by tendons connected in the upper segments of the legs.

9.4.5 Digestive system

Many birds possess a generalized digestive system capable of processing mixed diet but in few it is specialized to process specific type of food. The beak is the organ that collects food and its shape and structure reflect the diet type of a bird.

The tongue is small, sharply pointed and covered with a cornified epithelium. It varies in shape and function. For example in parrots it is muscular, in birds that feed on nectar it is tubular or brushy while it is vestigial in ostriches. Some birds swallow their food quickly and the tongue and mouth are poor in taste glands.

Esophagus extends from the pharynx to the stomach. It has mucous glands and is somewhat muscular. Its size depends on the size of the food swallowed. Species that swallow small food items such as insects have narrow and those that swallow large food items have wide distensible esophagus. In many species the esophagus has an enlarged part, the crop, where food is stored because their stomachs are too small to accommodate all the food. In many doves, pigeons and flamingos with young, the crop secretes "pigeon's milk" analogous to mammalian milk which they feed to their young. The crop allows a bird to gather food in hurry. It also allows seeds to soften before being subjected to digestion in the stomach.

All birds have two types of stomachs; an interior glandular proventriculus; and prosterial stomach or gizzard. Proventriculus secretes protein-digesting enzymes. The gizzard is lined with horny plates that crush and grind foods. Many seed eating birds swallow grit that aids the grinding action of the gizzard. In carnivorous birds the gizzard is a trap that prevents the sharp bones and other indigestible fragments from proceeding further down the intestines. Such items are rolled into elongated pellets and regurgitated by mouth. The two lobed livers is the largest gland in the body and its function is the same as that of liver in mammals.

The intestines are coiled and are the main organ of digestion and absorption of food. Birds show their reptilian origin by having intestines with circular muscles lying outside the longitudinal layer which is a reverse of the order in mammals. Intestines lead to the larger rectum. At the junction are the two slender caeca, blind pouches. Beyond the rectum is the cloaca, a common chamber into which urinogenital and digestive systems open.

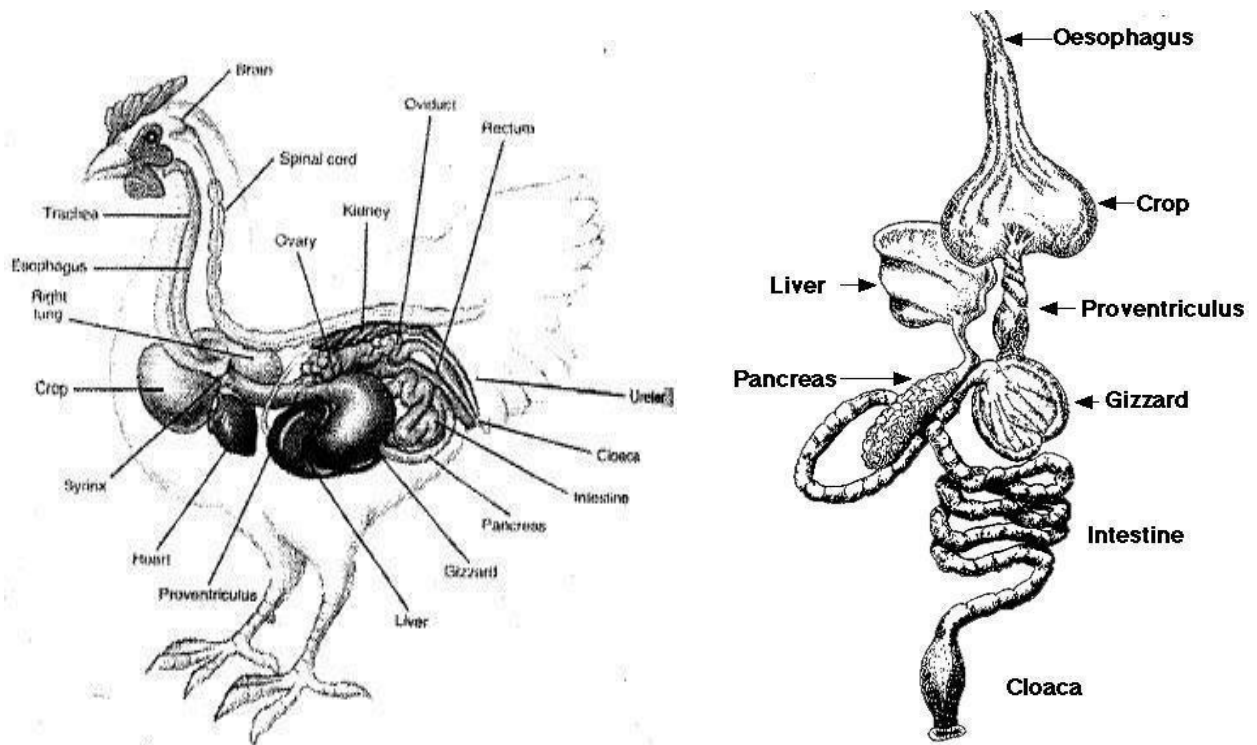


Figure 9.5a, Internal structures of a bird b, Digestive system

9.4.6 Circulatory system

A completely double circulation occurs in birds for the first time, since at no point is there any opportunity for oxygenated and unoxygenated blood to mix. Pulmonary veins return oxygenated blood from the lungs

To the left atrium. Both atria are thin-walled. The ventricles are completely separated as in crocodilians. Blood from the left ventricle is distributed over a considerable distance to all parts of the body and has therefore a heavy muscular wall. The relatively thin-walled right ventricle forces blood only a short distance through pulmonary arteries to the lungs. A single atrio-ventricular valve separates the right atrium from the right ventricle. Two valves, however, known together as the bicuspid valve, are present at the left atrio-ventricular aperture.

The main advance shown by the heart of the bird over the four-chambered crocodilian heart lies in the elimination of the left aorta. Only two vessels leave the heart in birds: a pulmonary trunk, or aorta, from the right ventricle, and a systemic aorta, corresponding to the right aorta of reptiles, from the left. A single set of three semi-lunar valves is present at the base of each. A well developed coronary system is present in birds. Coronary arteries arise from the systemic aorta. A venous, coronary sinus enters the right atrium.

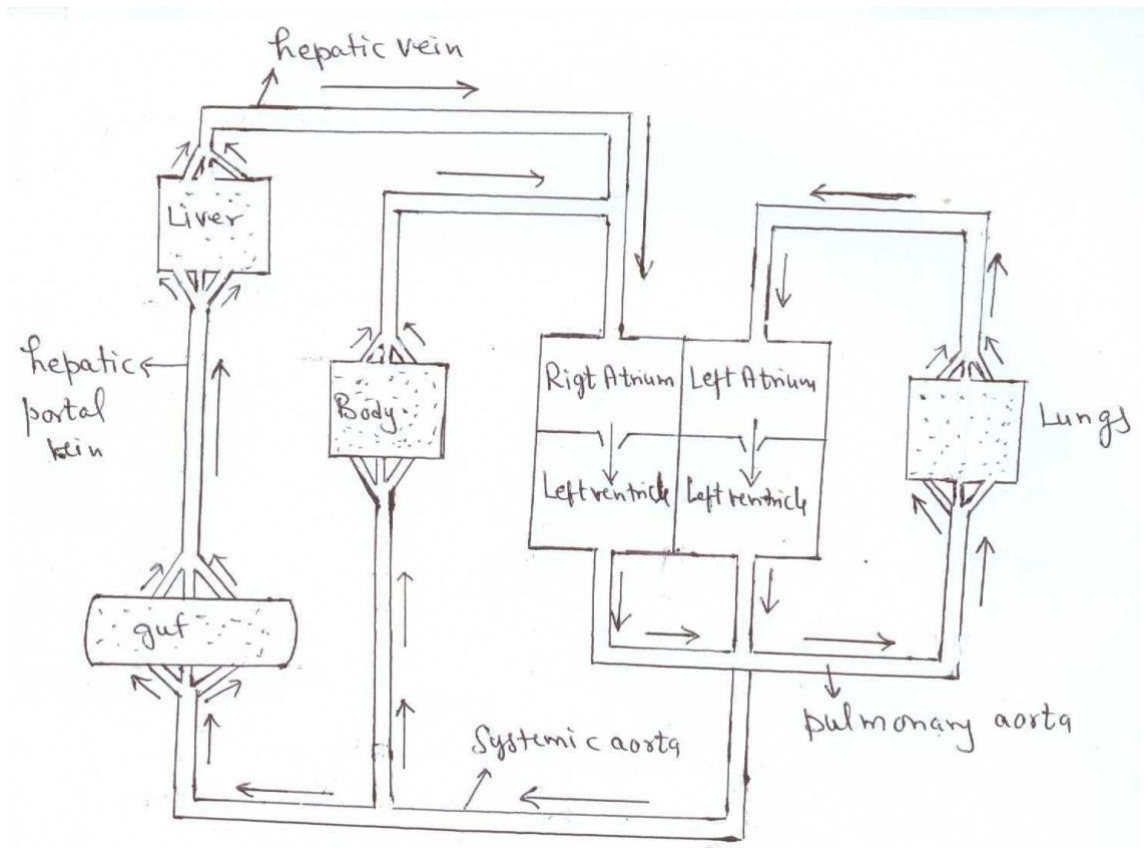


Figure 9.6 Schematic diagram of a circulatory system of a bird

9.4.7 Respiratory system.

Lungs are comparatively smaller than most mammals but are efficient because air flows through them in one direction unlike in mammals and reptiles where air moves back and forth. The Oneway flow helps to maintain much higher concentration of O_2 at the exchange surface than in other terrestrial vertebrate. The one-way flow is possible because the lungs consist of numerous tubules called parabronchi through which air flows in one direction.

In reptiles and mammals the bronchus divides again and again into smaller tubes ultimately ending in sacs or avicoli. In birds however, the bronchus passes through the ventral side of the lung and enters the posterior air sacs of the abdomen. Inside the lung each bronchus, called mesobronchus, gives off 4-6 secondary tubes called ventrobronchi that in turn subdivide into smaller units called parabronchi. These are small parallel tubes each several millimeters in diameter. From each

parabronchus extends outwards air capillaries and are surrounded by blood capillaries. It is between the air and blood capillaries that gaseous exchange take place.

Unlike the avoli of mammalian lung, the parabronchi are open on both ends. They connect to the ventrobronchi and dorsobronchi on the ventral and dorsal sides respectively. The dorsobronchi join the parabronchi with the mesobronchi that in turn joins with the posterior and anterior air sacs respectively. The air sacs act as air reservoirs. Air sacs have further functions: making the bird light and provide large surface area through which water evaporates thus losing excess heat.

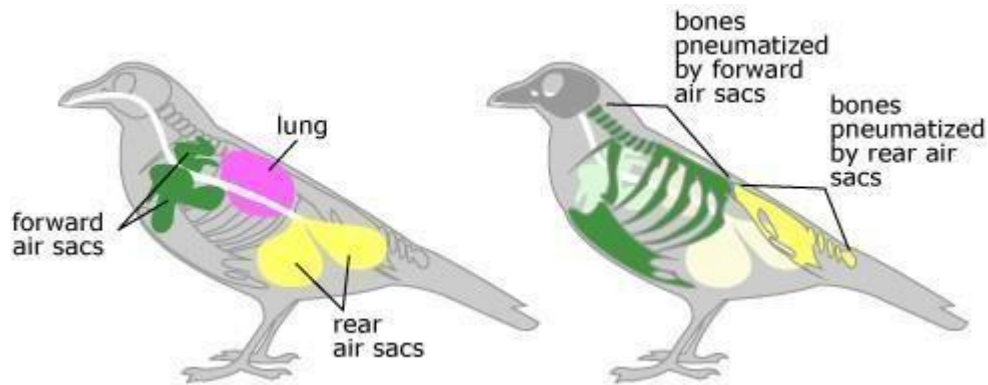


Figure 9.7 Air sacs of a bird

Respiration acts like a two cycle pump. During the first inhalation air passes through the lung into the posterior air sacs. During the first exhalation, first slug air moves to the lungs into the parabronchi where it gives up oxygen and receives carbon dioxide. During the second inhalation, a second slug of air is drawn into the posterior air sacs while the first moves to the anterior air sacs. During the second exhalation the second slug of air moves to the lungs and the first is expelled from the anterior air sacs to the bronchus and to the outside. This unidirectional flow of air explains bird's efficient extraction of oxygen and ability to fly at high altitudes.

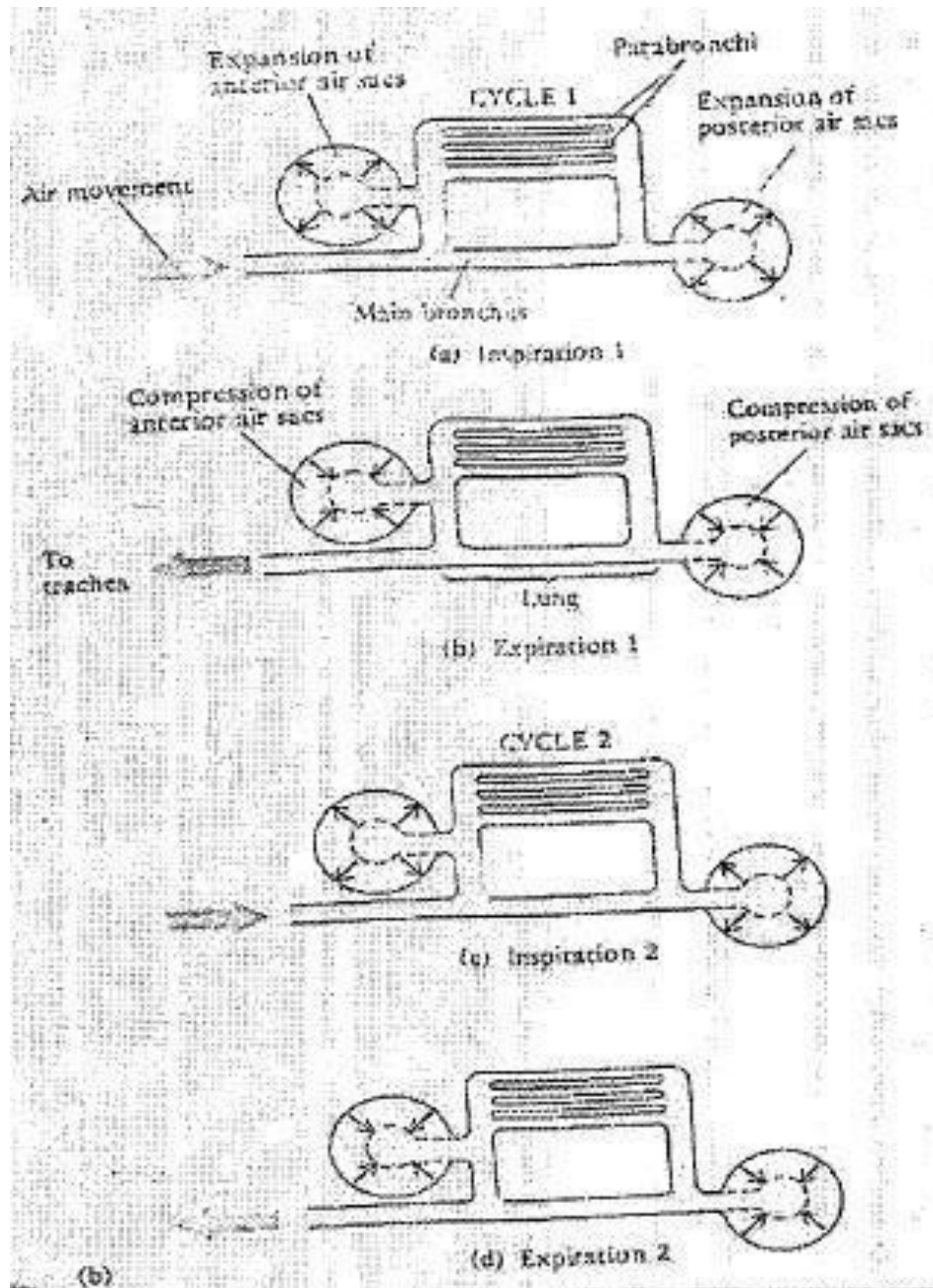


Figure 9.8 Respiration in birds; diagrams of movement of air

9.4.8 Excretory system.

The excretory system is similar to that of mammals. The kidneys are metanephric type. They are 3-lobed structures attached dorsally under the pelvis. From each kidney a slender ureter extends

posteriorly to the dorsal wall of the cloaca, there is no bladder. They excrete uric acid in a semisolid form. It passes out with feces as a whitish material on the latter. Most of the uric acid is secreted into the kidney tubules rather than entering by filtration. Excess salts that sea birds again are eliminated by salt excreting glands located above the eyes and discharged into the nasal cavities.

9.4.9 Reproductive system.

Male have two whitish oval testes attached to the anterior end of the kidneys. From Each testis is a convolutedly ductus deferens extending posteriorly parallel to the ureter. The ductus deferens enters the cloaca on a papilla close to the urinary opening. In the cloaca of ducks, geese and ostriches is a median copulatory organ (penis) for transferring sperms to the female cloaca (this organ is present in chicks when they hatch but disappear after few days- used for determining sex of young). Testes are fully enlarged during the breeding seasons. Sperm developed in the testes accumulate in the ductus deferens to be transferal to the female's cloaca during copulation.

Females have only the left ovary developed. The right ovary is commonly present, in a rudimentary form, and may enlarge to functions if the left is removed. The ovary is near the kidney and close by is the open expanded funnel, infundibulum, of the oviduct. The oviduct extends posteriorly to the cloaca. In birds that are not laying the ovary is small containing minute eggs. The duct is also small. During the breeding season, both ovary and oviduct are enlarged.

Each egg receives full quota of yolk before being released from the ovary. A mature egg is released from the ovary into the abdominal cavity where it is grasped by the funnel and enters the oviducts. It moves slowly down the oviduct by contraction of muscles in the wall of the oviduct. Fertilization takes place in the upper oviduct. The egg white, albumen, is added by gland in the mid part of the oviduct. The shell and shell membranes is secreted by glands in the posterior part, uterus, of the duct. After that, the egg is ready for laying.

Fertilization is internal and all birds lay eggs. Bird's eggs must be warmed or incubated for the embryo to grow. Either one or both parents incubate. In some birds like chicken and ducks, the young are well formed, covered with down feathers and able to walk, they are termed precocial. Those of other birds are blind, naked and cannot walk or fly and are termed altricial.

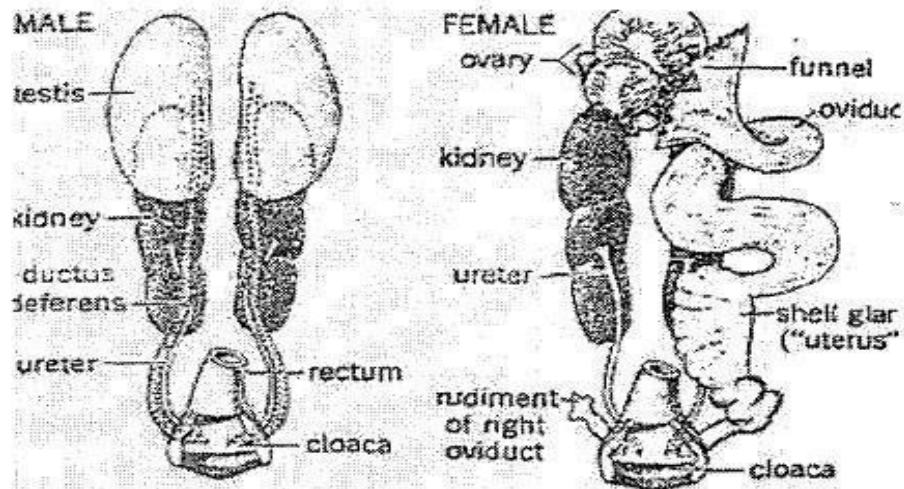


Figure 9.9 Reproductive systems of a bird

Activity 9.2



1. List three types of birds that have altricial and three that have precocial young.

9.4.10 Nervous system

The brain is highly developed. The brains are as large as those of mammals of same body size. The cerebellum is disproportionately large compared to that of mammals. The cerebral cortex is not as highly developed as in mammals. Since they most of their time off the growth the sense of smell is less important than other senses. Thus the olfactory parts of the brain are reduced in most species. Only ground birds such as the ostrich and carrion feeders have well developed sense of smell.

Sight is very important to flying and so eyes are very large making up to 15% of the weight of the head. Muscles that move the eyes are nonexistent so the bird moves the head to change the direction

The yolk sac, allantois and amnion all connect to the centre of the belly region of the embryo and the connecting cord stake is homologous to the umbilical cord of mammals. The yolk sac is reabsorbed during the course of development and the connection with other extra embryonic membranes is broken when the newborn hatches from the egg. The membranes are left behind with the shell.

The cleidoic egg enables the reptiles to live in drier habitats than amphibians and many are found in desert Some have readapted to aquatic life but all lay their eggs on land unless they can broad

them internally, like in sea snake. They are abundant in the tropics a few in temperate regions and absent in arctic regions.



9.5 Other Adaptations of Reptiles

Sensory system

Reptiles lack cerebral cortex that is responsible for the complex behaviour in mammals. Diurnal reptiles have retinas with numerous cones and have good colour vision. Most focus by changing the lens shape as do the mammals. Snakes however, move the lens back and forth. Most have eyelids and nictitating membrane that clear the cornea or foreign material. Snakes have no eyelids. The pineal eye of tuatara does not see but it can detect light intensity

Excretion

Reptiles like other amniotes have metanephric kidneys. Less water is needed to remove nitrogenous waste from the blood than is amphibians because they produce mainly uric acid in forming clystals that do not require water to remove. Some of the water removed by the kidney is reabsorbed by other parts of the kidney tubules, the urinary bladder and cloaca.

Activity 8.1



Revise the structure of the reptilian kidney discussed in lesson 3.

System of vision

Most have a perception of colour. Many large birds have better vision than man. The ability to distinguish objects as they become small and close together, resolution is high in birds than in mammals. A bird can also accommodate or change focus very rapidly since they must change from distant to near visions as they maneuver among the branches of a tree. The position of the eyes correlates with the bird's mode of life. For example ducks have laterally placed eyes and can see behind themselves as well as towards. Eyes of hawks are directed forwards.

Sense of hearing also is highly developed in most species. They have an inner ear with semi circular canals as in mammals. They have a well-developed cochlea. They can detect as wide range of frequencies as humans do and the ability to detect rapid change in frequency in greater. They can hear subsonic vibrations hundreds or thousands of kilometers away and can be used for navigation.

Revision questions



1. Explain why a bird requires much more food than a reptile of the same size.
2. Why are birds less variable in size and morphology than the mammals?

9.6 Summary



In this unit we have learnt that birds share many features with reptiles. Most features that distinguish from reptiles and other vertebrates are all geared to making them efficient flying machines. Many have to do with reduction in weight or efficient production and transport of power for flight.

Further Reading

1. Harris C.L 1996. Concepts in Zoology, 2nd Ed. HarperCollins College Publishers
2. Hildebrand M. 1995. Analysis of vertebrate structure, 4th Ed. John Wiley & Sons, New York.

LESSON 10: CLASS AVES: BIRDS

10.1 Introduction



Welcome to lesson 10. In lesson 9 you learnt how the morphology and anatomy of birds are adapted for flight. In this lesson we will discuss the evolution, diversity, ecology and behaviour of birds, also their importance to man.

Objectives



By the end of this lesson you should be able to

- Explain the origin of birds
- Outline the different methods of classifying birds
- Explain birds adaptations to their environments
- Discuss the reproductive and migratory behaviour in birds.
- Discuss birds interaction with humans

10.2



10.3 Evolution and Adaptive Radiation of Birds

Birds evolved from reptiles. The earliest known species is Archaeopteryx lithographical from Jurassic deposits of Germany. The fossil was uncovered in 1861 in a lithographic limestone quarry. Since the original discovery, four other fossils of Archaeopteryx have been discovered. This species had both reptilian and avian features and was small, the size of a pigeon. It had the following features.

- Jaws with teeth,
- Thick-walled bones

- Long bony tail
- Clawed fingers
- Abdominal ribs
- Trunk and sacral vertebrae not fused
- Large orbits
- Structure of fore limb resembled using of present day birds, 3 finger and elongate hand
- Covered with feathers.

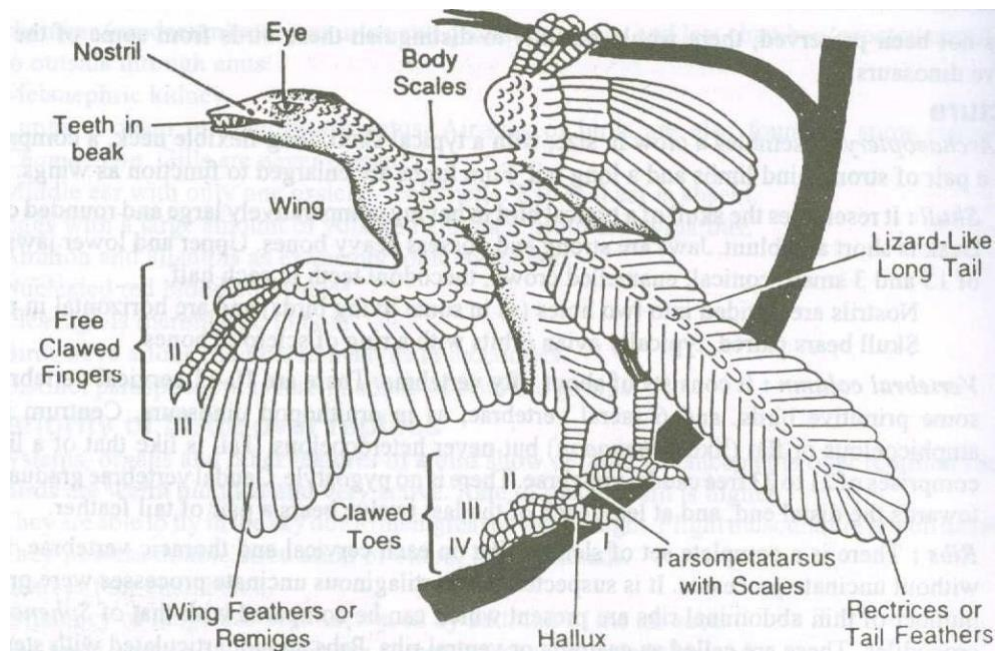


Figure 10.1 Restoration of *Archaeopteryx lithographica*

Probably this creature did not fly because wings were small compared to body size and the sternum to which flight muscles are attached in modern birds was not present. So if it flew it was for short distances. Archaeopteryx probably evolved from a group of early small bipedal (with two legs) dinosaurs. This creature did not fit in any class scheme of classification so it was put in its own subclass, Archaeornithes. All other birds belong to Subclass Neornithes

There is a lot of controversy about the origin of flight. Some paleontologists believe that bird ancestors were arboreal climbers. Later, feathers along the posterior edge of the forelimb elongated as an adaptation for gliding. Others believe that birds' ancestors ran rapidly along the ground and

feathers enlarged enabling them to use the fore limbs catch insects. Further enlargement enabled them to use wing feathers for flight.

The next fossil birds were discovered from Cretaceous deposits. They had no tail but had sternums and teeth. *Hesperornis* was large, diving species, with powerful hind legs and vestigial wings; *Ichthyornis* was a flying species the size of sea gull. Both species are placed in subclass Neornithes, together with modern birds but in their own super order, **Odontognathae**. Other super orders of subclass Neornithes are: Super Order Paleognathae, the flightless or ratites such as the ostrich, cassowary, emu, rhea and kiwi have keel less sternums and Super order Neognathae, all the remaining birds.



10.4 Types of Birds

Since bird's skeleton does not preserve well in fossils, it is not possible to know the history of the evolution of the various orders of living and extinct birds. Traditionally, birds have been classified using morphological characteristics such as the shape of beaks, skeleton, plumage and so on to determine evolutionary relationships. However, morphology is not a very reliable criterion for classification as birds that are not closely related may acquire similar structures as a result of adaptation to the same environmental conditions (convergent evolution). For example ostriches, herons and cranes all have long legs although they are not closely related. Likewise, closely related species may differ as a result of adapting to different conditions (divergent evolution).

Recently, taxonomists have turned to the use of DNA to determine which birds are closely related. DNA is useful to bird taxonomists because closely related birds have more similar DNA than do groups of birds that are distantly related. The new approach suggests relationships that differ somewhat from those suggested by morphology. For example, Vultures of the Americas, that were once thought to be closely related to those of other parts of the world, have been found to be more closely related to storks than to vultures of Europe or Africa.

Despite this discrepancy, both methods support a division of birds into two major groups: **ratites** and **carinates**. Most ratites are large flightless birds of the Southern Hemisphere. They include the ostriches of Africa, rheas of South America, emus and cassowaries of Australia and kiwis of New Zealand. They have sternums shaped like flat-bottomed boats (*ratitis* in Latin), without keels. The carinates have keels for attachment of flight muscles. All carinates are good fliers except for few that have become secondarily flightless like the penguins. There are 27 or 28 orders of extant birds.

Another method of categorizing the birds focuses on adaptive types, or lifestyles. This system groups together birds that live in similar environments or have similar methods for obtaining food.

Even among a given adaptive type, birds show tremendous diversity. Some of the widespread adaptive types of birds are discussed below.

Aquatic Birds

Aquatic birds obtain most or all of their food from the water. For example, the albatross, shearwaters, storm petrels, and diving petrels are aquatic birds. These birds spend much of their time over the open ocean, well away from land. Many other birds have aquatic lifestyles but live closer to land. They include the penguins, grebes, waterfowls, ducks, geese, swans and pelicans found on or near lakes or marshes.

Wading Birds

These spend most of their time on the shores. They include the herons, storks, ibises, spoonbills, and flamingoes. They wade in shallow waters. The length of their legs and beaks and the type of beak depend on how deep they wade and the food type respectively. Those that feed in deeper waters, for example the avocets and stilts, have long legs and long bills. Many long-legged, longbilled birds like storks and flamingoes, are adapted to live at the junction of land and water.

Birds of Prey

Birds of prey have hooked beaks, strong talons or claws on their feet and good eyesight and sense of hearing. They include the eagles, hawks, falcons and owls. They hunt and kill their prey. Scavengers that feed on dead animals are considered birds of prey. They include vulture, condors and marabou storks.

Running Birds

These are birds that spend most of their time running or walking on the ground. They include the flightless ratites and birds that feed on the ground such as bustards, grouses, turkeys and pheasants. Vegetation is an important part of the diets of running birds.

Perching Birds

Perching birds are typically small with their toes arranged such that it enables them to perch. More than half of all living species of birds are perching birds. Perching birds have been successful in all terrestrial habitats. They are divided into two main groups: the sub-oscines, which are mainly tropical and include tyrant flycatchers, and ovenbirds, and the oscines or songbirds, which make up about 80 percent of all perching bird species, among them the familiar sparrows, finches, warblers, crows, blackbirds, thrushes, and swallows.

Other Land Birds

Many other groups of birds thrive in terrestrial habitats. There are forest birds, tree-dwelling species, ground-inhabiting species and so one that do not fall to any of the categories outlined above.



10.5 Ecology of Birds

Ecology is the study of the relationships between living organisms and their environments. Here I will outline some of the components of birds' environment and how they affect the birds. Despite the power of flight birds tend to stick closely to the habitat to which they are adapted. They usually are adapted to the substrate, climate and food type.

Substrate

The physical characteristic of the land that a bird lands on can be deduced from its feet and legs. Birds adapted to firm, flat, open country have long powerful legs with short toes that are often reduced in number. Examples are ostriches, bustards and secretary bird that are common in the open grasslands plains of Kenya. Birds that do much scratching of the ground in search of food have powerful legs and feet with heavy blunt claws. Those that frequent marshes; like cranes, storks and herons; have long legs and necks. Some have webbed feet like the grebes and coots for swimming. "Jacanas have extremely long toes to help them walk on floating vegetation such as water lilies.

Birds of temperate region have special adaptations for snow. Some have feathers covering their feet and toes to provide support and insulation against cold. The feathers on the toes are shed after winter. Others grow fringe-like scales along the side of each toe during winter. They enable the bird to walk on snow. Like the feathers, they are shed after winter.

Birds that spend most of their time in water have webbed feet, powerful leg muscles, large uropygeal glands and dense waterproof plumage. Divers have marrow-filled bones rather than hollow bones.

This enables them to swim underwater efficiently. Penguins swallow stones for the same purpose.

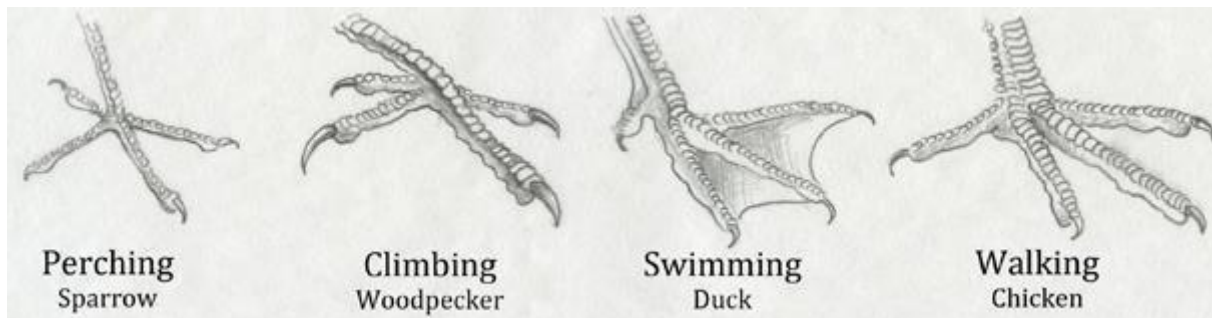


Figure 10.2 Adaptation of bird's feet to the mode of life Climate

The climate determines the distribution of birds. Climate may affect the birds directly or indirectly. For example rainfall affects birds indirectly by affecting the vegetation and food supply. It also affects directly. Prolonged cold rain may damage plumage of a bird. Wind may also worsen the effects of temperature or rainfall and nest sites are chosen with reference to prevailing wind.

Other organisms

Birds affect other organism in their environment. Herbivorous birds affect the distribution of plants that in turn affect the distribution of other animals. They also aid the distribution of plants by acting as transport agent of seeds. They aid in pollination of flowers thus helping the plants to propagate.

Activity 10.1



1. Observe five types of birds in your neighborhood give their local, common and scientific names if possible.
2. For each of the identified birds in No. 1, list their adaptations to the habitat in which they live.



10.6 Birds' Reproductive Behaviour

Each species has a characteristic time when it breeds. Prior to breeding the bird's gonads increase in size. This is controlled by endocrine glands. The breeding season for many species begins by males uttering characteristics song at frequent intervals. Males and females may pair for purposes of breeding. In some species males and females separate soon after copulation and may breed again

with different partners. Most species however are monogamous and remain with each other during the breeding season until the young leave the nest. Monogamy is common among altricial species. Monogamy also occurs in some precocial species. In some monogamous species, like the pied crow, pairing is for life while in others the pair remains together only one breeding season.

Cooperate breeding occurs in some monogamous birds in which only one monogamous pair breeds in a flock but other members of a flock help in raising the young. This occurs in poor habitats where resources can only support limited number of breeding pairs.

Polygamy (a bird of a particular sex mating with more than one mate during one breeding season) is common among birds. When a male breeds with more than one female, the type of polygamy is known as polygyny. Usually such males do contribute in raising the chicks. It is common in precocial species. Polygamy in which the female mates with more than one male is called polyandry. Each male incubates the eggs it has fertilized and raises the young with little help from the females. It occurs among plovers and jacanas.

Many terrestrial birds acquire territories at this time suitable to the needs of breeding. Some may have courting performance (usually males). Birds build nest before laying starts. In some species both sexes participate in nest construction but in others one sex builds the nest. Some water and terrestrial birds do not construct nest but lay on bare rock or on the ground but most species constructs nests. In some the nest is just a few bits of vegetation or just a depression in the soil. The main function of a nest is to hide the eggs from predators and protect them from cold or sun.

Clutch size (number of eggs per breeding period) is smallest in birds nesting in safe places and largest in those that nest in unsafe areas. Some eggs are protectively coloured so that they closely resemble the background and so not easily seen by predators. The ovoid shape of the eggs allows them to be closely packed and so covered by the incubating parent. Incubation period varies, for most small land birds it is 14 days, chicken 21 days, ostrich 42-60 days. The young of the altricial species require about a week after hatching to be able to leave the nest. In most monogamous species both parents incubate and one remains on the eggs in the nest while the other one brings it food.



10. 7 Birds voice

Most birds utter calls and songs. Calls are mono-syllabic and not necessarily species specific. Songs are multi syllabic and species specific. Calls and songs serve the following purposes

- Assemble gregarious species
- Advertise territory

- Attract males
- Directional between parents and young
- Warning of danger.

Calls and songs can be used to identify birds since they are species specific. Birds' song is analyzed by use of sonography that makes a tracing on paper (sonogram) of frequency plotted against time.

Much information in bird song is not evident to human ears but is the means by which birds of a species recognize one another.



10.8 Migration

One of bird's behaviour is their habit of seasonal movement from areas with adverse environmental conditions to those where conditions are favorable. The most familiar migrations are the north south over movements over the landmasses of the northern hemisphere. Though of lesser extent, some tropical birds migrate to breed in the outer tropics. Some migrate from temperate region during winter to the south where temperatures are higher.

The factors that determine the direction and course of migration are not well known. It is known that most species use established routes for migration and travel more or less on schedule, arriving and disappearing regularly. Their ability to follow a given route depends on;

- Ability to navigate by the observation of the sun and stars.
- Individual memory (juvenile return not to exact place but adults do) • Some birds follow obvious land marks such as coasts, rivers, mountain range
- Use the earth's magnetic fields to navigate.

Many factors interact to dispose certain birds to migrate;

- Increasing day length
- Increase in size of gonads
- Increase in fat deposits
- Increase in restlessness

Hereditary factors also may play a role, for example, inner rhythms that approximate a year's length. Once birds are ready for migration, favorable weather and other external factors trigger the

onset of migration. Most songbirds migrate at night stopping to feed and rest during the day. Large birds such as hawks and herons migrate during the day. Ducks and geese may migrate during day or night. Some species follow the advance of certain temperature lines so their arrival at a certain destination depends on the weather. Others arrive at a destination at the same time year after year. The distance traveled and the route taken is consistent for each species but varies from species to species.



10.9 Birds and Humans

Birds have been of ecological and economic importance to humans for thousands of years. Archaeological sites reveal that prehistoric people used many kinds of birds for food, ornamentation, and other cultural purposes. The earliest domesticated bird was probably the domestic fowl or chicken, derived from jungle fowls of Southeast Asia. Domesticated chickens existed even before 3000 BC. Other long-domesticated birds are ducks, geese, turkeys, guinea-fowl, and pigeons.

Today birds are a cheap and easy source of protein. They are over 60 breeds of domestic fowl raised for poultry and eggs. Also other birds including turkeys, Guinea fowls, pigeons, ostriches, rheas and so on are consumed. Many birds are pets, for example the parrots. Some like ostriches are raised for ornamental plumage and leather. In some communities, birds' feathers are used for religious and magical rites. Also man uses birds for information. Fishermen may use water birds to find fish and farmers may use the movement of birds as indicators of coming rain and so start planting.

Birds are indicators of environmental quality. In the 19th century, coal miners brought caged canaries with them into the mines, knowing that if the birds stopped singing, dangerous mine gases had escaped into the air and poisoned them. Birds provided a comparable warning to humans in the early 1960s, when the numbers of peregrine falcons in the United Kingdom and raptors in the United States suddenly declined. This decline proved to be caused by organochlorine pesticides, such as DDT, which were accumulating in the birds and causing them to produce eggs with overly fragile shells. This decline in the bird populations alerted humans to the possibility that pesticides can harm people as well. Today certain species of birds are considered to be indicators of the environmental health of their habitats. An example of an indicator bird is the northern spotted owl, which can only reproduce within old growth forests in the Pacific Northwest.

Birds' conflict with man

In general, birds pose little direct danger to humans. A few birds, such as the cassowaries of New Guinea and northeastern Australia, are capable of killing humans with their strong legs and

bladelike claws, but actual attacks are extremely rare. Many birds become quite aggressive when defending a nest site; humans are routinely attacked, and occasionally killed, by hawks engaging in such defense.

Birds pose a greater threat to human health as carriers of diseases. Diseases carried by birds that can affect humans include influenza and Chlamydiosis. Chlamydiosis arise by breathing contaminated dust from droppings of infected birds. Salmonella food poisoning in man is caused by eating contaminated poultry or eggs. Birds commonly harbor blood sucking ticks and other arthropods that are vectors of viral diseases in man.

Negative impacts by birds on humans are primarily economic. The most spectacular economic effect is the damage caused by *Quelea*, *Quelea quelea*, in Africa including Kenya. This bird can destroy thousands of acres of grain such as rice and wheat. In 1890, it caused famine in Tanzania and in 1952 it caused a loss of 100,000 bags of wheat in Kenya. The familiar mouse birds can destroy seedlings such as beans resulting in great yield losses. Hawks and eagles often kill chicken, lambs and other small animals.

Another hazard caused by birds is at airports. If a high-speed jet engine strikes a bird an accident may result. There has been plane disasters caused by birds. At Jomo Kenyatta Airport, there have been reports of such birds being sucked into the planes as they take off.

Man's destruction of birds

Although birds, with some exceptions, are tremendously beneficial to humans, humans have a long history of causing harm to birds. Studies of bone deposits on some Pacific islands, including New Zealand, suggest that early humans hunted many hundreds of bird species to extinction. Island birds have always been particularly susceptible to predation by humans. Because these birds have largely evolved in the absence of land-based predators, they are tame and in many cases flightless. They are therefore easy prey for humans and the animals that accompany them, such as rats. The dodo, a flightless pigeon-like bird on the island of Mauritius in the Indian Ocean, was hunted to extinction by humans in the 1600s.

Today, the major threats to birds have been the destruction and modification of their habitats. Conversion of natural habitats into farmland replaces natural vegetation with monocrops. This results in reduction in the diversity of food and habitat for birds. In Kenya clearance of forests threatens the existence of birds such as the Sokoke owl that are endemic to the local forests. The fragmentation of habitats into small parcels is also harmful to birds, because it increases their vulnerability to predators and parasites.

Pesticides, pollution, and other poisons also threaten today's birds. These substances may kill birds outright, limit their ability to reproduce, or diminish their food supplies. Oil spills have killed thousands of aquatic birds, because birds with oil-drenched feathers cannot fly, float, or stay warm. Acid rain, caused by chemical reactions between airborne pollutants and water present in the atmosphere, has decreased the food supply of many birds that feed on fish or other aquatic life in polluted lakes. Many birds are thought to be harmed by selenium, mercury, lead, and other toxic elements present in agricultural runoff and in drainage from mines and power plants. Global warming, an increase in the earth's temperature due to a buildup of greenhouse gases such as carbon dioxide, is another potential threat to birds.

Many laws have been enacted to protect birds. More than 120 countries signed the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), which went into effect in 1975. This treaty regulates or restricts the trade of endangered birds, including many parrots. Other means of protecting birds include the creation of sanctuaries and captive breeding programs. Sanctuaries for birds exist all over the world. An example is the Sawia National Park in Kitale for cranes and Hell's Gate for eagle. Some endangered birds are bred in settings such as zoos and specialized animal clinics and later released into the wild. Recently two such bred eagles were released into Hell's Gate. Many countries, including Kenya, are finding they can reap economic benefits, including the promotion of tourism, by protecting the habitats of birds and other wildlife. The protection of the earth's birds will require more than a single strategy. Many endangered birds need a combination of legal protections, habitat management, and control of predators and competitors. Ultimately, humans must decide that the bird's world is worth preserving along with our own.

Activity 10.2



1. List three species of birds hunted for food by man in Kenya.
2. List three species of birds that are crop pests and the crops they attack.

Revision questions



1. Explain the following mating systems:
 - Polyandry and
 - polygyny
2. Give two examples of birds that exhibit each of the above mating systems.

10.10 Summary



In this lesson we discussed the criteria used in classifying and grouping birds. We also discussed some of the interesting behaviour of birds and also the economic importance of birds to man.

1. Harris C.L 1996. Concepts in Zoology, 2nd Ed. HarperCollins College Publishers
2. Hildebrand M. 1995. Analysis of vertebrate structure, 4th Ed. John Wiley & Son New York.

LESSON 11: CLASS MAMMALIA

11.1 Introduction



Welcome to lesson 11. In this lesson we will learn about the most advanced vertebrates, the mammals.

11.2 Objectives



By the end of this lesson you should be able to

- Define the term mammals
- List the characteristics of mammals
- Outline the orders of mammals



11.3 Characteristics

External features

Mammals are a very diverse group. Most are terrestrial although some are adapted to aquatic habitats. Few can fly. They range in size from very small (1.5g to about 100 tons). Do you remember the largest and smallest mammal mentioned in lesson 10? Some, for example the duckbilled platypus and spiny anteater, lay eggs. Most mammals produce few young but give a lot of parental care. Most can maintain body temperature to almost a constant level irrespective of environmental temperature. Thus they are found in even areas with extreme temperature like the polar region, deserts and mountains ranges. Endothermy and care of young are the most important features of mammals and most of their characteristics are related to these two attributes. Like the birds, there are two features unique to the mammal, possession of hair and mammary glands (even males). No other vertebrate have these features. The following features distinguish mammals from other vertebrates

- Body usually covered by hair that is molted periodically. Skin with many glands (sebaceous, sweat, scent and mammary).
- Limbs four usually beneath the body. Each foot usually with 5 toes. Foot adapted for walking, running, climbing, burrowing, swimming or flying. Toes with claws, nails or hoofs.
- Skull is of synapsid type and has large braincase. Teeth present on both jaws and different in relation to food habits.
- Three auditory ossicles in the middle ear and spiral cochlea in the inner ear.
- Neck vertebrae usually seven, tail when present usually long and mobile.
- Heart completely divided into four chambered, two atria and two ventricles.
- Respiratory and digestive passages nearly completely separated in the oral and pharyngeal regions. A diaphragm separates lungs and heart from the abdominal cavity and together with the ribcage plays a significant role in lung ventilation.
- Urinary bladder presents. Nitrogenous wastes usually eliminated in fluid as urea.
- Males have copulatory organs and fertilization is internal. Tests of most mammals either lie permanently within the scrotum or descend into the scrotum during the reproductive seasons.
- Except primitive species, those that lay eggs, the ovaries are small and produce few eggs with little yolk.
- Mammals are viviparous except monotremes. Mammary glands are always present in females
- They are homoeothermic.



11.4 Structural Features Body Covering

The skins of mammals produce hair, various horny and cornified structures and contain many glands.

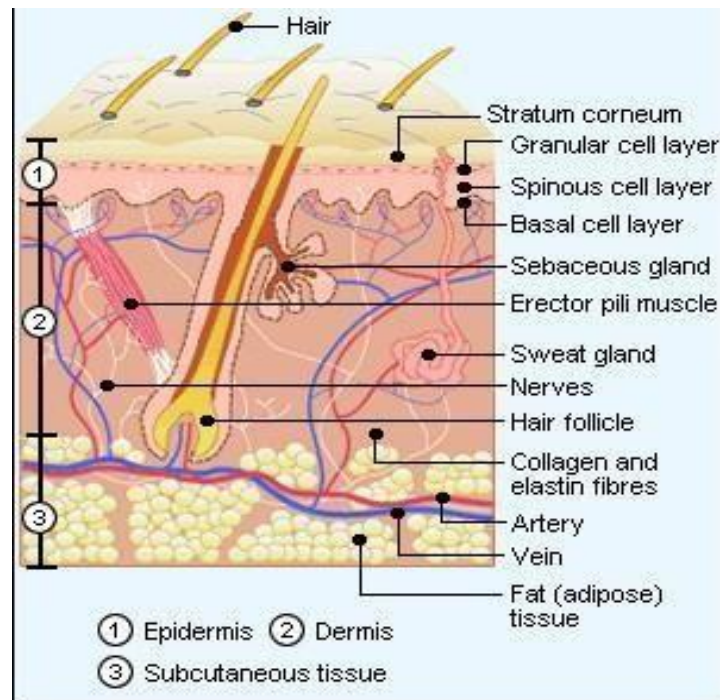


Figure 11.1 shows the structure of a mammalian skin.

Glands

Sweat glands are unique to mammals. Some species have more sweat glands than others. Whales have no use for them and they are lacking. Sebaceous glands are also unique to the mammals. Their oily secretions lubricate the hair and prevent excess drying of the skin. Scent glands serve for defense, recognition or attraction. They may be found in the anal region, face, feet or any other part of the body. Mammary glands too are found only in mammals. They secrete milk to suckle to the young. Each mammary gland sends numerous ducts to the surface that emerge at an elevated point, nipple, which the young can hold.

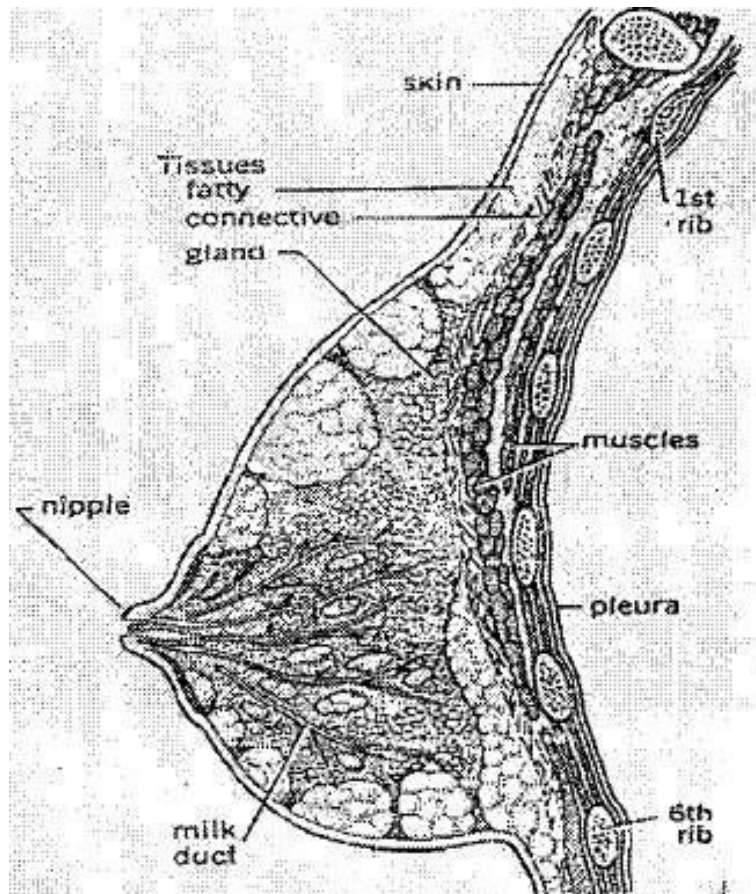


Figure 11.2 Structures of a mammary gland

Hair

A hair covers most mammals. Each hair grows from a papilla at base of a tubular pit, hair follicle, embedded in the skin. It is formed of epidermal cells accumulating at the base and pushing older cells out from the follicle. Like the feathers of birds and scales of reptiles, hair consists of dead, keratinized epidermal cells. However, hair is not homologous with scales or feathers, but appears to have evolved independently. Each follicle is joined to a small sebaceous gland that provides an oily secretion, sebum, to lubricate the hair. Hairs drop out individually and new ones grow the same follicles.

An individual hair strand is covered with cuticular scales over a tubular cortex that surrounds an inner medulla. The coloration of hair and so of the pelage depends on the type of pigment granule presents in the cortex, black, brown red or white when there is no pigment. The size, shape and overlapping pattern of scales on the hair vary from species to species. So they can be used in identification of species.

Like the feathers of birds there are specialized hairs. Guard hairs are the relatively long and straight.

They give the pelt its apparent coloration and texture. They are usually grouped in twos and threes. Whiskers are extra long and coarse. They are tactile organs, organs of touch. Quills are the heaviest hairs. They are hollow but can be very stiff. They are found in porcupines and hedgehogs. Those of a porcupine have the cuticle at the tip modified into barbs. Figure 10.2 shows the structure of hair.

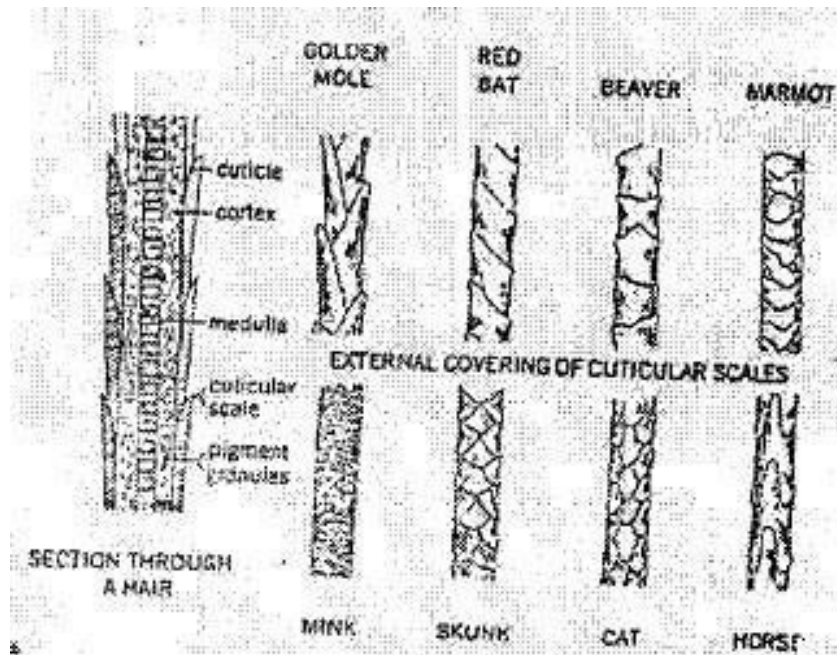


Figure 11.3 Mammalian hair; a section through the hair and external coverings of hair for different animals

Hard structures on the skin

The skin also produces a variety of hard structures used in defense, predation and/or social aggression. Claws are keratinized sheaths that cover the pointed end of toes. In ungulates the claws are modified to form hoofs. Ungulates also have structures on the head: horns or antlers. The two structures are usually confused. They are different in

- Antlers consist of dermal bone that is initially covered by skin, velvet. A horn has a core of bone projecting into a hollow sheath of keratin.
- Antlers are shed and re-grow annually while horns are permanent.
- Antlers occur usually in mature males but horns generally occur in both sexes. Antelope, cattle, buffalo, goat and sheep have horns while deer and caribou have antlers. Rhinoceros

horn is different from that of animals mentioned earlier in that it consists of fibers of keratin. Those of the giraffe consist of bone outgrowth.

Skeleton

Vertebral column forms an arched back. The spines of thoracic vertebrae are inclined backwards while those of the lumbar vertebrae are inclined forwards. Elbow and knee are closer to the trunk with the elbow pointing posteriorly and the knee pointing anteriorly so that the legs can extend down to the ground. Most mammals have a sacrum formed by the fusion of three sacral vertebrae that forms a strong articulation between the pelvic and vertebral column. Tail, when present, is used for balancing. However, the tail may be used for other purposes, for example it is used for locomotion in whales.

Teeth

Mammalian teeth are fixed in sockets in the jaws and are definite in number, 32 for man. They are differentiated in form and function according to the kind of food a species uses. Different types of teeth perform different functions during food processing. Such teeth are termed, heterodont dentition. Most other vertebrates have homodont dentition in which all of the teeth are similar usually simple cones of different sizes used primarily in seizing food items. The different kinds of mammalian teeth are

- Chisel-shaped incisors at front of each jaw used for nipping and cropping cutting.
- Single canine tooth usually long for tearing
- Premolars and molars, cheek teeth, for grinding. In primitive mammals each premolar tooth bears 3 small cones called cusps arranged in a triangle. The triangle of cusps is called trigon in an upper molar and a trigonid in a lower molar. These are mirror images of each other. Crushing of food occurs when the primary cusp of the trigon falls upon the lower heel or talonid located on the posterior surface of the trigonid. Shearing-action occurs when parts of the trigon and trigonid slide past each other. In order to crush or shear, upper and lower teeth must come together or occlude in a precise manner.

Mammals are born without teeth so they are suckled until teeth appear. A first set incisors, canines and premolars develop and the young begins to feed for themselves. These teeth, milky teeth, are later replaced with permanent ones. Molar teeth appear as a mammal matures and jaws enlarge. They are not replaced.

Some mammals have teeth that are adapted to perform functions other than biting or chewing. In elephants, the incisors are enlarged to form the tusks.

Draw and label the different types of mammalian teeth.

Respiration

Like the birds mammals are very active and require an efficient respiratory system. The evolution of pulmonary alveoli increases the respiratory surface of the lungs enabling greater exchange of gases, oxygen and carbon dioxide. Evolution of diaphragm increases the efficiency of ventilation.

Palate separates the mouth cavity from the air passage so mammals can breath continuously even when they are still manipulating food in he mouth (food or air cross only in the laryngeal part of the pharynx) In other vertebrates, the nasal cavity opens into the mouth cavity but in mammals they does not.

Reproduction and care of young

Mammalian embryos develop in warm environment. In viviparous mammals they develop in the uterus where temperatures are controlled. In oviparous species such as platypus and anteater they brood their eggs and the newborn is feed on milk form mother's mammary glands. In the viviparous species placenta grows between the mother and embryo and it is though it that food and gases are exchanged between mother and the foetus. In marsupials, young are born when very small and have not completed development. They remain attached to the mother's nipples in a pouch that also keeps them warm.

In males the two testes lie in a scrotum, a skin covered double extension of the abdominal cavity below the anus. From each testis spermatozoa are collected in a network of minute tubules in the epididymis to enter the sperm duct or ductus deference. The two ducti defentia enter the base of the urethra, common, urinogenital canals though the male copulatory organs, penis, that serves to transfer sperms into the vagina of a female during copulation. '

In females there have two small ovaries that lie behind the kidney. Ova ire released from the ovary to the funnel or ostium joined to the fallopian tube, Oviduct. The tube joins the uterus. Fertilization occurs in the oviduct. The fertilized eggs rest against the wall of the uterus and become implanted. The faetal membranes of the developing embryo from the placenta. Females experience estrus cycle, marked by cellular changes in the uterus and vigina and behaviour.

Period of gestation vary with the size of species e.g. domestic rabbit 30-32 days guinea pig 68 etc. The number of young produced at one time (litter) is inversely proportional to size- elephant, goat, humans have litter size of one 1 or 2 lions 3, and rodents as many as 8. Some have antiracial young

(helpless blind) e.g. rabbit, mice and others have precocial young for example goat, donkey, zebra (able to move about soon after birth). Age of maturity varies. Mice may breed few weeks after birth, while larger species take years to reach maturity.

In most species males are polygamous, mating with any receptive female. This type of polygamy as we saw in birds is called polygyny. A few species such as wolves are monogamous with male mating with only one female at a time and may be involved in parental care. Polyandry, female mating with many males at the same time, is rare among mammals. It occurs among naked mole rats, *Heterocephalus glaber*.



11.5 Adaptations of Mammals

Mammals have adapted to a wide range of environmental conditions. One of the most important adaptations is that they are endothermic, meaning that they maintain their body temperature within a narrow range despite changes in the environment. Polar bears survive on Arctic ice, while Arctic foxes can sleep on open snow in temperatures as low as -68°C . Camels and kangaroo rats live in deserts, and can tolerate blazing temperatures that would kill many animals from cooler habitats.

Endothermy

It is probable that ancestral mammals had lower metabolic rate and simpler set of thermoregulatory controls than present day ones. Probably their temperatures were not very different from the environmental temperature. Probably they were active at night and moved to cool areas during the day. Probably they were insectivorous. Mammals with such characteristics are the hedgehogs. They are nocturnal, insectivorous and probably have occupied this niche throughout their evolutionary life. Their temperature is only slightly higher than that of a surrounding.

When reptiles declined in Cretaceous period, mammals evolved the ability to be active at day times when ambient temperatures would be higher. To maintain a body temperature as low as that of their nocturnal ancestors, $25-30^{\circ}\text{C}$, would require evaporative cooling and thus would have meant great loss of water. Thus, they evolved higher body temperatures, $35-40^{\circ}\text{C}$. For small mammals to maintain this temperature, they would require more energy than a similar sized reptile would. Thus their metabolic rate increased to three times that of reptiles of similar size under similar conditions. Mammals maintain constant body temperature by regulating heat loss at body surface and rate of heat production. Each species has a characteristic thermal neutral zone, a range of ambient temperature at which body temperature can be maintained with little change in metabolic rate.

Adaptation for cold stress

- Mammals in cold areas have thick coat of hair, the pelage (all hairs on animal are collectively called pelage). In winter they lose their summer pelage and develop a much thicker undercoat for winter. In addition to undercoat, the pelage includes coarse guard hairs that protect the undercoat.
- Some mammals migrate to warmer areas.
- Some hibernate during winter i.e. maintain temperature at a lower thermostat setting. The animal uses food reserves stored and metabolism is low and tempers maintained lower.

Adaptation for heat stress

Mammals that live in hot climates have adaptations that help keep them cool. Some of ways they manage to this are

- Evaporative cooling but without excess loss water.
- Small desert mammals are active at night
- Hair in some animals reduce heat gain
- Some can tolerate higher temperatures as long as they keep the brain temperature lower.



11.6 Origin of Mammals

Mammals evolved from a group of early reptiles known as synapsids. They originated before birds during the Triassic period. Earliest mammals like reptiles are placed in order Pelycosauria and were medium size. Their limbs were attached at right angles to the body, not beneath like present day mammals. Some had fins on their backs that helped maintain a higher temperature by absorbing heat.

Pelycosaurs were replaced by more advanced synapsids of order Therapsida in late Permian. Their limbs were beneath the body mouth and nasal cavities separated, braincase enlarged, teeth differentiated for various functions with cheek teeth adapted for chewing. Therapsides fall into two groups

- Diacynodonts, mainly herbivores
- theriodonts, mainly carnivores

Most therapsids became extinct as soon as the dinosaurs appeared but their close relatives, the mammals, became more successful. There is no fossil record linking them with the dinosaurs but there is no doubt that they had a common ancestry.

The earliest true mammals, appearing over 200 million years ago, were only 5cm (2 in) long and probably spent _at least some of their lives in trees. These tiny mammals resembled shrews with four roughly equal short legs and sharp claws. Since that time mammals have evolved an extraordinary variety of body shapes and sizes. Of the approximately 4600 species of mammals alive today, most move about on all four legs. But modern mammals also include animals that hop on two legs, ones that live permanently in water, and ones that can fly. These swimming and flying species include the world's largest mammal-the blue whale, which can grow over 30m (100ft) long and also the smallest, the Kitti's hog nosed bat. Discovered in 1973 in the forests of Thailand, this tiny bat is about the size of a bumblebee, and weighs just 2g (0.07 oz).



11.7 Types of Mammals

Among the mammals, there are three major variations in reproductive systems. This is the basis for dividing them into subclasses and infraclasses.

class:	Mammalia		
subclass:	Prototheria	Theria	
infraclass:		Metatheria	Eutheria

Members of the **subclass *Prototheria*** lay eggs like most non-mammalian vertebrates. However, they feed their newborn with mammary gland secretions like all other mammals. They lack nipples, but the skin over their mammary glands exude milk for their babies. The *Prototheria* are also referred to as **monotremes**, which literally means that they have one opening for excretion and reproduction. This is similar to birds and reptiles. The *Prototheria* are also similar to reptiles in some aspects of their skeletons. Notably, their legs are on the sides of their bodies rather than underneath them. This results in a reptile-like gait. There are only three surviving rare species groups of *Prototheria*. These are the Australian platypus and 2 echidna (spiny anteater) species of Australia and New Guinea.



Platypus (subclass *Prototheria*)



Echidna (subclass *Prototheria*)

To find out a little more about the strange lives of monotremes, select the "Echidna Reproduction" button below:

All other living mammalian species, including humans, are in the **subclass *Theria***. They have in common the fact that they give birth to live young. Therian mammals apparently did not evolve from the *Prototheria*. The relatively primitive prototherian reproductive system evidently evolved after their evolutionary line separated from the other early mammals.

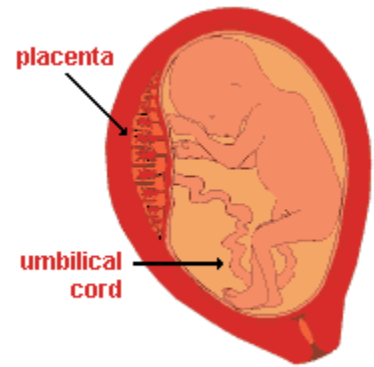
The oldest **infraclass** of therian mammals is the ***Metatheria***, or the **marsupials**. Their young are born very immature and cannot live without further development in the mother's pouch. The word marsupial comes from *marsupium*, the Latin word for purse. Marsupials include kangaroos, koalas, opossums, and many other similar animals. Most of them are native only to Australia and New Guinea.



Kangaroo, koala, and opossum (infraclass *Metatheria*)

Most mammal species, including humans, are in the **infraclass**

Eutheria. They are also referred to as **placental mammals**. Eutherian mothers carry their unborn children within the uterus where they are nourished and protected until an advanced stage is reached. This is made possible by the umbilical cord and placenta which connects the fetus to the uterus wall and enables nutrients and oxygen to get to the offspring as well as provides a means of eliminating its waste. At the same time, the placenta functions as a barrier to keep the blood cells and



other components of the immune systems of the mother and her fetuses separate to prevent their destruction. Human fetus in utero

Giant pandas are an exception among the placental mammals. Their babies are born at only 1/4 the size predicted for the general placental mammal pattern. Marsupial babies are born at an even more immature stage because their rudimentary placentas are comparatively inefficient in nurturing fetuses.

Placental mammals have been extremely successful in out-competing monotremes and marsupials for [ecological niches](#). This is mostly due to the fact that their babies are born more mature, which increases their chances of survival. This is particularly true of herbivores that are predated on by carnivores. Marsupials give birth to early stage fetuses. Placental mammals give birth after fetuses are much more developed. The downside is that pregnant placental mammals must consume significantly more calories to nurture their fetuses and themselves, especially during the second half of their pregnancies. Like monotremes and marsupials, placental mammals feed their babies with milk from their mammary glands. Species that have multiple births at the same time generally have more mammary glands. The number ranges from 2 in primates, goats, sheep, and horses to 18 in pigs.

Placental mammals are found on all continents, in the air, and in the seas. Primates, cats, dogs, bears, hoofed animals, rodents, bats, seals, dolphins, and whales are among the dominant placental mammal groups today. Nearly 94% of all mammal species now are placental mammals (5,080 species out of 5,416).



Whale, dolphin, monkey, and zebra (infraclass Eutheria)

The next tutorial in this series, *The Primates*, investigates all of the Linnaean classification categories below the infraclass level for humans, apes, monkeys, and some other closely related animals. This will take us from the "order" level down to "species."

I would like to introduce you to the Salient features and distribution of prototherians, Metatherians and Eutherians and their relationship with each other.

Prototheria (Order-Monotremata)

Distribution, Salient features and Affinities Salient features of Prototheria.

" Prototheria, a subclass of the class mammalia, includes primitive egg-laying mammals. Living prototherians are included in a single order, Monotremata. These are often called as 'living fossils'. They form a connecting link between the mammals and their reptilian ancestors. They are called as unfinished mammals.

Distribution

The monotremes are restricted to the Australian region including Australia, Tasmania, New Guinea and neighbouring islands. The living monotremes include two genera of echidnas (**Tachyglossus**, the echidna; **Zaglossus**, the proechidna) and one genus of platypus, the **Ornithorhynchus**.

Salient features

1.Habits and habitat: All are nocturnal, burrowing and oviparous or egg laying. The echidnas are terrestrial and ant and termite feeders. The platypuses inhabit rivers and icy mountain streams and feed on molluscs, crustaceans and worms.

2.External features : Living monotremes are of small size. Body is covered with spines and hairs (echidnas) or hairs only (platypuses). The head is produced into a snout (echidnas) or a beak like that of a duck (platypuses). Tail is long and flat in platypus and vestigial or absent in echidnas. Echidnas contain rudimentary external ears (pinnae) while platypuses are without it.

120 large sweat glands are modified to act as the mammary glands. Nipples are lacking. The young licks milk from pores along the mammary line. The milk contains lactose and casein. Male monotremes carry a poison spur on the ankle of each hind leg that delivers toxic but nonfatal venom from a gland located under the skin of each thigh.

3.Exoskeleton : It includes hairs, spines, beaks and strong claws (adapted for digging the burrow). Hands and feet of platypuses are webbed. Skin is glandular.

4.Body cavity and diaphragm : The body cavity is divided into an anterior thoracic and a posterior abdominal cavity by a typical mammalian diaphragm.

5.Endoskeleton :

(a) Skull: Skull is dicondylic . Many bones are fused. Orbits are continuous with temporal fossae. Small prefrontal, postfrontal and separate pterygoids are present (reptilian features). The lateral wall of the cranium is formed by an anterior extension of the periotic, and not by the alisphenoid, as it is in therian mammals.

The lower jaw consists of a single dentary bone. The quadrate, articular and tympanic form the part of ear. The tympanic bone forms a partial ring around the tympanum. The ear ossicles include a large malleus, small incus and elongated stapes (like the columella of reptiles). The whole ear apparatus is not enclosed in a bulla (bulla is present in modern mammals).

The adult monotremes are without teeth (young platypus contains ridged teeth which are replaced by horny structures in the adult.)

(b) Vertebral column: Vertebrae are reptile-like. The cervicals bear separate ribs (as in reptiles). There is an atlas-axis complex and seven cervical vertebrae. Vertebral epiphyses are indistinct or absent. Cervicals are without zygapophyses.

Ribs are unicephalous, contain only the capitulum. The ribs articulate only with the bodies of the vertebrae, not with the transverse processes.

(c) Girdles and limbs : Limbs and girdles are remarkably reptilian. The pectoral girdle has separate clavicles and a median interclavicles. The coracoid region includes large and separate coracoid and precoracoid. There is no spine on the scapula.

The ventral portion of pelvic girdle contains a large epipubic bone.

The humerus is large and broad and held horizontally. The fibula is expanded at its upper end in the platypus (like the ulna of other mammals).

6. **Digestive system:** Adults are toothless. Rectum opens into a cloaca. The lining of the stomach is horny and without glands in echidnas.

7. **Respiratory system :** They respire through lungs.

8. **Larynx:** The larynx is mammalian. Platypuses make growling sounds and echidnas coo like a ringdove bird.

9. **Circulatory system;** The heart and single left aortic arch resemble those of modern mammals. There are no renal portal veins. They are imperfectly warm-blooded. Body temperature varies between 24-28C°. RBCs are enucleated.

10. **Nervous system :** The brain is relatively large and built on the mammalian plan but there is no corpus callosum. The roof of the cerebral hemispheres is large, and smooth in platypus, and convoluted in echidnas. The distribution of sensory and motor areas is different from other mammals.

The cochlea, in the internal ear, is only partially coiled but contains organ of Corti. It carries a lagenar macula at its tip, as in reptiles.

11. Urinogenital system : Kidneys are metanephric. The rectum, ureters and genital ducts (vas deferens or sperm ducts and vaginas) open into a common cloaca (reptilian feature) which opens to outside through a single aperture. The order is named monotremata (single aperture) on the basis of presence of a single aperture at the posterior end of the body.

In males, the testes are abdominal (no scrotum). The penis is situated at the base of the cloaca. It is erectile and carries a groove throughout its length. It carries only the sperms, not urine. In females, right ovary is reduced and there are two vaginas.

12. Development: The females are oviparous. Fertilization is internal. The eggs are covered with soft shells and are pea-sized (in echidnas) or 1.3 x 2 . 0 cm (in platypuses). Eggs contain yolk. Cleavage is meroblastic. There is no uterine gestation. Newly hatched young are very immature and are fed with milk. The embryo carries a tubercle or the caruncle on the head and an egg-tooth to break open the egg shell.

13. Other features : In the adrenal glands of echidnas, cortical and medullary tissues are separate and not arranged one within the other.

The chromosomes include microsome, as in sauropsida (reptiles and birds). These are not found in other mammals. The male is heterogametic but there is no Y-chromosome (present in all other mammals). ,

Affinities

Prototherians are regarded as primitive mammals since they contain a number of ancestral (or reptilian) features. Monotremes are fully mammalian in some respects, but have retained many reptilian characters also. They are called as unfinished mammals.

A. Reptilian affinities

The reptilian characters of the monotremes are as follows :

1. The skull shows many reptilian or transitional features. Lacrymal is absent, the jugal is reduced or absent. Traces of a reptilian prevomer are present. There is no tympanic bulla. Pterygoid is separate. Stapes is elongated like the columella of reptiles.
2. The vertebrae are without epiphyses.
3. Cervical vertebrae bear separate ribs.
4. Ribs are unicephalous or single headed, the tuberculum is absent.
5. Pectoral girdle carries separate and large coracoid and precoracoid bones. Scapula is without spine. Clavicles are separate and the interclavicle is median.
6. Pelvic girdle possesses a prepubic bone.
7. Alimentary canal contains a cloaca. There is a single posterior aperture in the body.
8. Ureters and the male and female genital ducts open into the cloaca, which opens to outside through an aperture.
9. Corpus callosum is absent.
10. The cochlea of the internal ear contains a lagenar macula at the tip.
11. Testes are abdominal.
12. The females are oviparous. They lay yolky eggs covered with a soft shell. Cleavage is meroblastic. _
13. The chromosomes include microsomes.

B Mammalian affinities

Prototherians show the following mammalian features :

1. Body is covered with hairs.
2. External ears or pinnae present.
3. Skin is glandular and consists of sweat and sebaceous glands,
4. Mammary glands are present. They secrete milk.
5. Skull is dicondylic.

6. Middle ear is formed by three ear ossicles (malleus, incus and stapes).
7. Lower jaw is formed by the dentary alone.
8. There are seven cervical vertebrae.
9. Heart is four chambered and only the left aortic arch persists.
10. Red blood cells are non-nucleated.
11. Larynx is typically mammalian.
12. Diaphragm is typically mammalian.
13. A small intestinal caecum present.
14. Brain is of mammalian type. Cerebral hemispheres are large and convoluted.
15. They are warm-blooded. They regulate their body temperature (Griffiths 1978).

Conclusion

The prototherians show a peculiar mixture of mammalian and reptilian characteristics. It seems that they separated from the main mammalian stock at an early stage. They have not changed much since the Jurassic. They are not ancestral to the modern mammals. The present status of the living prototherians is that they are extremely primitive in many respects and highly specialized in relation to their habits and habitat. The ant-eaters and aquatic platypuses specialized along totally different lines for two different modes of life. **Systemic position**

Metatherians are supposed to be 'imperfect mammals'. They show certain resemblances with the egg-laying, reptile-like mammals or the prototherians but they are definitely more advanced and very similar to Eutherians. They undoubtedly diverged from some early stage of evolution of the main mammalian stocks (Tyndale-Biacoe 1973). Many of their features are specialized and they do not represent a stage on the way to placental evolution. They represent a specialized side-branch of Eutherian evolution. They are classified in a separate infraclass Metatheria which is classified in the subclass Theria along with the infraclass Euthera.

Distribution

Discontinuous distribution: During Cretaceous period the metatherians inhabited much of the earth. Today they are restricted to the Australian region and Central and South America only.

The first marsupial (opossum like) fossils of the Upper Cretaceous and Eocene (about 100 million years old) periods are found in Europe and North and South America. These continents were probably connected together at that time by the time placental (Eutherian mammals) appeared. Australia and South America were separated from the other continent and metatherians became extinct in Europe and North America which when still connected South America remained isolated until the formation of the Panama bridge in the Pliocene period and most of its marsupial fauna became extinct. After this time. In North America also there were no marsupials between middle Miocene and the Pleistocene, when opossums (*Didelphis*) appeared, probably from South America. Australian marsupials only became threatened by the advent of man from Europe and the placental he brought with him. However, the large kangaroos flourished.

2. Isolation and adaptive radiation in Australia: The opossums of North and South America show that the marsupial life can continue effectively in competition with placentals (eutherian mammals). However, marsupials evolved and diversified in Australia in isolation. There are only 108 species of placental mammals in Australia. Out of these, there are 50 species of bats and 48 species of murid rodents there are no other placentals. Against this, there are 124 species of marsupials these have diversified into numerous types, such as arboreal, fruit-eating, grazing, gnawing, digging, burrowing, ant-eating, insectivorous and carnivorous. It will be interesting to see how these marsupials stand up to competition with placentals in future. Carnivores, ruminants, lagomorphs rodents and primates have recently become firmly established in Australia and it is not surprising that some of the corresponding marsupial types are already becoming rare or extinct.

Salient features of Metatheria.

1. Habits and habitat: Marsupials are terrestrial, arboreal, burrowing and rarely aquatic. They are nocturnal, diurnal, air-breathing, herbivorous, carnivorous and omnivorous. They are warm blooded, viviparous and pouched mammals.
2. External features : The body is covered with soft hairs or fur. External ears or pinnae are well developed. Tail is long and often prehensile. They contain mammary glands with teats or nipples. Mammary glands are formed from the sebaceous glands. Female marsupials usually have a pouch or marsupium on the ventral side of the abdomen. It encloses the nipples and opens either forward (in kangaroos) or backward (in opossums and Tasmanian wolf). Some species contain only a pair of skin folds (no pouch) around the nipples, and some species lack the folds and pouch both.
3. Exoskeleton : They contain hairs and claws. Skin is richly glandular.
4. Endoskeleton:
 - (a) Skull; The cranium is small and the top of the skull is flat. The orbit and temporal fossa remain fully confluent and there is no post-orbital bar. The bony palate is incomplete posteriorly. There are large holes in its palatine portion. The jugal reaches up to the glenoid articulation of the jaw. The lower jaw consists of a single dentary which has a typical inturned inner angle for the attachment of large pterygoid muscles.

Other special features of the skull, not found in placental mammals (eutherians), are :

- i. Apertures for the passage of ophthalmic branch of trigeminal nerve, optic nerve and nerves eye muscles are not separated from each other.
- ii. Lacrymal bone extends outside the orbit.
- iii. Auditory region has no bulla (found in eutherians).
- (iv) The alisphenoid bone sends a wing-like process over the middle ear.

- b) **Vertebral column:** It is similar to that of placentals. There are no cervical ribs. The thoracic region consists of about 13 rib bearing vertebrae and 7 lumbar
- c) **Girdles :** The pectoral girdle has no interclavicle, but the clavicles are large. The coracoid is reduced (as in placentals), the scapula is large and bears a spine. The pelvic girdle consists of epipubic bones (sometimes wrongly called 'marsupial bones'). There are two large and flat bones that extend forward and outward and support the abdominal muscles and pouch. These are reduced in Tasmanian wolf
- d) **Limbs :** The hands carry five clawed digits, but the number of toes is often less than five and they may bear hoof-like structures .

5) **Dentition :** The dentition is unique in the following respects :

- (i) Marsupials are almost monophyodont, i.e., their teeth come only once. Only the last upper premolar is shed and replaced by a molar.
- (ii) Adult marsupials contain 3 premolars and 4 molars in each half jaw (4 premolars and 1 molar in placentals).
- (iii) Most marsupials have more than three incisors in each half jaw (eutherians never possess more than three).
- (iv) Number of incisors is different in upper and lower jaws (always equal in both jaws in eutherians)

The basic dental formula of marsupials (still found in opossums) is

I 3, C 1, PM 3, M 4
 4 1 3 4

6) Digestive, respiratory, circulatory and excretory systems are typically mammalian (eutherian).

7) **Brain :** The brain shows some reptilian features. The cerebral hemispheres are much smaller and smooth. The olfactory bulbs are large. The hemispheres do not cover the cerebellum there is no corpus callosum. The cerebellum is small and simple.

The cochlea of the internal ear is spirally coiled.

8) **Reproductive system** : The males have a well developed penis, which is bifid at the tip in several species. In such species the clitoris in females is also double, this arrangement ensures fertilization of eggs in both oviducal tubes. The testes are extra-abdominal. They lie in a scrotal sac.

The female reproductive tract is peculiar. The vagina branches into two lateral vaginae which extend anteriorly and meet together in the middle line to form a vaginal sinus. The two uteri open into the sinus separately. The vaginal sinus often extends posteriorly to open into the urinogenital sinus that opens into the cloaca. The cloaca is short in females. The openings of rectum and urinogenital sinus are surrounded by sphincters. In many species the vaginal sinus ends blindly until the young is about to be born. At the time of delivery the vaginal sinus opens to outside through a pseudovagina or birth canal. This usually closes after the delivery but in most macropids, forms a permanent median vagina. The sperms enter through the lateral vaginae (Note: The female contains two vaginae and two uteri)

9) **Reproduction and development**: The marsupials show a viviparous condition that is not closely similar to those found in eutherians. The egg is yolk-y and covered with albumen and a shell membrane. Cleavage is unequal. Marsupials have either a rudimentary placenta or no placenta at all. The embryo depends upon the yolk for its development but gets oxygen and some nourishment by way of close contact between its chorion and the uterine wall. Birth occurs after a short uterine gestation period (opossum, 13 days, red kangaroo 33 days; the marsupial mouse, 30- 33 days) the young are born in a highly immature stage but contain clawed fore limbs. The baby clutches the fur of mother through the claws and reaches the pouch of the mother and takes a nipple of the mammary gland in its mouth. The nipple swells within the mouth of the baby and the baby cannot leave it now. The baby completes its development lying attached to the nipple. The sides of the lips of the baby grow together around the nipple which extends into his pharynx. Milk is pressed out by a special muscle attached to the epipubic bone.

The mother remains unaware of the birth of a baby and does not recognize the young one. After birth the baby enjoys long protection in the pouch but the mother does not look after it. The baby

stays in the pouch for 8-9 months. The fully developed baby leaves the pouch but may return into it for safety.

Affinities

The metatherians resemble the protodienans and eutherian mammals (placentals) they possess a mixture of primitive and advanced characters and have developed many adaptations also.

A. Affinities with prototherians Similarities with prototherians

Metatherians resemble the primitive prototherians in having the following characteristic features

1. Presence of a cloaca, in which rectum and urinogenital ducts open
2. Presence of epipubic (or marsupial bones attached to the pubis, and ring-like tympanic
3. Absence of a tympanic bulla
4. Near absence of corpus callosum in the brain
5. Absence of true allantoic placenta

Differences from prototherians

1. Metatherians are viviparous
2. Their mammary glands are formed by sebaceous glands
3. They possess nipples or teats in the mammary glands.
4. Ribs are two-headed or bicephalous.
5. Vertebrae have epiphyses
6. There are no interclavicles, no cervical ribs and coracoids are not separate.
7. They contain teeth in the adult
8. Testes are extra-abdominal

Eutheria (Placental Mammals) Distribution, Salient features and Affinities

Eutheria is the infra class of class mammalia. They are commonly called placental mammals (but this name is not accurate because some marsupials also have placenta). Placenta is an organ of exchange between maternal blood and foetal blood, Eutheria is by far the largest of the three mammalian groups and includes all the common mammals except the Antarctica

Distribution

Placental mammals populate all continents except Antarctica. Mostly they live on land, but some (eg., Whales, dolphins, seals, sea lions etc.), are secondarily adapted to live in water, and bats are able to fly.

Salient features

Eutherian mammals are- or were, until they were greatly reduced in numbers by humans - the most important grazers and browsers in most terrestrial ecosystems. Grazing and browsing have been evolutionary features, for instance enough to select for the spines, tough leaves, and difficult to eat growth forms found in many plants - a striking example of coevolution.

1. Eutherians are more developed at birth than marsupials, and no external pouch houses them after birth.
2. Nutrients are supplied to growing offspring and wastes are passed to the mother for excretion through placenta.
3. Eutherians lead an active life. Their senses are acute and the brain is enlarged due to expansion of the foremost part- the cerebral hemispheres. These have become convoluted and have expanded to such a degree that they hide many parts of the brain from view. The brain is not fully developed for sometime after birth, and there is a long period of dependency on the parents, during which the young learn to take care of themselves.
4. Eutherians have separate anal and urogenital openings.
5. They are viviparous. Eggs are microscopic with practically no yolk, and no shell.
6. There are well developed milk glands with teats
7. All embryonic membranes are found in eutherians
8. They are true placental mammals with well developed placenta.
9. Period of gestation is long. It is shortest in mouse - 29 days and longest in elephant - 22 months.

Affinities

Eutherians are advanced mammals. They show affinities to metatherians and have developed many adaptations to live in water and fly in air besides living mostly on land

A. Affinities with metatherians

1. External ears or pinnae are well developed
2. Mammary glands are sebaceous with teats
3. They possess epiphyses between the vertebrae.
4. Cervical vertebrae are without ribs. Ribs are bicephalous.
5. Cochlea is spirally coiled
6. Heterodont type dentition.
7. Testes are extra-abdominal and lie in scrotum
8. They are viviparous. Ova small with little or no yolk.

Differences from metatherians

1. Eutherians give birth to fully developed young ones
2. Eggs have no egg shell and the cleavage is equal
3. A true allantoic placenta develops uterine gestation is considerably long
4. There is no marsupium.
5. Palate has no vacuities.
6. The orbit is separate from temporal fossa
7. They possess a tympanic bulla
8. There are no epipubic bones
9. They have no cloaca. Anus, urinary bladder and genital apertures are separate.
10. The corpus callosum is well developed
11. Cerebral hemispheres are large and highly convoluted
12. Penis is not bifid and carries both urine and sperms

13. The paired uterus unit and open into A single vagina, which opens outside 14. Ureteres open into urinary bladder

B. Affinities with Prototherians Eutherian mammals resemble only in two aspects

1. There IS no marsupium.
2. Palate has no vacuities.

Differences with Prototherians

Since prototherians are primitive reptile - like oviparous mammals, they differ with eutherians which are advanced viviparous mammals, in all aspects.

Activity 11.2



Outline the ecological and economic importance of mammals found in your area.

Revision questions



1. Explain the main differences between birds and mammals.
2. Explain the differences between placental and marsupials mammals.
3. List the characteristics that distinguish mammals from other vertebrates.

11.8 Summary



In this lesson, we have seen that mammals are the most advanced vertebrates. The most important evolutionary feature responsible for their success is their ability to regulate their body temperature and care of the young.

PRACTICALS

PRACTICAL 1

Introduction



Welcome to practical 1. The aim of this practical is for you to be familiar with some of the lower chordates, their external features and some of the internal features: You will be able to see the features that define chordates: notochord, nerve cord and gill slits, in amphioxus.

Objectives



At the end of this practical you should be able to

- Observe the external and features of tunicates and cephalochordates (amphioxus).
- Observe the chordate features in amphioxus.
- Draw and label the structures features observed above.
- Note the similarities and differences between acorn worm and the lower chordates.

Materials

In this practical you will need

- Microscope
- Preserved specimens of tunicates and amphioxus
- Dissected specimens of amphioxus
- Well labeled diagram of acorn worm

Procedure

Hemichordata (acorn worm)

Observe the diagram of acorn worm and note the different body regions.

Observe the head region. What feature do you see on the body surface especially the head?
Examine the point where the proboscis attaches to the collar. Note the mouth and the proboscis pore.

Examine the trunk. Note the arrangement of the pharyngeal pores and count them (if possible).
Observe the rear tip and note the anus.

Copy the well-labeled diagram.

Urochordata (tunicates)

Observe the preserved specimen of tunicate. Note the inhalant and the exhalant siphons.

Examine its bottom part. What are the special features that attach the animal to the substrate?

Observe the diagram of a vertical section of a tunicate Draw and label the different parts. Note the oral tentacles, atrium, pharyngeal slits and intestine.

Cephalochordata (amphioxus)

Examine the whole specimen and make a large labeled diagram of its external features including oral cirri, anus, atriopore and the dorsal, ventral and caudal fins.

Observe the dissected specimen and note the wheel organ, pharynx. Pharyngeal slits and the notochord.

List the similarities and differences between acorn worm and amphioxus.

PRACTICAL 2

Introduction



Welcome to practical2. During this practical you will observe the, external features of agnathans and compare them with those of true other fishes.

Objectives



At the end of this practical you should be able to

- Observe external features of preserved specimen of lamprey.
- Compare the external features with those of amhioxus (practical 1). • Observe transverse sections of various regions of the body
- Observe longitudinal section of the body.
- Observe the external features of lamprey larva, ammocoetes and compare them with those of adult lamprey.

Materials

- Preserved specimens of lamprey larva and adult.
- Dissected specimens of lamprey
- Microscope

Procedure

Lamprey

Observe the specimen and make a sketch showing its external features. Note the following

- Body form that is elongate and bilaterally compressed at the posterior,
- The colour of the skin. Does the skin have scales, as is the case with the more fishes?
- The gill slits. How many?
- The fins. Note that it does not have the pelvic nor pectoral fins.

Examine the anterior end and make a sketch of what you see. Give a brief description of the sucker, mouth and teeth.

Compare the external features of lamprey with those of amphioxus. List the similarities and differences between them.

Ammocoetes (lamprey larva)

Examine the specimen provided. Note the shape of the body and compare it with that of the adult lamprey.

Examine the anterior part of the larva. Note the buccal cirri. Does it have teeth and tongue? Note the presence of gill slits. Observe the dorsal part of the animal. Note the eye, nostril and the auditory organ.

On the rest of the body note the presence of the dorsal, ventral and caudal fins. Note the spinal cord and notochord both of which lie ventral to the dorsal fin.

List the similarities and differences between the ammocoetes and the adult lamprey.

Longitudinal section of lamprey

Observe the dissected specimen with the help of hand lens. Note the following

- The notochord, a gelatinous rod
- Spinal cord above the notochord
- The gill slits opening to the out side individually
- The s-shaped heart

The transverse section of lamprey

Examine the transverse section of the brachial region using a hand lens. Note the following features

- Spinal cord
- Notochord
- Gill pouches
- The dorsal aorta just below the notochord
- The esophagus just below the dorsal aorta
- The ventral aorta below the esophagus

Examine the transverse section of the trunk region of a lamprey and note.

- Notochord
- Spinal cord
- Dorsal aorta
- Posterior and cardinal veins
- The kidney
- Testis/ovary
- Intestine, a small tube ventral to the gonads

Hagfish

Hag fishes are not very common and it may not be provided. But, if is provided observe the specimen and draw and label the external features Note the tooth-like plates on either side of the protrusible tongue. Note the median nostril, which connects by a duct to the pharynx (allows water to enter gill pouches when food is in the mouth). Note that gill pouches do not open individually to the outside but lead to one or more ducts that lead to the surface

PRACTICAL3

Introduction



This practical is meant to give you the skill of dissecting vertebrates. It should also enhance.

Objective



At the end of this practical you should be able to

- Dissect a preserved specimen of dogfish
- Observe and draw the external and internal features of a dogfish.
- Observe and draw the structure of a dogfish scale.

Materials

You will need the following materials for this practical: dogfish, microscope, dissecting tray and dissecting kit.

Procedure

The preserved dogfish will be presented on a dissecting tray. Observe the specimen and note the following

- General body shape
- Body colour
- Shape of the tail and tail fin
- Position and number of fins
- Presence of scales
- Number of gill slits
- Position of eyes, nostrils
- Shape of mouth, teeth
- Claspers in males
- Two openings to the cloaca of females.

Make a large well-labeled diagram showing the features observed above.

Cut a small skin and boil in dilute potassium hydroxide for two to three minutes. Then observe it under a microscope. You should be able to see the tooth-like placoid scales. Draw and label the structure of the scale.

The scale resembles the vertebrate tooth in structure,

Follow the steps outlined below to dissect the dogfish.

- Wash the dogfish with tap water to remove formalin.
- Place it on the dissecting tray with the ventral surface facing up.
- Using the dissecting scissors in your dissecting kit cut the body wall starting from the cloaca to the point the pectoral fins are attached to the body.
- Make transverse incisions at the cloaca and the pectoral girdle. Pin each flap of skin on each side of the fish onto the tray.
- You should be able to see the internal organs but if you can't cut through the pectoral girdle slightly.
- Examine and draw the digestive system and note the various divisions, mouth, pharynx, stomach and intestines. You will compare the digestive system with that of a bony fish after we dissect one in the next practical.
- Note the liver, a large brown structure, and the whitish gall bladder attached to the liver. The spleen is brownish and is attached to the stomach.
- Cut from the pectoral girdle to the tip of the lower jaw and remove the skin till you see gill openings. Cut through the gill pouches of one side.
- Examine the gills and note the gill pouches, gills. Remove the gills and examine them and make well-labeled diagram of them.
- Note the position of the heart.
- Try to identify the circulatory and urinogenital systems.
- Draw the internal features showing the digestive system, circulatory and urinogenital systems

Questions

- List the differences between a dogfish and lamprey's external morphology.
- Give a possible explanation for the similarity between mammalian tooth and, placoid scale.

PRACTICAL 4

Introduction



This practical is similar to practical 3 but we shall be dissecting a bony fish, Tilapia, instead of a cartilaginous fish.

Objectives



- Observe and draw the external features of the fish provided
- Dissect the specimen, observe and draw the internal structures
- Examine and draw the scales
- List the morphological and anatomical differences between this specimen and the one you had for practical 3.

Materials

Same as for practical three

Procedure

Observe the specimen provided and note

- The body shape • Color.
- Note that scales cover the body.

With the aid of a hand lens observe the arrangement of scales on the body and comment on it.

Remove a scale and observe it under the microscope. Note that it is nearly circular, thin and bony.

Note the circular lines on it. Some of the lines are thick and their number depends on the age of the fish. These are known as cycloid scales.

Examine the head and note

- The large eyes without eyelids
- The two nostrils in front of the eyes • The operculum on each side of the head
- Position of the mouth.

On the trunk note the following;

- Two lateral lines on each side of the body, the ventral is shorter than the dorsal one.
- Dorsal, anal and ventral fins (unpaired). Note the number of spines and rays in the fins.
- Note the paired fins, pectoral and pelvic. Note that the pectoral fins are more anterior than in dogfish
- Note the shape of the caudal fin.
- Anal opening and the urogenital papilla posterior to the anus (with one opening in males and two in females).

Examine the head and mouth and note the jaws and teeth on them. Describe the size and arrangement of teeth on the jaws.

Lift the operculum with forceps and observe the gills). How many are they? Note the following on the trunk:

Dissection

Put the fish on a dissecting tray lying on one side. Cut through the body wall starting from the anus up to the upper ventral line.

Proceed cutting forward above the ventral lateral line up to the pectoral arch (i.e. parallel to the posterior edge of the operculum).

Cut through the pectoral arch then down parallel to it up to the mid ventral line.

Turn down the flap of the body wall cut through the body wall along the mid-ventral line until you reach the anus. Remove the flap.

Note the body cavity is divided by the pericardio-peritoneal or transverse septum into anterior pericardial cavity enclosing the heart and a posterior large peritoneal cavity containing the rest of the viscera

Observe the mouth, esophagus, and stomach. Note the shape of the stomach, its cardiac portion extends backwards into a blind sac, caecum, which stores food. Note the large coiled intestines. It starts with the duodenum, which passes through the left lobe of the liver and receives the hepatopancreatic duct from the gallbladder and pancreas. The liver is large and bi-lobed and dark in color

Note the position of the gallbladder, pancreas and spleen.

Make a large labeled diagram to show the features observed above.

The respiratory system

Remove the operculum. Count the number of gills. Note the way the gill filaments hang down the gill arches into the brachial cavity. Remove a gill. Note that two rows of gill filaments are attached to the gill arch. Note the gill rakers on the anterior end of the gills (tooth-like). These protect the gill filaments from damage by hard particles. Draw a gill arch and show the arrangement of gill filaments.

Urinogenital system

Remove the air bladder to expose the kidneys. Examine the urinogenital system, which consists of two elongate kidneys, which are fused their entire length. Note the mesonephric duct from each kidney.

The two ducts unit shortly after leaving the kidneys to form a common mesonephric duct which opens to the outside at the genital papilla. There is a small urinary bladder.

Reproduction system

Examine the gonads. These are either two testis or two ovaries. In both cases, they are elongate bodies connected to the exterior by a thin strand. Ovaries have granulated walls due to the presence of eggs.

Circulatory system

Note the position of the heart. It is ventral to the gills. It has three chambers only: a sinus venosus and an auricle in front it (both are thin-walled.) and thick-walled ventricle below the auricle. Note the main arteries.

Skeleton

Observe the mounted skeleton. Note the many bones of the skull. Note that the trunk vertebrae have very long ribs. The caudal vertebrae has haemal arch.

Questions

1. List the differences between Tilapia and dogfish
2. Describe the lateral line in fish.

PRACTICAL 5

Introduction



Welcome to practical 5. The aim of this practical is for you to be familiar with the external and internal features of the amphibians that adapt them to terrestrial life. Although adult amphibians

are terrestrial their larvae are not. We will see how an aquatic larva is transformed to a terrestrial adult.

This practical is easy to carry out and amphibians are fairly common. You could carry out the practical on your own but it is advisable to organize with another student so that you do it together.

Objectives



- Observe and draw the external features of representatives of the three orders of
- amphibians: Urodela, Anura and Gymnophiona
 - Name the features that adapt each to their mode of life
 - Dissect a frog (procedure can be found most basic biology books including those for high school). Copy the procedure.
 - Observe and draw the internal structures of the frog.

Materials

The following specimens are required: Toad, frog, salamander and caecilian. Also provided are mounted skeleton and developmental stages of a frog. You are expected to examine them and make well-labeled diagrams.

Procedure

Salamander, (Order: Urodela)

These are more similar to ancestral amphibians than are toads and caecilians. They are distributed mainly in the temperate region.

Examine the specimen provided and note the long trunk, tail and 4 limbs that are about equal in size. At first glance a salamander looks like a lizard. However, at a closer look you will notice the following differences between them;

- While the body of a lizard is covered by scales that of a salamander is not.
- Lizard's limbs terminate in horny claws but those of salamanders do not have claws. Draw and label the external features of salamanders

Toads and frogs, (Order: Anura).

These occur in all continents except Antarctica. Note their short trunks without tail, the long strong hind limb (both are adaptations for jumping. How?

Examine the external structures of a frog and a toad. Note the difference in the skin and feet of the two. Explain the reason for the differences noted. List any other differences between toads and frogs. Examine the palms and soles of the frog and note the tubercles.

Dissection of a frog

You can catch a frog or toad from a pond or dam. During the rainy season, you could collect toads by putting a light outside. Toads will be attracted by insects that themselves are attracted by light (especially flying termites).

Kill the toad by snapping its spinal cord at the nap and follow the steps outlined below;

- Lay the toad on its back in the dissecting tray.
- Make a longitudinal cut through the body wall from the rear tip to the mouth.
- Make two transverse cuts. One behind the fore limbs and the other behind the hind limbs.
- Fix the flaps of body wall and pin them onto the tray.
- Cut the muscles without injuring or damaging the organs.
- Trace the digestive system from the mouth, to the esophagus, stomach and intestine

Observe the mounted skeleton of a frog. How many vertebrae are there? Observe the leg bones and note the long femur and tibio-fibula and the extended ilium and urostyle (all adaptation for jumping). The absence of tail allows the hind legs and feet to be extended synchronously during a leap.

Additional thrust is provided by the ilium and urostyle.

Observe the mounted developmental stages of a frog. Note the following landmarks

Stage 1 (6 days) - On hatching the tadpole has an ovoid head-and-body, a slender compressed tail, horny jaws, long coiled intestines and 3 pairs of external gills.

Stage 2 (9 days) - External gills replaced by internal gills

Stage 3 (12 days) - Operculum covers the gills.

Stage 4 (21/2 months) - Hind legs appear but the fore legs are hidden under the operculum.

Stage 5 (21/2- 3 months) - The fore limbs appear.

Stage 6 (3+ months) - Larva under go metamorphosis -lungs develop, mouth widens, tail is reabsorbed and intestines become shorter. In other words the larva is transformed into a toad.

Caecilians (Order: Gymnophiona)

Observe the specimen provided. Take care not to damage it. These animals are not very common so you should not touch or poke the animal with sharp objects.

Note the elongate body and the annular folds (making it appear like an earthworm). Note on the head the nostrils, vestigial eyes and a pair of tentacles in a pit between the nostril and the eye. Note the position of the cloaca; it is not terminal. Draw and label the external features of the caecilian.

PRACTICAL 6

Introduction



Aim to be familiar with various reptiles and their external features. You could do this practical on your own.

Objectives



By the end of this practical the student should be able to

1. List the characteristics of each order of reptiles
2. Distinguish lizards from snakes.

Lizards (Order Squamata)

Examine the specimen provided and note the following: Body is divided into head, short neck, trunk and tail. Body covered with horny scales.

Mouth (terminal), nostrils and eyes with eyelids (how many?)

Check behind the eyes for tympanic pit and tympanic membrane (precursor of external ears).

Note the two pairs of limbs are attached to the side of the body (thus they hardly lift the body off the ground). How many digits are on each limb?

Note the position of the cloaca between the hind limbs.

Study the diagram of the internal features.

Note: mouth has jaws each having small strong teeth. The mouth has a tongue which is forked at the end. Pharynx is short but the esophagus is long. Stomach is a large sac. Intestines are differentiated into duodenum, ilium and rectum. The ilium is coiled but is not long compared with that of Tilapia. Between the ilium and rectum is a short caecum. Liver, gallbladder, pancreas and spleen are all present (note their position).

The respiratory system consists of larynx, which opens to the pharynx by the glottis. Trachea is long and supported by incomplete cartilaginous rings. Trachea divides to two bronchi that lead to the lungs. The two kidneys are very posterior in position and extend into the tail region posterior to the cloaca. A single ureter extends from each kidney and opens into the cloaca. The urinary bladder is absent in many species but where present it is a ventral sac.

Male organs are two testes in the middle of the body cavity far from the kidneys. From each testis arises a number of vasa afferentia which form an epididymis. This is small and gives off a vas deference that opens to the cloaca.

Female organs are two ovaries larger than the testes. Opposite each ovary an oviduct opens into the caelom and extends backwards as a fold tube into the cloaca.

Snakes (Order Squamata)

Observe the specimen provided. Does it have eyelids? Tympanic membrane?

In life specimen (which we do not have), a most notable feature is the forked tongue often seen darting from the mouth (it is an organ of smell and touch). Note the absence of limbs (pythons have rudiments of limbs). So snakes move by undulations of their long trunks and tail.

Examine the scales covering the body. Note any difference between dorsal and ventral scales.

Observe the skeleton provided. On the head note the backward slant of teeth. Specialized teeth or fang (what is its function?). The fang is fixed in cobras and it allies. In rattle snakes and other vipers, it is folded back when not in use.

Turtles (Order Chelonia)

Observe the specimen provided. Note the oval shell encasing the body (this a layer of plate-like bones in definite pattern over which there is a covering of cornified scutes. The dorsal convex portion is known as carapace and the flatter ventral part is plastron. The head and limbs protrude between the two parts of the shell. Observe the jaws and note that teeth are absent but in their place are cornified sheaths used to cut and crush food. There are horny claws on the toes. In the tortoise, the feet are stumpy but in marine turtles it is flipper-like-- for swimming.

Crocodile (Order Crocodalia)

Note the distinctive head, neck, mouth and tail. Also the short limbs with toes tipped with horny claws. Note the large lateral eyes and small nostrils near the tip of the snout. Ear openings are

situated behind the eyes. Note the large mouth with stout teeth and a flat tongue in the floor of the mouth cavity.

The body is covered by leathery skin covered with rectangular horny scales.

PRACTIAL 7

Introduction



Welcome to Practical seven. The aim of this practical is for you to see the morphological adaptations of birds for flight.

Objectives



By the end of this practical you should be able to

- Describe the different types of feathers
- Infer the diet of a bird from the shape of its beak
- Infer the habitat of a bird from the type of feet
- Outline how a birds skeleton is adapted for flight

Procedure

External characteristics

Observe the external features of a pigeon. Comment on its general body shape. Note the position of nostrils, the hidden ear, beak and its shape, outer covering of legs, position of toes on the feet.

Feathers

Observe the different type of feathers provided (contour, down, filoplume and bristle). Make drawing. Use a hand lens to observe the fine structures of the contour feather, draw what you see.

Observe the feathers of a sunbird or starling. What colour are they? How does this compare with the colours of the live bird?

Beaks

Observe the beaks of finch, hawk, fly catcher, Snipe, duck and woodpecker. Draw their beaks and explain how each reflects the bird's mode of feeding. Most of these birds can be found in most parts of the country and you can see the beaks if you use a pair of binoculars. Alternatively this can be observed on skins in any of the national museums and also in Zoology Department museum.

Legs

Examine the legs and feet of duck, finch, woodpecker, jacanas and flamingo. How does each reflect the habitat and/or feeding habits of the bird?

Skeleton

Observe the mounted skeleton of a bird. Note the structure of the fore limbs. In what ways are they adapted for flight? Note the breastbone. How is it different from that of mammals? What other features of the skeleton show that the animal is adapted for flight?